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Artificial Intelligence and Machine Learning for Official Statistics

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Workpackage 13

Proof of Concept (PoC) to generate synthetic data in official statistics, applying AI/ML techniques

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1 Introduction

The AIML4OS project brings together national statistical institutes and partner organisations to explore how artificial intelligence (AI) and machine learning (ML) can be used in a responsible and effective way in official statistics. Within this broader framework, Work Package 13 (WP13) focuses on the use of AI/ML techniques to generate synthetic data for statistical purposes. Synthetic data provide the chance to improve access to microdata, support research and innovation, enable software development and training, and allow wider data dissemination while still respecting confidentiality requirements.

The goal of WP13 is to examine how different methods for generating synthetic data can be applied in official statistics, and under what conditions they strike a good balance between analytical usefulness and the protection of sensitive information, the so-called utility/privacy trade-off. To do this, the work combines a review of existing methods with practical testing and comparisons across a range of national use cases. These use cases span different statistical domains, data types, purposes, and access settings, including datasets designed for code development, public use files, scientific use files, and methodological benchmarks.

So far, the work has included several activities. First, partners have exchanged knowledge on current approaches to synthetic data generation, as well as methods for assessing data utility and measuring privacy risks. This helped build a shared understanding of key concepts, available techniques, and the specific needs of official statistics. Second, the groundwork for the Proof-of-Concept (PoC) phase was laid by identifying suitable country use cases, defining the purpose of each synthetic dataset, selecting appropriate generation methods, and outlining how both utility and disclosure risk would be evaluated. Third, partners have developed and documented their PoCs, applying a range of methods such as sequential synthesis, Classification and Regression Trees (CART)-based approaches, R packages *simPop* and *synthpop*, deep learning methods such as Conditional Tabular Generative Adversarial Networks (CTGAN), Tabular Variational Autoencoders (TVAE), differential privacy-based approaches, and other statistical or machine learning techniques. Different evaluation strategies have also been used, including distributional comparisons, propensity score measures, and privacy risk indicators such as replicated uniques, Targeted Correct Attribution Probability (TCAP), Distance-to-Closest-Record (DCR), Distance-based Synthetic-data Closeness (DiSCO), membership inference attacks, and Risk Assessment of Privacy Inference Disclosure (RAPID).

WP13 is linked to other work packages within AIML4OS. The work on synthetic data contributes to the broader methodological discussion on AI/ML in official statistics and provides practical insights into quality, privacy, feasibility, and operational considerations.

This document presents a first overview of the work carried out in WP13. It outlines the main purposes and methods of synthetic data generation, discusses approaches to evaluating utility and privacy risks, and describes the country use cases developed so far. It should be seen as an initial version of Deliverable 13.2. The final version, planned for March 2027, will provide a more comprehensive summary of the PoC results, deeper comparisons across countries, further methodological insights, and more detailed conclusions and recommendations.

2 Overview of Synthetic Data Purposes and Methods

2.1 Purpose of Synthetic Dataset

Synthetic datasets play an important role in various domains, providing a privacy-protecting alternative to real data while retaining essential structural characteristics for research, development, and analysis. Their purpose depends on the specific use case, striking a balance of utility and confidentiality. From enabling software development and testing to facilitating public data sharing and scientific research, synthetic data can fulfil multiple functions, each with distinct requirements in terms of statistical fidelity, privacy protection, and accessibility.

Below, we outline the key purposes and types of synthetic datasets, highlighting their characteristics and intended applications. In categories a. through e. the requirements for and therefore the privacy risks associated with synthetic datasets is gradually increasing.

a. ***Structural dataset:*** a dataset containing the same variables, categories and univariate distributions as the original data but is not intended to produce statistics. This dataset can be used for developing and testing software tools, data-processing workflows, and statistical methodologies, as well as a preparatory file before access to the original microdata.

   Primary Use: development and testing of software tools and statistical methodologies, both internally and externally, and preparatory work for researchers. For example, researchers and developers can use the structural dataset to develop and test code, analytical workflows, or statistical methods that will later be applied to the original data. Such a synthetic dataset can also help researchers define their projects or prepare their methods before requesting secure access to the real data.

   Detail Level: variables mirror the original dataset, but the amount of meaningful information is minimal.

   Risk/Utility Trade-Off: basically, zero risk, but also very limited utility.

b. ***Public use file (PUF):*** a dataset that can be made freely available, e.g. on a public web site. It can be used for educational / training purposes.

   Primary Use: teaching, demonstrations of new methodology and preliminary data exploration by a wide audience.

   Detail Level: higher aggregation than original but reflecting core statistical characteristics. The possibility for advanced analysis is limited, and multivariate structure is accounted for.

   Risk/Utility Trade-Off: lower risk of disclosure due to less detail, but still useful enough for general tasks.

c. ***Scientific use file (SUF):*** a dataset than can be used by registered researchers to gain insights. The access is not open, and researcher agree to certain condition, e.g., to not use additional datasets to identify individuals***.***

Primary Use: research and in-depth analysis aimed at discovering insights. Another use case might be hackathons, datathons or data challenges where participants also agree to certain rules how to handle the dataset.

Detail Level: closer to the original data in terms of statistical fidelity, especially for multivariate structures. Multivariate structure should be well reflected as well.

Risk/Utility Trade-Off: higher risk than a PUF because of richer detail. Balances the need for meaningful research outcomes with privacy safeguards.

d. ***Special purpose scientific use file:*** a SUF variant tailored to be focused on well-defined known research questions. E.g. to study the relationship between specific variables or matching with another dataset.

   Primary Use: targeted research where the utility can be enhanced because the variables and relationships of interest are known in advance.

   Detail Level: similar to an SUF but optimized for specific analyses, potentially allowing richer detail where needed and omitting detail where possible.

   Risk/Utility Trade-Off: similar to SUF.

e. ***Perfect synthetic data / digital twin for safe microdata sharing:*** a synthetic dataset that is almost a "twin" of the original dataset but reflecting expectations that synthetic data are inherently non-personal/non-confidential. In this case, there is no need to predefine a research area as all properties of original data are replicated in the synthetic data and no details are removed.

   Primary Use: potentially for any purpose where near-complete fidelity is required, from general analysis to advanced machine learning.

   Detail Level: full coverage of variables and relationships, enabling most analyses (including ML training) to yield results essentially identical to those on real data.

   Risk/Utility Trade-Off: the disclosure risk of such datasets must be carefully assessed, especially when advanced machine learning methods are used, as these may inadvertently reveal information about the training data in some situations.

2.2 Classification of Methods to Generate Synthetic Dataset

This section categorizes approaches to synthetic data generation intro two groups that can be distinguish, **deep-learning-based methods** and **all others**. This distinction is useful, since using deep learning method with its high number of parameters increases the risk of memorization drastically (Belkin et al. (2019); Ricciato (2026)).

2.2.1 Deep-Learning-Based Methods

Deep learning approaches are used in several country use cases to capture complex, high-dimensional relationships in the data. These methods typically rely on neural network architectures and latent representations, enabling flexible modelling of nonlinear dependencies and mixed data types.

Generative Adversarial Networks (GANs)

- Conditional Tabular GAN (CTGAN): specifically designed for tabular data with mixed variable types, they incorporate conditioning mechanisms for categorical variables;
- Recurrent GAN (RGAN): mentioned as a potential approach for structured or time-series data.

Variational Autoencoders (VAEs)

- Tabular VAE (TVAE): learns probabilistic latent representations to generate new samples.

Hybrid / ML-enhanced frameworks

- Deep-learning models combined with privacy-preserving techniques. Example: use of TensorFlow Privacy for differentially private training.

2.2.2 Non-Deep-Learning-Based Methods

Non-deep-learning methods are widely used across the country use cases and remain the dominant approach in production-oriented statistical environments. These methods rely on statistical modelling, classical machine learning, and rule-based approaches.

Sequential and Regression-Based Approaches

These methods generate variables sequentially, modelling each variable conditional on those previously generated.

- Fully Conditional Specification (FCS) / Sequential synthesis;
- CART (Classification and Regression Trees);
- Regression-based models.

Examples of software implementations include R packages *synthpop* and *simPop*, which support several of these synthesis models.

Statistical Simulation Methods

- Copula-based models;
- Empirical distribution-based synthesis.

Classical Machine Learning Methods (non-deep)

- Random Forest (RF)-based approaches (e.g., with feature selection with the Boruta algorithm);
- Extreme Gradient Boosting (XGBoost).

Rule-Based and Constraint-Based Methods

- Logical constraints / rule-based synthesis;
- Rejection sampling.

Partial Synthesis Approaches

- Approaches in which only selected variables, typically sensitive variables, are synthesised while the remaining variables are retained from the original dataset.

2.3 Methods for Utility Measures Assessment

Measuring the utility of synthetic data is essential to assess whether the data are suitable for their intended purpose. Following Drechsler and Haensch (2024), utility measures can broadly be distinguished into fit-for-purpose, global, and outcome-specific measures. The country use cases considered in WP13 show, however, that this classification needs to be interpreted broadly, since utility may refer not only to statistical similarity but also to structural validity, suitability for code development, preservation of aggregated outputs, or performance in specific analytical tasks.

The following categories cover the utility assessment approaches used across the WP13 country use cases:

2.3.1.1 Structural and fit-for-purpose utility

These assessments verify whether the synthetic dataset satisfies the basic structural requirements associated with its intended use. Depending on the purpose of the dataset, this can include:

- preservation of variable names, types, formats, value domains, categories and coding schemes;
- presence of all categories required for developing and testing software or analytical code;
- validation against editing rules, logical constraints and hierarchical relationships;
- checks of household or other relational structures;
- basic plausibility checks and descriptive statistics such as means, standard deviations, ranges and frequencies.

This type of assessment is particularly important for structural datasets, where analytical similarity to the original data may be less important than ensuring that software and code developed using the synthetic data can subsequently be applied to the original data. The Portuguese use case, for example, explicitly evaluates whether variables, formats, types, coding schemes and all relevant categories are retained for code-development purposes.

2.3.1.2 Distributional utility

Distributional utility evaluates how well the statistical distributions of the original data are reproduced. Assessments may be performed at the univariate, bivariate or multivariate level and include:

- graphical comparisons such as histograms, boxplots, empirical or cumulative distribution functions;
- comparison of means, variances, proportions and other descriptive statistics;
- overlap measures for categorical distributions and numerical ranges;
- distance measures such as Hellinger or Bhattacharyya distance;
- comparison of correlation coefficients or complete correlation matrices;
- dimension-reduction approaches such as Principal Component Analysis (PCA) or Factor Analysis of Mixed Data (FAMD) to compare multivariate structures.

These assessments range from simple sanity checks to more formal measures of distributional similarity.

2.3.1.3 Global utility

Global utility measures summarise differences between the joint distributions of the original and synthetic datasets. A commonly used approach in the WP13 use cases is propensity-score-based

assessment. Original and synthetic records are combined and a classifier is trained to distinguish between them. If the two datasets are similar, classification should be difficult.

Measures include:

- propensity score mean squared error (pMSE);
- standardized pMSE or pMSE ratio, which account for the expected value of pMSE and, depending on the implementation, model complexity.

The pMSE and related measures are used in several country studies, including France, Germany and Spain. Germany additionally applies the pMSE at microdata level while explicitly distinguishing this from assessments of aggregated outputs.

2.3.1.4 Information-loss measures

Utility can also be expressed in terms of the amount of information lost through synthesis. Such measures quantify changes in individual variables or relationships between variables. Examples used in the country studies include:

- Information loss per variable (IL_variables), which measures differences between original and synthetic observations using distance functions adapted to nominal, ordinal and continuous variables;
- Information loss in the correlation structure (IL_correl), which quantifies changes in the correlation structure of continuous variables

These measures complement distributional and global utility measures by expressing the distortion introduced through synthesis directly as information loss. They are used in the Polish use case.

2.3.1.5 Outcome-specific and task-based utility

Outcome-specific utility assesses whether synthetic data support the same substantive or analytical conclusions as the original data. The relevant assessment depends on the intended analytical use and may include:

- comparison of regression coefficients, signs, significance levels and confidence intervals;
- Confidence Interval Overlap (CIO);
- comparison of estimates or indicators such as employment rates, unemployment rates or income statistics;
- comparison of predictions or classification performance when models are trained on synthetic rather than original data.

For predictive tasks, models can be trained on synthetic data and evaluated on real test data, with their performance compared with models trained on real data. Metrics used in the country studies include classification accuracy, precision/recall and F1-score. The Italian use cases, for example, use classification models specifically as a task-based measure of whether statistically relevant relationships have been preserved.

2.3.1.6 Aggregate- and table-level utility

Where the intended statistical outputs are aggregated tables rather than microdata analyses, utility should also be assessed directly at the level of those outputs. This can include:

- comparison of contingency-table cell counts or proportions;
- distributional distances between original and synthetic tables;
- comparisons across different table dimensions and levels of geographical aggregation.

The German use case, for example, computes Hellinger distances between normalized original and synthetic contingency tables and explicitly distinguishes these table-level measures from microdata-

level utility measures. This distinction is particularly relevant in official statistics, where dissemination frequently takes the form of aggregated tables rather than microdata.

2.3.1.7 Structural consistency and rule-based utility

For datasets containing logical, hierarchical or relational structures, utility also depends on whether these structures remain internally consistent. This can be assessed through:

- numbers or proportions of records violating predefined editing rules;
- preservation of hierarchical or nested variables;
- consistency of household structures;
- preservation of survey filter logic and structural missingness.

For example, the Austrian use case derives a quality measure from the proportion of synthetic records violating predefined editing rules, while the French use case assesses logical filters, nested variables and survey weights.

This revised taxonomy is not intended to imply that every synthetic dataset should be assessed using all categories. Rather, the choice of utility measures should be driven by the intended purpose of the synthetic data. Structural datasets may primarily require structural and fit-for-purpose assessments, whereas PUFs and SUFs generally require stronger distributional, global and outcome-specific evaluation.

2.4 Privacy Risk Assessment

2.4.1 Metrics for Privacy Risk Assessment

Assessing disclosure risk in synthetic data is inherently more complex than evaluating data utility. While utility can be measured by comparing statistical properties between original and synthetic datasets, privacy risk requires the explicit definition of intruder models, attack scenarios, and assumptions about the information available to a potential adversary. In the context of official statistics, privacy risk assessment aims to evaluate whether the release of synthetic microdata may enable the re-identification of individuals or the inference of sensitive attributes. This requires considering both the data generation process and the intended data release setting (e.g., fully or partially synthetic data, microdata vs aggregated outputs).

Among the main attack scenarios considered in synthetic data disclosure risk assessment are attribute inference and membership inference.

Attribute inference occurs when an intruder uses the synthetic data, possibly together with external information already known about a person, to infer a sensitive attribute of that person. For example, if the intruder knows some non-sensitive characteristics of an individual, such as age group, area of residence, education level or employment status, the synthetic data may reveal that individuals with a similar profile are strongly associated with a specific income class, health condition or other sensitive attribute. This is problematic when the synthetic data preserve very specific relationships from the original data, because they may allow the intruder to draw conclusions about real individuals even though the released records are synthetic.

In a membership inference attack, the intruder tries to determine whether a specific individual was included in the original dataset used to train or generate the synthetic data. This can also be sensitive, for example if the mere presence in the source dataset reveals information about the individual, such as being part of a survey on a specific health, social, or economic condition. Testing synthetic data

against these attack scenarios is therefore useful to assess whether, and to what extent, the generation process protects individuals from the disclosure of sensitive information.

In what follows, we first summarise the methods that have been explicitly investigated within the country use cases developed in WP13. We then describe a broader set of disclosure risk metrics drawn from the literature, ensuring that all relevant methodological approaches are consistently documented and linked to existing research.

2.4.2 Methods Investigated in the Country Use Cases

The following privacy risk assessment methods have been explicitly investigated within the WP13 country use cases (see chapter below "Country use cases"). Each method is briefly described and supported by references from the synthetic data literature.

These methods should be interpreted as risk-detection tools: they can provide evidence of the presence of disclosure risk when problematic patterns are observed, but a low value, or even the absence of detected risk under a given metric or attack model, should not be interpreted as proof that the synthetic dataset is risk-free.

Replicated Uniques (repU)

Replicated uniques measure the proportion of records that are unique in both the original and synthetic datasets. This metric captures identity disclosure risk by identifying cases where rare or unique combinations of quasi-identifiers are reproduced in the synthetic data. Such replication increases the likelihood that an attacker could isolate and potentially re-identify individuals. The measure is particularly relevant for datasets with high dimensionality or sparse attribute combinations, where uniqueness is more common (Nowok et al. (2016); Raab et al. (2024, 2026)).

Disclosive in Synthetic, Correct in Original (DiSCO)

DiSCO evaluates attribute disclosure risk by identifying cases in which a sensitive attribute can be uniquely inferred within the synthetic dataset and is also correctly matched in the original data. It focuses on high-confidence inference scenarios, where the synthetic data provide unambiguous signals about sensitive attributes. As such, it directly operationalises attribute inference risk under a deterministic attacker model (Raab et al. (2024, 2026)).

Disclosive in Synthetic and Disclosive in Original (DiSDiO)

DiSDiO extends DiSCO by restricting the analysis to records that are already disclosive in the original dataset. This allows separating disclosure risks that are intrinsic to the original data structure from those introduced by the synthesis process. It is particularly useful for distinguishing whether synthetic data amplify existing risks or introduce new ones (Raab et al. (2024, 2026)).

Within Equivalence Class Attribution Probability (WEAP)

WEAP measures the probability that a sensitive attribute can be inferred within an equivalence class defined by quasi-identifiers in the synthetic data. It represents the attacker's confidence at the synthetic data level, before validation against the original data. WEAP is typically used as a preliminary step in TCAP-based frameworks to identify high-risk records where inference is deterministic (Taub et al. (2018); Pilgram et al. (2025)).

Targeted Correct Attribution Probability (TCAP)

TCAP measures the probability that an attacker can correctly infer a sensitive attribute for a target individual, given knowledge of quasi-identifiers and focusing on high-risk cases where the synthetic data provide a unique prediction (i.e., WEAP = 1). The metric combines identification of high-confidence inference scenarios with validation against the original data, making it a key measure of realistic disclosure risk under targeted attack assumptions (Taub et al. (2018); Pilgram et al. (2025)).

Distance to Closest Record (DCR)

DCR evaluates disclosure risk by computing the distance between each synthetic record and its nearest neighbour in the original dataset. Small distances indicate high similarity, which may result from memorization or overfitting in the generative model. This metric provides a continuous measure of disclosure risk and is particularly useful for detecting local copying behaviour in high-dimensional data (Snoke et al. (2018); Yao et al. (2025)).

Risk of Attribute Prediction Induced Disclosure (RAPID)

RAPID evaluates disclosure risk by measuring how accurately sensitive attributes can be predicted using models trained on synthetic data. It directly links privacy risk to predictive performance, aligning the assessment with real-world analytical use cases. This makes it particularly relevant for evaluating risks in machine learning applications (Templ et al. (2026)).

2.4.3 Additional Privacy Risk Assessment Methods

The following methods are part of the broader synthetic data privacy risk assessment literature but were not explicitly implemented in the WP13 country use cases. They provide complementary perspectives, particularly in probabilistic linkage and attribution risk modelling.

Expected Match Risk (EMR)

EMR estimates the expected probability that an intruder can correctly match records between synthetic and original datasets, considering uncertainty about whether individuals are present in the dataset. It reflects the average success probability of a linkage attack under realistic assumptions about incomplete information (Reiter (2005); Drechsler (2011); Pilgram et al. (2025)).

True Match Rate (TMR)

TMR measures the proportion of correct matches among all true matches, reflecting how effectively an attacker can identify correct links between datasets. It is commonly used alongside EMR and the False Match Rate (FMR) to evaluate the performance of linkage attacks (Reiter (2005); Drechsler (2011)).

Differential Correct Attribution Probability (DCAP)

DCAP extends attribution-based risk measures by incorporating different assumptions about intruder knowledge and uncertainty regarding record inclusion. It provides a more flexible and realistic evaluation of disclosure risk across multiple attack scenarios (Drechsler & Reiter (2010); Pilgram et al. (2025)).

2.4.4 Attack-based Privacy Assurance Framework for Synthetic Microdata in Official Statistics

2.4.4.1 Introduction

This specifies a practical, attack-based framework to obtain high assurance that released synthetic microdata has low disclosure risk in the context of official statistics.

Fully synthetic data often breaks one-to-one correspondence between original and synthetic rows, so classical record linkage may be less informative by itself; however, **attribute inference** and **membership disclosure** remain central risks, particularly when the synthetic data preserves strong predictive relationships or when the generator overfits.

The release policy is a static release, so only one synthetic dataset is released. This limits the knowledge on the generator and overfitting.

However, disclosure risk also depends on several additional factors, including the size of the released dataset, the extent of information made available on the generation mechanism, and whether synthetic data are released periodically over time. Repeated releases, or releases that are strongly linked to the same underlying population, may create additional opportunities for linkage, comparison, or reconstruction attacks.

It is also important to stress that synthetic data can be exposed to different types of privacy attacks. Showing that a synthetic dataset is robust with respect to one or more attacks does not, by itself, guarantee that it is also robust against other attacks that have not been considered. For this reason, the framework is organised around a set of complementary attack families, each capturing a different disclosure scenario.

The framework therefore clusters attacks into five families:

1. Linkage / identity attacks;
2. attribute inference attacks;
3. membership inference attacks;
4. memorization / copying attacks;
5. reconstruction attacks.

Linkage / identify attacks

The attacker aims to isolate a unique individual (singling out) and/or correctly link records across datasets (linkability). These attacks attempt to link synthetic records to original individuals by exploiting similarity in feature space, typically using distance-based or nearest-neighbour matching rules. Their effectiveness depends on how closely synthetic data reproduce local structures in the original dataset. They provide a realistic approximation of re-identification risk in practical settings (Snoke et al. (2018); Giomi et al. (2022); Domingo-Ferrer & Torra (2004)). **Measures:** repU (replicated uniques), distance/nearest-neighbor proxies (DCR / Nearest-Neighbour Distance Ratio (NNDR) / Identical Match Share (IMS)).

Attribute Inference Attacks

The attacker aims to infer a sensitive attribute given quasi-identifiers and the released synthetic data. Attribute inference attacks simulate adversarial scenarios in which an attacker, given partial

information (quasi-identifiers), attempts to infer unknown sensitive attributes using the synthetic data or models trained on them. These attacks directly assess the predictive power of synthetic data with respect to sensitive variables and are central to evaluating inferential disclosure risk (Domingo-Ferrer & Torra (2003); El Emam et al. (2020); Slokom et al. (2023)). **Measures:** DiSCO/DiSDiO, DCAP/TCAP/WEAP, and **RAPID** (categorical + continuous), which explicitly measures record-level inferential disclosure under attacker models trained only on the released synthetic data.

Membership Inference Attacks (MIA)

The attacker aims to infer whether a target individual's record was included in the training data used for generation. Membership inference attacks aim to determine whether a specific individual was included in the training dataset used to generate synthetic data. These attacks typically exploit differences in model behaviour between training and non-training samples, such as overfitting or confidence gaps. They are especially relevant for deep learning-based generative models, where memorization risks are higher (Shokri et al. (2017); Stadler et al. (2021)). **Measures:** MIA performance metrics, such as the area under the receiver operating characteristic curve (AUC) and the true positive rate (TPR) at a specified false positive rate (FPR) using modern no-/black-box MIAs such as **DOMIAS** (a density-based membership inference attack based on overfitting detection) and **TAPAS** (Toolbox for Adversarial Privacy Auditing of Synthetic Data) style threat-modelled attacks, and (where relevant) multi-table MIAs.

Memorization / copying attacks

The attacker aims to detect whether the generator effectively copied training data locally, through exact matches or near-copies, and to quantify the privacy risk induced by copying. These approaches evaluate whether generative models have learned general patterns from the training data or have instead memorized specific training examples. Memorization can lead to privacy risks when synthetic data reproduce or closely approximate original records. Assessment methods include analysing discrepancies between training and test behaviour, evaluating model generalization, and performing targeted data extraction attacks. High levels of memorization are associated with increased disclosure risk, particularly in high-capacity generative models (Carlini et al. (2023); Steier et al. (2025)). **Measures: Data Plagiarism Index (DPI)** and DPI-based MIA, plus IMS as a simple leakage detector.

Reconstruction attacks

The attacker aims to reconstruct sensitive information, a column, or even parts of the original dataset through aggregate or statistical reconstruction methods, possibly without direct copies. These attacks extend classical attribute inference by exploiting learned representations, model outputs, or feature importance structures derived from synthetic data or trained models. Rather than relying solely on quasi-identifiers, they leverage complex patterns captured by machine learning models to infer sensitive information. These approaches are particularly relevant in modern data synthesis pipelines based on high-capacity models and represent an emerging direction in privacy risk assessment for synthetic data (Stadler et al. (2021); Pilgram et al. (2025)).

2.4.5 Framework

The detailed framework will be released in March 2027 as part of the deliverable D13.2.

3 Country Use Cases

The country use cases developed within WP13 provide a diverse set of practical implementations of synthetic data generation in official statistics. They cover a wide range of statistical domains, data sources, and methodological approaches, reflecting both the heterogeneity of national contexts and the different objectives associated with synthetic data production.

Across the use cases, synthetic data are generated for multiple purposes, including the creation of structural datasets, the development of scientific use files (SUF), and the production of public use files (PUF). These different objectives imply varying requirements in terms of statistical fidelity, level of detail, and acceptable disclosure risk, and therefore lead to the adoption of different synthesis and evaluation strategies.

The purpose of generating synthetic data varies across these datasets, including creating structural datasets**,** enabling scientific use files, and producing public use files**,** such as those available on European Union Statistics on Income and Living Conditions (EU-SILC) and the Labour Force Survey (LFS). The *Human Capital Balance* dataset originates from a survey by the Polish Agency for Enterprise Development and is also intended for public use.

Several use cases compare or combine deep learning and traditional approaches (e.g., France, Italy), but sequential synthesis remains a benchmark. CART-based methods (e.g., synthpop) are widely used as reference approaches.

Choice of method depends on use case

- Public Use Files (PUF): often deep learning or hybrid methods to capture structure;
- Structural datasets: simpler statistical methods;
- Partial synthesis: rule-based and CART methods.

Trade-offs:

- Deep learning → higher flexibility, higher risk;
- Non-deep learning → higher interpretability, easier deployment.

Germany and Austria strongly rely on **CART-based sequential synthesis**, emphasizing:

- Transparency;
- Interpretability;
- Operational feasibility.

Synthpop and *simPop* is used in multiple countries for **population-level synthesis**, especially in census-like contexts. These methods are preferred when:

- Interpretability is required;
- Production stability is important;
- Computational resources are limited.

Table 1, below, provides a comparative overview of the country use cases, summarising the statistical domains, the main data characteristics, the generation methods adopted, and the approaches used for utility and privacy risk assessment.

Table 1 - Overview of the country use cases.

FRANCE		**Domain** Social statistics (Labour Force Survey)	**Purpose** Public use file (PUF) for pedagogy and reproducibility
Non-Deep Learning methods	Sequential and regression-based synthesis (CART)	**Utility Assessment** Global utility (pMSE), Distributional utility (univariate & bivariate comparisons, PCA/FAMD), Outcome-specific and task-based utility (CIO)	**Privacy Risk Assessment** repU, DiSCO, TCAP
Deep-Learning Methods	CTGAN, TVAE		

GERMANY		**Domain** Statistics of road traffic accidents (police-based microdata)	**Purpose** PUF / special purpose SUF (safe tables)
Non-Deep Learning methods	Partial synthesis using sequential CART models (synthpop)	**Utility Assessment** Global utility (pMSE, pMSE ratio), Aggregate- and table-level utility (Hellinger distance (tables)), Distributional utility (correlation analysis)	**Privacy Risk Assessment** TCAP, WEAP, group disclosure validation
Deep-Learning Methods			

PORTUGAL		**Domain** Business statistics (Structural Business Survey / Integrated Business Accounts System (SCIE))	**Purpose** Structural dataset for code development (safe environment preparation)
Non-Deep Learning methods	Sequential and regression-based synthesis (synthpop), Empirical distribution-based synthesis (synthesizer)	**Utility Assessment** Structural and fit-for-purpose utility (preservation of variable categories and value domains)	**Privacy Risk Assessment** repU, DiSCO, DiSDiO, TCAP
Deep-Learning Methods			

SPAIN		**Domain** Social statistics (synthetic census based on Encuesta de Condiciones de Vida (ECV) Census, Atlas data)	**Purpose** Public use file (PUF) for income and demographic analysis
Non-Deep Learning methods	Sequential and regression-based synthesis (simPop) + calibration	**Utility Assessment** Global utility (pMSE), Distributional utility (correlation matrices, distributional comparisons (income by age/sex))	**Privacy Risk Assessment** Qualitative assessment (design-based, low-risk assumption)
Deep-Learning Methods			

POLAND		**Domain** Social statistics (Human Capital Balance survey and simulated data)	**Purpose** Public use file / methodological experimentation
Non-Deep Learning methods	Sequential and regression-based synthesis (synthpop), DP-based synthesis	**Utility Assessment** Global utility (Standardized ratio propensity mean squared error (S_pMSE)), Information loss measures, Outcome-specific and task-based utility (regression-based comparisons)	**Privacy Risk Assessment** No synthetic-specific metrics; focus on Statistical Disclosure Control (SDC) / Differential Privacy (DP)-based reasoning
Deep-Learning Methods			

ITALY - USE CASE 1		**Domain** Social statistics (survey microdata)	**Purpose** Public use file (PUF) for analytical use
Non-Deep Learning methods		**Utility Assessment** Outcome-specific and task-based utility (classification-based accuracy comparison (real vs synthetic))	**Privacy Risk Assessment** DCR, Membership Inference Attacks
Deep-Learning Methods	CTGAN, TVAE		

ITALY - USE CASE 2		Domain Synthetic benchmark ("Toy world" controlled setting)	Purpose Methodological benchmark for utility–privacy trade-off
Non-Deep Learning methods	Incremental XGBoost, Empirical distribution-based synthesis (synthesizer)	**Utility Assessment** Outcome-specific and task-based utility (classification accuracy comparison across scenarios)	**Privacy Risk Assessment** Attack-based framework (attribute inference, memorization)
Deep-Learning Methods	CTGAN, TVAE		

AUSTRIA		Domain Social statistics (census-like microdata)	Purpose Scientific use file (SUF)
Non-Deep Learning methods	Sequential and regression-based synthesis (simPop, synthpop)	**Utility Assessment** Distributional utility (graphical comparisons, Hellinger distances), Structural consistency and rule-based utility (proportions of records violating predefined editing rules)	**Privacy Risk Assessment** RAPID
Deep-Learning Methods	Tabular Auto-Regressive Generative Network (TabularARGN)		

France

3.1.1 Statistical Domain and Input Data

Access to microdata is almost exclusively restricted to a public of sworn experts, mainly researchers from public institutions. Publishing such files, which contain several hundred variables and therefore detailed characteristics about responding units, for the public is extremely challenging. However, making available individual-level data restricted to the most characteristic variables of the dataset would allow users to reproduce certain aggregates from official statistics or enable students to learn how to apply the models taught in class.

In this work, we propose to reproduce an extract of the French sample of the Labour Force Survey. The objective is not to provide the more than eight hundred variables present in the production dataset, but rather to generate a replica of the sample using a few dozen variables. The quarterly sample includes approximately 90,000 respondents, and the variables are almost exclusively categorical variables such as:

- Survey variables of interest: employment status (ACTEU), hours worked, type of job search, etc.;
- Socio-demographic characteristics of respondents: sex, age, education level, location (region);
- Household characteristics: household identifier to link individuals, family relationships, etc.;
- Weighting variable: the calibrated sampling weight allowing users to reproduce official aggregates.

3.1.2 Purpose of the Synthetic Data

The objective is to produce a synthetic dataset in the form of a public use file (PUF) made available to the public. Such a dataset can serve several purposes:

Pedagogy: providing students with data enabling them to conduct realistic socio-economic analyses using models learned in class.

Reproducibility: allowing users to reproduce some key statistics produced by official statistics.

The promise of open data: opening data has become a major requirement for official statistics whenever possible. Providing synthetic data is one way to respond to this societal demand.

Naturally, the anonymity of survey respondents must not be compromised by this dissemination, and the quality of the released data must be sufficient to justify a dedicated publication.

3.1.3 Methods for Synthetic Data Generation

The generation of synthetic data relies on learning the underlying probability distribution of the original data. Three major families of generative models are compared in this work:

- sequential approach based on Fully Conditional Specification, relying on Classification and Regression Trees (CART);
- Generative Adversarial Network adapted to tabular data (CTGAN);
- Variational Autoencoder adapted to tabular data (TVAE).

3.1.3.1 Sequential approach

The Fully Conditional Specification (FCS) method, popularized by Van Buuren et al. (2006) for multiple imputation, simplifies the problem of estimating a high-dimensional multivariate joint distribution. Rather than directly modelling the joint distribution, sequential synthesis decomposes it into a series of univariate conditional distributions. For a vector of variables, the joint density is approximated by the product of conditional densities. The generation process is sequential: the first variable is generated independently, then the second one is generated conditionally on the first one, and so on. The order of the variables (visit sequence) is crucial for reconstructing correlation structures.

To estimate each conditional density, we use the CART algorithm proposed by Breiman et al. (1984).

Since the visit order of variables is important in this approach, a score based on each variable's contribution to overall variance was used to construct an ordering that significantly improves the utility of the generated data. The score is computed for each variable as a combination of its contribution to the different dimensions and the percentage of variance explained by those dimensions in a factorial analysis.

3.1.3.2 CTGAN

Introduced by Goodfellow et al. (2014), Generative Adversarial Networks (GANs) rely on a zero-sum game between a generator, which produces synthetic data from latent noise, and a discriminator, which attempts to distinguish generated data from real data. Xu et al. (2019) adapt this architecture to tabular data, overcoming typical GAN limitations for such datasets, particularly due to multimodality and mixed variable types.

3.1.3.3 TVAE

Proposed by Kingma and Welling (2022), Variational Autoencoders (VAE) approach generation from a probabilistic perspective. Unlike classical autoencoders that learn a deterministic compressed representation, VAEs learn the parameters of a probability distribution in the latent space. The model seeks to maximize the marginal likelihood of the data.

3.1.4 Methods for Utility Measurement

We distinguish between global utility measures, which evaluate the quality of the entire dataset, and specific measures, which target analyses.

3.1.4.1 Global propensity-based measures

These measures evaluate overall dataset utility by attempting to distinguish synthetic records from original ones. To compute the propensity score, we merge the original dataset and the synthetic dataset into a single dataset. A classification model is trained to predict the origine of each observation using the other variables. For each individual, the model produces a propensity score as the estimated probability that the observation belongs to the synthetic dataset. The pMSE (propensity score Mean Squared Error) introduced by Woo et al. (2009) is then computed

If the synthetic data closely resemble the original distribution, it should be difficult for the classifier to distinguish them, resulting in a low pMSE. Conversely, a high pMSE indicates poor similarity and therefore lower utility.

3.1.4.2 Specific measures

However, pMSE does not capture all aspects of utility, such as the ability of synthetic data to reproduce specific analyses.

3.1.4.3 Univariate analyses

A common approach consists of comparing marginal distributions, often visually through empirical distributions.

Several measures compare probability distributions, including Bhattacharyya and Hellinger distances. These distances lie between 0 and 1. Values close to 0 indicate strong agreement between synthetic and original data.

3.1.4.4 Bivariate analyses

Bivariate analysis studies the relationship between two variables. Depending on variable types:

- Two numerical variables: Pearson correlation coefficient;
- One numerical and one categorical variable: correlation ratio;
- Two categorical variables: Cramér's V.

3.1.4.5 Multivariate analyses

Multivariate analyses depend strongly on the analyst's objective. We compute two types of multivariate analysis: Comparison of projected individual clouds and Confidence Interval Overlap (CIO) of odds ratio computed from a logistic regression.

A Principal Component Analysis (PCA) or Factor Analysis of Mixed Data (FAMD) can be performed. Individuals are projected onto the first factorial plane, and the resulting cloud of points is compared with that obtained from the original dataset. The objective is for the shapes of the clouds to be similar and for the projected variables to appear in comparable locations.

Proposed by Karr et al. (2006), the CIO measure compares the overlap between confidence intervals obtained from regression models estimated on the original and synthetic datasets.

3.1.4.6 Reproduction of the intrinsic structure of the data

Data produced by official statistics are highly structured: variables have restricted value domains, and some variables are hierarchically nested. Moreover, survey questionnaires often include conditional questions, meaning that some questions are asked only depending on previous answers. Consequently, certain variables may legitimately contain missing values. For example, an employed individual will not answer a question about the type of job search conducted during the reference week. It is therefore important to assess to what extent generation methods can reproduce such logical structures automatically.

3.1.5 Methods for Privacy Assessment

3.1.5.1 Replicate Uniques

Identity disclosure occurs when an attacker is able to link a synthetic record to a unique individual in the original population. This happens when the data generation process has exactly reproduced individuals from the original dataset. To assess this risk, we rely on the rate of replicated uniques, using the repU metric implemented in the synthpop package (G. Raab, 2024). A high rate indicates that the model copied rare profiles, increasing the confidence of an attacker.

3.1.5.2 DiSCO indicator

The DiSCO (DISclosive in Synthetic and COrrect) metric measures the proportion of individuals for whom sensitive attribute inference is both unique in the synthetic data and correct in the original data. Deterministic relationships (logical rules) are excluded from the calculation.

3.1.5.3 Probability of correct attribution: TCAP

Proposed by Taub et al. (2018), the TCAP (Targeted Correct Attribution Probability) assesses the confidence an attacker could place in their inferences. The attack proceeds in two stages:

- identification of high-risk targets (WEAP): The attacker first computes the attribution probability within the equivalence class in the synthetic dataset. The attacker focuses only on records where the WEAP = 1, corresponding to cases where the information is "certain" in the synthetic dataset (l-diversity = 1);
- verification of the inference (TCAP): The TCAP then measures whether this synthetic certainty holds in the original data.

Interpretation:

- TCAP ≈ 1: attacker guessed correctly → critical risk;

- TCAP ≈ 0: attacker is misled → privacy preserved.

3.1.5.4 Attack scenarios

We also plan to test the robustness of synthetic data protection through alternative attack scenarios. We will test the Anonymeter tool proposed by Giomi et al. (2022), which performs three types of attacks: singling-out, linkability, inference. Additional scenarios proposed by project collaborators will also be tested.

Tools

Synthpop package

The synthpop R package (Nowok, Raab, Dibben, 2016) is a reference tool for synthetic data generation. It implements sequential synthesis using different models (CART, parametric models, random forests, bagging, etc.). It also allows control over visit sequence, prediction matrix, constraints on variables.

Synthetic Data Vault (SDV) library

The SDV project provides the SDMetrics library for evaluating synthetic data quality. The SDV library includes several generative models, including CTGAN and TVAE, which are used in this project. It provides numerous hyperparameters, training diagnostics, and built-in utility and risk metrics.

MLFlow

MLFlow is a platform designed to track machine learning experiments and model deployment. It centralizes metrics, results, and visualizations, making it particularly useful for comparing models or managing hyperparameter searches (e.g., grid search for CTGAN or TVAE).

TableEvaluator

TableEvaluator is a Python package designed for quickly performing visual comparisons between two datasets. It provides distribution comparisons, correlation analysis, PCA projections, metrics such as mean absolute error (MAE) and root mean squared error (RMSE). Although very convenient, the package is poorly maintained, which limits its use as a long-term solution.

3.1.6 Preliminary Results

3.1.6.1 Experimental setup

The experiment relies on the public-use individual file of the French Continuous Labour Force Survey (Enquête Emploi en Continu, EEC), first quarter of 2022 — about 89,970 individuals and more than 70 variables — chosen specifically to make the work reproducible. Three preprocessing decisions shaped the comparison. First, variables with more than 30 modalities were dropped to limit overfitting and computational cost in the tree-based generator, leaving 65 variables covering activity status, working hours, socio-demographic characteristics and household context. Second, structural missing values — the “not applicable” answers produced by the survey’s filter questions — were recoded explicitly as out-of-domain values (−8 for numeric variables, a “999” modality for categorical ones), because the deep-learning models do not handle missingness natively; this forces the generators to learn the association between a condition (e.g. being unemployed) and the recoded value. Third, only the aggregated levels of the French activity classification (Nomenclature d'activités française, NAF), using the aggregated NAF10 and NAF21 levels were retained, to test whether the models preserve parent–child consistency without the extreme cardinality of the finest level (600+ modalities).

Two tuning steps preceded generation. For the sequential CART method, the visit order of the variables was optimised: ordering them according to a factor-analysis indicator — built from each variable's contribution to the dimensions of a Factor Analysis of Mixed Data, weighted by the share of variance explained — improved the global utility score (pMSE) by roughly 40–68 % over the default order while also reducing runtime. For CTGAN and TVAE, a grid search over batch size, learning rates / network architecture and number of epochs was run to minimise the pMSE; the best configurations both used a batch size of 100 (CTGAN: generator and discriminator learning rates of 1×10^{-4}; TVAE: two layers of 256 units for both encoder and decoder).

Three generators were then compared: a sequential method based on classification and regression trees (CART, via the synthpop R package), a conditional tabular GAN (CTGAN) and a tabular variational autoencoder (TVAE), the latter two from the SDV library. All models were evaluated under the same risk–utility framework.

3.1.6.2 Statistical utility

Univariate fidelity. On marginal distributions, CART reproduces the original almost exactly (for instance the six-class age variable), whereas the neural models introduce visible distortions: CTGAN tends to "rejuvenate" the population (younger classes overrepresented, the 65+ class strongly underrepresented), while TVAE underestimates the youngest and overstates the intermediate classes. The relative ranking of the two neural models is not stable across variables — on usual weekly hours worked, TVAE is the weakest and CTGAN the more coherent of the two.

Bivariate fidelity. Comparing correlation matrices, the difference matrix for CART is almost entirely flat (errors close to zero), confirming that the sequential approach captures pairwise relationships well. The CTGAN and TVAE difference matrices are dominated by errors frequently reaching 0.3, and both fail to reproduce the dense correlation blocks and the cross-relations present in the original.

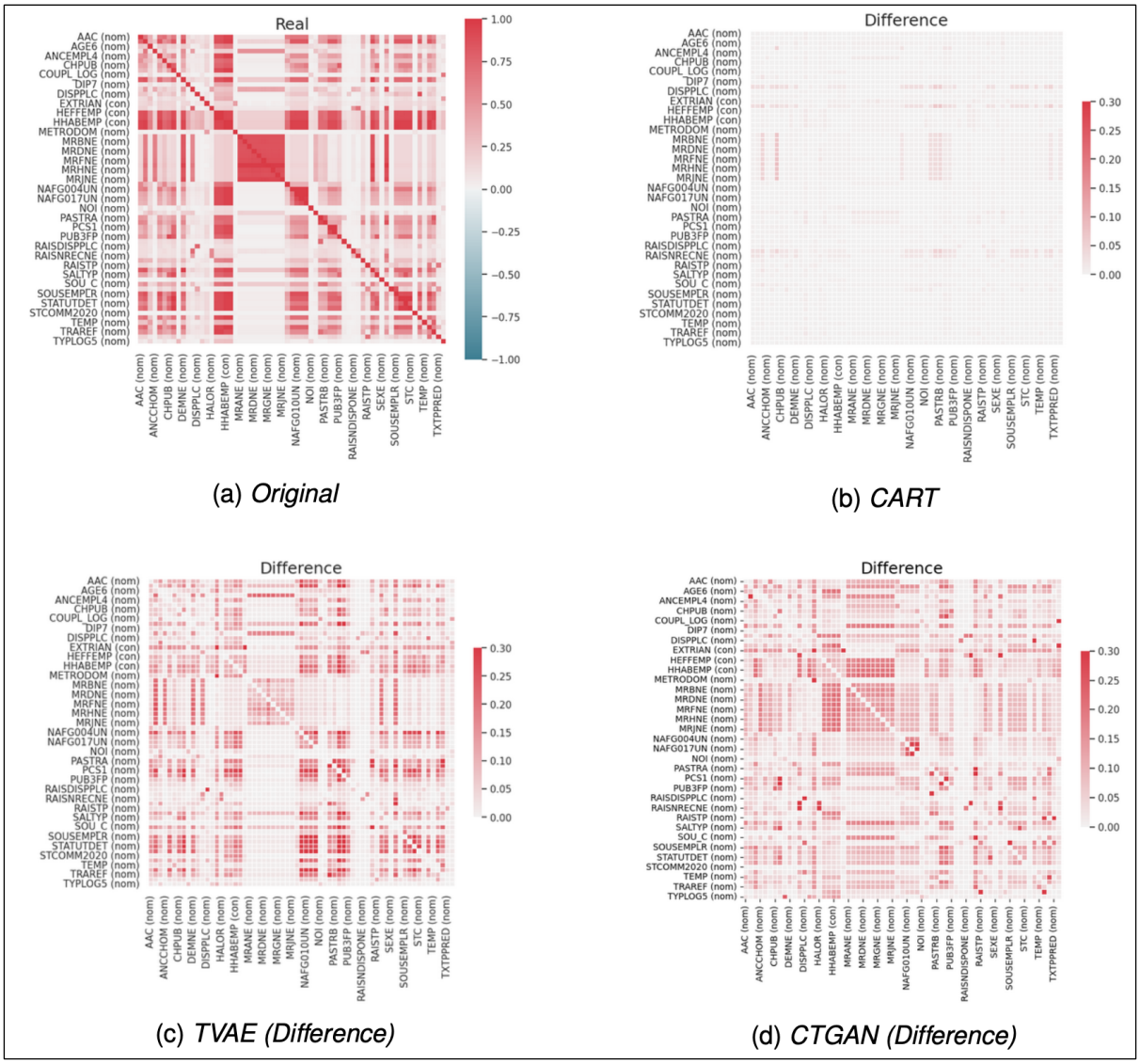


Figure 1: Original correlation matrix (a) and Correlation difference matrices for each model tested

Analytical validity. A logistic regression of unemployment on sex, age, diploma and location (metropolitan / overseas) provides the most decision-relevant test. The odds ratios and 95 % confidence intervals estimated on CART-synthesised data are close to those obtained on the original, with largely overlapping intervals: analytical conclusions are preserved. The neural models, by contrast, shift several coefficients substantially — the odds ratio for diploma level DIP76, for example, is estimated above 1.5 against roughly 1.1 in the original — and, more seriously, they alter statistical significance in both directions. The confidence-interval-overlap (CIO) measure confirms this: CART achieves high positive overlap for most diploma modalities (e.g. 0.932 for DIP76), whereas the neural models are almost systematically negative, with extreme cases such as −10.156 for TVAE on the same modality. Notably, no model — not even CART — achieves positive overlap on the age coefficients, which points to a systematic shift there, although CART remains the least biased.

Global utility and cost. Table 2 reports the overall propensity-score utility measure (pMSE) and the training time. CART obtains the best (lowest) pMSE and, although the gap with the deep-learning models is narrower than on other datasets, its computational advantage is decisive — about 38 minutes against nearly 8 hours for CTGAN. (All models were trained on a central processing unit (CPU) rather than a graphics processing unit (GPU)).

Table 2 - pMSE and training time (single generated dataset).

Method	pMSE (lower is better)	Training time
CART	0.1055	~38 min (2,310 s)
CTGAN	0.1361	~7 h 50 (28,260 s)
TVAE	0.1695	~1 h 43 (6,200 s)

3.1.6.3 Structural coherence and official-statistics constraints

Logical filters. Eleven business rules were defined such that, given the internal consistency of the original file, no individual should violate them. The contrast is categorical (see table 3): CART produces zero violations on all eleven rules, whereas CTGAN and TVAE each generate several thousand inconsistent records — a level of noise that would compromise the usability of the file without heavy post-processing.

Table 3 - Number of inconsistent individuals per filter (out of 88,589 observations) - first five filters shown.

Filter	CART	TVAE	CTGAN
Filter 1	0	4,459	318
Filter 2	0	22	3,614
Filter 3	0	3,443	215
Filter 4	0	7	6
Filter 5	0	1,311	340

Nested variables. The same pattern holds for the hierarchical NAF activity variables: CART misclassifies only a small share of records (about 0.7 %), readily corrected in post-processing, while CTGAN and TVAE fail to maintain the nesting for several thousand records.

Survey weights. With no special treatment of the weight variable, CART conserves the total weighted population almost exactly (+0.11 %), whereas TVAE loses about 7 % of it — a failure that distorts the representativity of the file (see table 4). The weighted employment and unemployment rates by age and sex confirm this: CART reproduces them with near-perfect overlap, while TVAE introduces large biases, most visibly for the 15–24 group, where the employment rate is heavily underestimated and the unemployment rate inflated.

Table 4 - Sum of survey weights and relative gap to the original.

Dataset	Sum of survey weights	Relative gap
Original	13,580,598	—
CART	13,595,546	+0.11 %
TVAE	12,626,612	−7.03 %

An important qualifier: these neural-model results reflect a “raw” use of the algorithms. The SDV library allows a metadata object to guide learning (forcing variable types, declaring relational constraints, encoding rules), which could in principle improve structural coherence. In a production setting, however, this carries two practical costs — a heavy dependence on human expertise to translate the eleven filters and the multiple NAF levels into formal constraints, and the risk of omitting an implicit business rule. CART’s strength is that it detects these structural breaks intrinsically through recursive partitioning, satisfying most of the coherence constraints natively, without upstream metadata engineering.

3.1.6.4 Disclosure risk

Re-identification was assessed on a six-variable key (sex, age, status, diploma, location, one-digit occupation). CART’s rate of replicated uniques (repU = 0.104 %) is about twice that of the neural models (0.062 % for CTGAN, 0.050 % for TVAE), as shown in table 5. This is the logical counterpart of its finer preservation of the data structure: better fidelity entails reproducing a larger share of rare profiles. The absolute level nevertheless remains very low for all three models. Attribute-disclosure analysis on activity status points in the same direction — CART preserves the links between sensitive attributes and identifying keys more strongly (DiSDiO ≈ 1.33 % against 0.48 % for TVAE).

Table 5 - Uniques and re-identification risk (% of records, six-variable key). UiO / UiS: uniques in the original / synthetic file; UiOiS: unique in both; repU: replicated uniques.

Method	UiO	UiS	UiOiS	repU
CART	0.373	0.395	0.209	0.104
CTGAN	0.373	0.646	0.233	0.062
TVAE	0.373	0.445	0.115	0.050

3.2 Germany

3.2.1 Statistical Domain and Input data

The road traffic accident statistics describe accidents that occur as a result of road traffic on public roads and public spaces in Germany and that lead to personal injury, fatalities, or property damage. Their primary purpose is to provide a current, comprehensive, and reliable data basis for analysing the structure, causes, and development of road traffic accidents over time.

These statistics are based exclusively on accidents recorded by the police. They therefore cover all officially documented traffic accidents on public roads and squares in which personal injury or property damage has occurred. On this basis, key indicators such as the number of road fatalities and injured persons are compiled and published monthly in regular official communications.

Beyond these headline figures, the available data allow for differentiated analyses by a wide range of characteristic. These include information on accident victims by age, the type of vehicle involved (for example cars, motorcycles, or buses), and detailed accident causes. Both driver-related factors, such as driving under the influence of alcohol or excessive speed, and external conditions—such as icy roads, fog, or collisions with wild animals—are systematically recorded. In addition to motor vehicle accidents, incidents involving pedestrians and cyclists are also fully included in the statistical domain

(see https://www.destatis.de/DE/Themen/Gesellschaft-Umwelt/Verkehrsunfaelle/_inhalt.html#sprg478574).

3.2.2 Purpose of the Synthetic Data

Publishing official statistics while protecting the confidentiality of individuals and organisations is a core responsibility of national statistical offices. The German Federal Statistical Office (Destatis) draws on an established set of statistical disclosure control (SDC) methods to meet this obligation. These include cell suppression with secondary suppression or coarsening to prevent disclosure-by-differencing, as well as the Cell Key Method (CKM), which introduces controlled noise into table cells as an alternative to outright suppression. While these techniques are well-suited to many publication contexts, they reach their practical limits when applied to particularly complex data environments.

The road traffic accidents dataset represents precisely such a case. It is a large-scale dataset of over 500 mixed-type variables, compiled from police reports filed after every accident on a public road in Germany. These reports are collected and quality-checked at the state level before being consolidated at the Federal Statistical Office. The dataset's complexity is compounded by the fact that two partially overlapping data products are released in parallel: a set of pre-published tables covering sensitive variables at coarse aggregation levels, and the openly accessible German Accident Atlas (available at: https://unfallatlas.statistikportal.de/).

The German Accident Atlas provides geocoded information on non-sensitive variables. The co-existence of these releases creates a challenging disclosure environment. Published table margins severely limit the scope for secondary suppression of internal cells, while the fixed margins equally reduce the protection offered by CKM noise, as they allow value ranges for protected cells to be reconstructed by a determined attacker.

Given these constraints, a synthesis-based approach offers a more viable path forward. Rather than modifying the already-released non-sensitive variables, only the sensitive attributes are synthesised,

leaving the rest of the dataset intact. The resulting synthetic microdata can then be used by domain experts as the basis for generating publication-ready output tables, preserving analytical usability while reducing the risk of confidential information being disclosed.

This setup, however, introduces a practical evaluation challenge. Because the precise tabular structures that will eventually be produced from the synthetic microdata are not known at the time of synthesis, it is not possible to assess table quality directly in advance. This raises the question of whether evaluation metrics computed at the microdata level can function as reliable early indicators of the utility and disclosure risk of future aggregated outputs — a question that sits at the heart of this use case.

3.2.3 Methods for Synthetic Data Generation

The synthesis approach applied in this use case follows a partial synthesis strategy, meaning that only a targeted subset of sensitive variables is replaced with synthetic counterparts, while all non-sensitive and already-published variables are retained in their original form. This design choice reflects both the publication context — where non-sensitive variables are already in the public domain — and the goal of preserving as much analytical structure as possible. To capture variability in the synthesis process and support downstream uncertainty estimation, five independent synthetic versions of the dataset are produced.

Scope and target variables. Working with the full accident dataset directly is not practical given its breadth of several hundred variables. A preparatory reduction step therefore brought the working dataset down to 85 variables by removing redundant attributes. From this reduced dataset, nine variables were designated as synthesis targets — seven categorical and two numerical — based on their sensitivity classification. All other variables, including those already covered by pre-published tables or the open geocoded data product, remain unmodified in every synthetic version.

Choice of synthesis method. Each sensitive variable is synthesised in turn using the CART-based engine provided by the R package synthpop (Nowok et al. , 2016). In this sequential framework, each target variable is modelled conditional on previously synthesised sensitive variables as well as on a selected set of non-sensitive predictors. CART fits a decision tree to the observed data and generates synthetic values by sampling from the leaf nodes, making it naturally suited to datasets with mixed variable types and complex interaction structures, without requiring explicit distributional assumptions (Reiter, J.P, 2005 and Drechsler, J. & Reiter, J.P., 2011).

Compared with generative models such as GANs or VAEs, classical machine-learning approaches like CART are more transparent: the tree structure and splitting rules can be inspected and documented, which may provide a higher explainability of the anonymization process. In addition, CART-based synthesis is typically less demanding in terms of computational resources and tuning complexity than high-capacity neural models, which makes it more practical for repeated production use in a statistical office environment.

Managing high dimensionality and cardinality. A central technical challenge in this dataset is the large number of categorical variables, several of which have many distinct levels. For tree-based methods, high-cardinality categorical predictors are computationally costly because the number of candidate splits grows rapidly with the number of categories (Hastie et al, 2009). Three measures were introduced to manage this.

First, for each of the nine target variables, the Boruta algorithm (Kursa et al, 2010) — a random-forest-based wrapper method — was used to identify the subset of variables that carry genuine predictive relevance for that target. This pre-selection step bounds the number of predictors entering each CART model and substantially reduces computation time while retaining variables that matter for synthesis quality.

Second, categorical predictors were binned to reduce their effective cardinality, with maximum cardinality thresholds set at varying levels — for example 10, 15, or 20 categories — in order to examine how different degrees of binning affect the utility and disclosure risk of the resulting synthetic data. As with the other dimensionality-reduction measures, binning was applied exclusively to variables in the predictor role; the variable being synthesised was always retained at its original level granularity, so that the synthetic output preserves the full category structure of each target.

Third, in specific situations where a categorical predictor combined very high cardinality with strong relevance for a given target, target encoding (Micci-Barreca, D. , 2001) was applied — converting the categorical predictor into a numeric representation derived from its empirical relationship with the target variable. As with binning, encoding was restricted to predictors and never applied to the target itself.

Synthesis order and plausibility constraints. The sequence in which variables are synthesised has a direct bearing on model complexity and output quality, particularly when later variables can be conditioned on earlier synthetic outputs (Raab et al, 2017). Variables with especially large numbers of categories were therefore placed toward the end of the synthesis sequence, limiting how much of their complexity propagates into the models for earlier variables.

For most variables, plausibility constraints were handled through the rule functionality built into synthpop, which supports conditional value assignments of the form “if condition then assign value”. Three high-cardinality target variables, however, were subject to constraints that restrict plausible outputs to a subset of admissible categories rather than a single fixed value — a situation the built-in rule mechanism is not designed to handle. These variables were moved to the final positions in the synthesis order and generated through a rejection sampling procedure using externally fitted CART models from the rpart package (Therneau, T., & Atkinson, B.,2022). Candidate values were drawn from the model and evaluated against the relevant plausibility rules; only values satisfying all constraints were accepted, ensuring that the synthetic records are both statistically plausible and logically consistent.

3.2.4 Methods for Utility Measurement

In synthetic data evaluation, utility metrics are typically classified as global (distributional) and outcome-specific measures (Drechsler&Haensch, 2024). Global measures compare the overall distribution of the original and synthetic data, while outcome-specific one focus on the similarity of outcomes from specific analyses (for example, regression coefficients). In other words, global utility looks at distributional fidelity, whereas outcome-specific utility assesses task-oriented output similarity.

Most synthetic-data research has developed metrics at the microdata level, because data synthesis itself occurs on individual records. However, national statistics agencies often release only aggregated tables (cross-classifications of variables) rather than raw microdata. Thus, we differentiate between microdata-level and table-level metrics. Microdata-level metrics exploit record-level detail (all attributes of individual accidents), whereas table-level metrics compare contingency tables of

aggregated counts. This distinction is necessary because not all metrics applicable to microdata can be computed from aggregated data alone. In our road traffic accident data, we therefore evaluate utility on both levels to understand their relationship and to suit a scenario where only aggregated tables are published.

Microdata-level utility metrics. For the microdata, we focus on a global utility measure that can handle mixed data types. Specifically, we use the propensity score mean-squared error (pMSE) (Woo, M. J. et al., 2009). The pMSE is derived from a classification exercise: original (n_o) and synthetic (n_s) records are combined into a single dataset and a CART model is trained to predict which records are synthetic. For each record i , the model produces a probability estimatep_i of belonging to the synthetic portion, and $c = \frac{n_s}{n_o+n_s}$ denotes the overall share of synthetic records in the combined dataset (here $c = 0.5$). The pMSE is then the average squared deviation of these estimates from c , formally $\frac{1}{n_o+n_s}\sum_{i=1}^{n_o+n_s}(p_i - c)^2$.

When the synthetic data closely resemble the original, the classifier cannot separate the two groups reliably, propensity scores cluster around c , and the resulting pMSE is small. A larger value signals that distributional differences between the two datasets are detectable. We computed pMSE on the full accident dataset (85 variables).

Because the raw pMSE can inflate with more complex propensity models, we also report the pMSE ratio (Snoke, J. et al., 2018). This ratio divides the observed pMSE by its null expectation under perfect synthesis, yielding an expected value of 1 when the synthetic data are generated by the true underlying model. In practice, larger pMSE or pMSE-ratio values indicate lower utility (greater discrepancy). Using the pMSE ratio helps account for model complexity effects on the pMSE measure.

Table-level utility metrics. For aggregated data (cross-tabulated tables of accident counts), we built a comprehensive set of contingency tables and applied a distributional distance metric. To construct a representative set of test tables, six variables were selected from the accident dataset — three that were left unmodified and three that were synthesised — alongside the geo code as an additional dimension. Every possible cross-tabulation of between two and five of these six variables was then formed, yielding tables of four different dimensionalities. This process was carried out at three spatial aggregation levels — state, district, and municipality — producing a broad collection of tables that varies systematically in both dimensionality and geographic granularity. Only tables containing at least one synthesised variable were retained for evaluation, resulting in the counts shown in table 6.

Table 6 - N-way tables with six variables. Only tables with at least one synthetic variable are considered.

Table dimensions	**2**	**3**	**4**	**5**	**Total**
Number of tables n(d)	12	19	15	6	52

To quantify table-level utility, we compare the probability distributions of original versus synthetic cell counts using the Hellinger distance. Before computing distances, we normalize each table's cell counts so that they sum to 1. The Hellinger distance between two distributions and is defined via the Euclidean norm of the difference of their square root vectors, i.e., $H(P,Q) = \frac{1}{\sqrt{2}}\sqrt{\sum\left(\sqrt{p} - \sqrt{q}\right)^2}$, where p and q represent original and synthetic cell counts of a table. Crucially, it is bounded between 0 and 1: a value of 0 indicates the distributions are identical, and a value closer to 1 indicates maximal divergence.

We compute the Hellinger distance separately for each aggregation level (state, district, municipality) and for each table. These distances summarise how the joint frequency distributions of original and synthetic data differ across all selected variables. Lower Hellinger values indicate that the synthetic tables closely match the original tables' patterns, reflecting higher utility. All tabular utility results reported in our experiments are based on these aggregated tables and their Hellinger distances.

3.2.5 Methods for Privacy Assessment

Measuring disclosure risk is generally more complex than measuring utility. Unlike utility, which can be assessed by comparing statistical outputs, risk must be defined via explicit intruder scenarios tailored to the data and publication context. When synthesizing key identifiers, the emphasis lies on preventing re-identification. In practice, this means risk evaluation depends on assumptions about what an attacker knows (e.g. values of certain key variables) and what they are trying to infer (the target), as well as on how the data were synthesized (Raab et al., 2024) note that there is "no standard" method for disclosure risk – checks tend to be "ad hoc" and highly contextual. In summary, unlike broad utility measures, privacy risk metrics must explicitly model the adversary's knowledge and the data release process, which inherently complicates their evaluation.

We again distinguish microdata-level and table-level risk metrics, as we did for utility. Synthetic data generation typically operates on individual records, so most existing privacy measures focus on microdata. However, official statistics often release only cross-tabulated tables. Thus, we assess both levels: microdata metrics gauge record-level disclosure given all synthetic microdata, while table metrics gauge risks in the published aggregated tables.

Microdata-level risk metrics. We use the Targeted Correct Attribution Probability (TCAP) metric (Taub, J., & Elliot, M., 2019) for the microdata. TCAP is defined under an intruder model where the attacker knows the key variables of a target individual (and that the target is in the data) and tries to infer the individual's sensitive attribute. In practice, the TCAP computation proceeds in two stages. First, synthetic records are screened for key combinations that leave no ambiguity about the target value within the synthetic dataset — that is, all synthetic records sharing a given key combination point to the same target value (Within Equivalence Class Attribution Probability (WEAP) = 1). Second, for each such key combination, the corresponding records in the original data are located and the proportion sharing the same target value is calculated. This proportion is the TCAP score for that combination. A score of 1 indicates that an attacker using that key combination would correctly infer the sensitive attribute with certainty, while lower values reflect partial or no disclosure. The TCAP ratio then summarises this across the dataset by expressing how many of the synthetic disclosure events (WEAP = 1) translate into genuine disclosures in the original data (Kolb, S. X. N. et al., 2024), i.e., $TCAP\ ratio = \frac{\#\ unique\ combinations\ with\ TCAP = 1}{\#\ unique\ combinations\ with\ WEAP = 1}$.

Table-level risk metrics. For the aggregated tables, we focus on so-called group disclosure (GD) events. A GD arises when the composition of a table cell is sufficiently homogeneous with respect to a sensitive attribute that an attacker with knowledge of cell membership can infer that attribute with high confidence. For example, if every accident recorded in a particular cell shares the same value on a sensitive variable, any individual known to belong to that cell is effectively re-identified on that attribute without any further information being needed.

To operationalise this, disclosive cells are identified in each synthetic table according to the group disclosure criterion, and the corresponding cells in the original table are examined. The table-level risk measure is the share of synthetic disclosive cells that correspond to genuine disclosive cells in the

original data — in other words, the probability that a GD flagged in the synthetic table reflects a real disclosure rather than an artefact of the synthesis. Cases where the dominant target value already accounts for 90% or more of original records in a cell are excluded, as correct attribution in such cases would require no special knowledge and amounts to a naive guess.

3.2.6 Preliminary Results

Synthesis Configurations. To systematically evaluate how synthesis parameter choices affect data quality, 21 distinct synthesizer configurations were examined. Each configuration was defined by two parameters: the number of predictor variables included during synthesis, varied from 5 to 20, and the maximum cardinality to which categorical predictor variables were binned prior to fitting, set at 10, 15, or 20 levels. Together, these parameters govern the trade-off between the richness of the synthesis model and the computational cost of running it via the CART-based procedure in synthpop.

For each of the 21 configurations, five independent synthetic datasets were generated, producing 105 synthetic microdata sets in total. Replicating the synthesis process across multiple runs allows metric variability attributable to randomness in the synthesis itself to be separated from variability attributable to configuration choices.

To assess both utility and disclosure risk at the tabular level, six variables were selected for cross-tabulation — three non-synthesised and three synthesised, together with a geocode — and all possible tables of 2 to 5 dimensions containing at least one synthesised variable were constructed (see Table 1). Tabular metrics were then computed across this full set of tables at three geographic aggregation levels: state (Bundesland), district (Kreis), and municipality (Gemeinde)

Microdata Utility: pMSE and pMSE ratio. Global utility at the microdata level was assessed using the propensity score mean squared error (pMSE) and the normalised pMSE ratio.

The pMSE reflects how successfully a CART classifier can separate original from synthetic records: a lower value indicates greater distributional similarity and therefore higher utility. As shown in figure 2 (left), pMSE declines markedly as the number of predictors increases from 5 to around 12, after which further improvements become marginal. This saturation pattern is consistent with the feature selection logic underlying the synthesis: the Boruta algorithm ranks predictors by importance, so the configurations with fewer predictors already include the most informative variables, and adding further predictors beyond a certain point contributes only incrementally. The effect of predictor cardinality is detectable but considerably weaker than the effect of predictor count across the range tested.

The pMSE ratio adjusts for the dependence of the raw pMSE on the complexity of the propensity model by expressing the observed pMSE relative to its expected value under correct synthesis — the scenario in which original and synthetic records are exchangeable draws from the same underlying distribution (Snoke, J. et al, 2018). A ratio of 1 represents this ideal baseline. In contrast to the pMSE, the pMSE ratio did not exhibit a consistent improvement with increasing predictor count (figure 2, right). Instead, it showed pronounced sensitivity to predictor cardinality, with substantial variability across configurations at the same predictor count. This behaviour stems from the sensitivity of the null expectation — estimated by pairwise resampling of the synthetic datasets — to distributional differences between independently drawn synthetic samples. Despite this variability, the pMSE ratio provides a conceptually more transparent benchmark for synthesis quality because it separates genuine distributional differences from the model complexity effects that tend to inflate the raw pMSE.

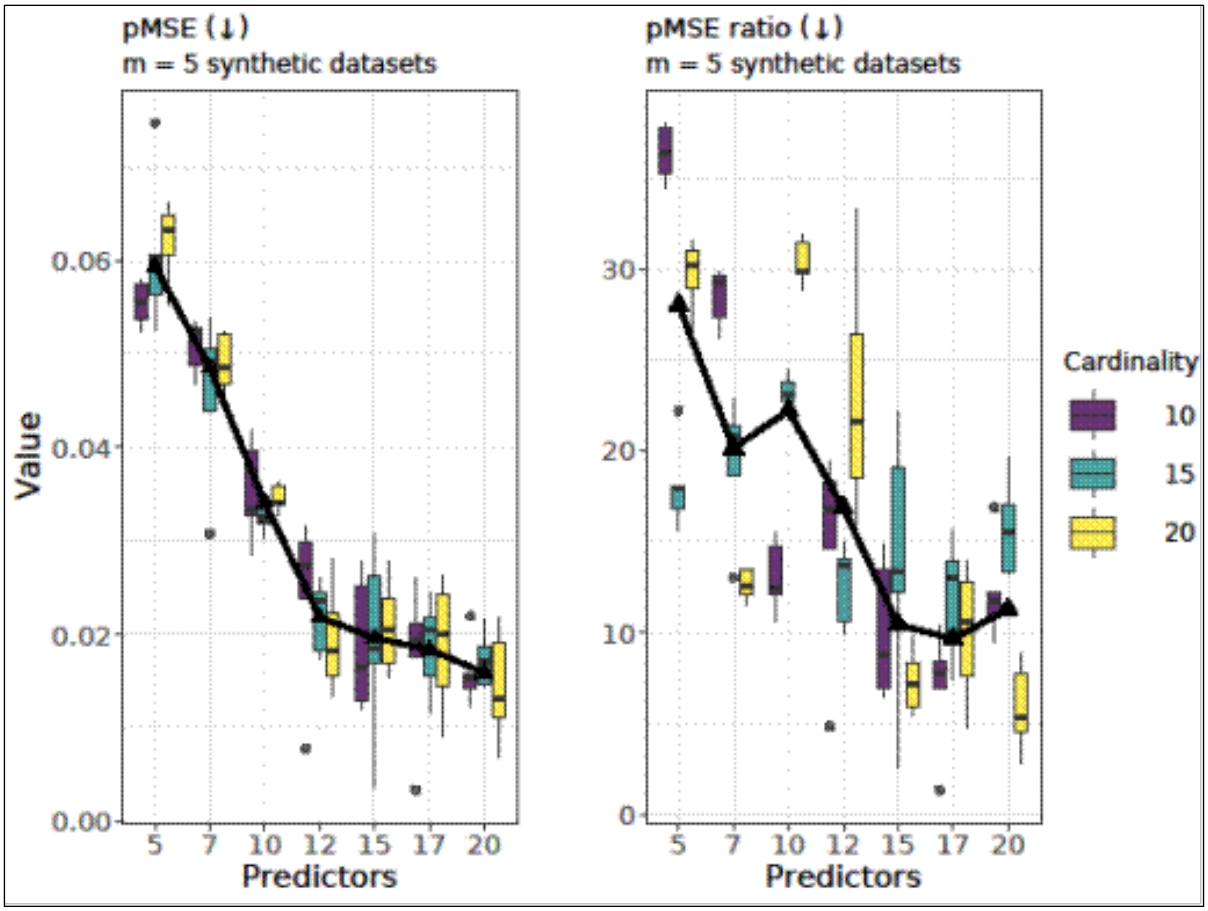


Figure 2: Comparing original and synthetic microdata. It shows the global utility pMSE and pMSE ratio in relation to the number of predictors and variable cardinality. ↓ lower value represents a better utility.

Tabular Utility: Hellinger distance and maximum relative improvement. Utility at the tabular level was quantified using the Hellinger distance between normalized cell count distributions of original and synthetic tables, which ranges from 0 (identical distributions) to 1 (maximal divergence).

A consistent pattern emerges across all 52 tables: Hellinger distances grow with table dimensionality and with geographic granularity (figure 2, left). At the state level, even five-dimensional tables produce relatively modest distances, while at the municipality level — particularly for four- and five-dimensional tables — distances are substantially larger. The underlying mechanism is data sparsity: fine-grained geographic breakdowns produce many cells with very few or zero observations, so even small absolute differences in cell counts generate large relative distributional divergences. Because CART synthesis operates at the level of individual records, distributional fidelity at highly disaggregated levels is inherently limited by the number of records that fall into each local geographic stratum.

This finding has a direct practical implication for statistical agencies considering fine-granularity publication: enforcing high distributional accuracy at the municipality level may require the synthetic data to approximate the original data so closely that the privacy-protective benefit of synthesis is substantially diminished.

The maximum relative improvement metric H(d) captures, for each dimensionality level, the largest achievable reduction in Hellinger distance across synthesizer configurations relative to the worst-performing one. The results (figure 2, right) show that improvement potential is greatest for lower-dimensional tables at coarser aggregation levels and diminishes for high-dimensional tables at the

municipality level. This confirms that the utility gains achievable through synthesizer tuning are not uniform across the space of possible publication tables, and that optimising a single synthesis configuration for the full range of tabular outputs is inherently difficult.

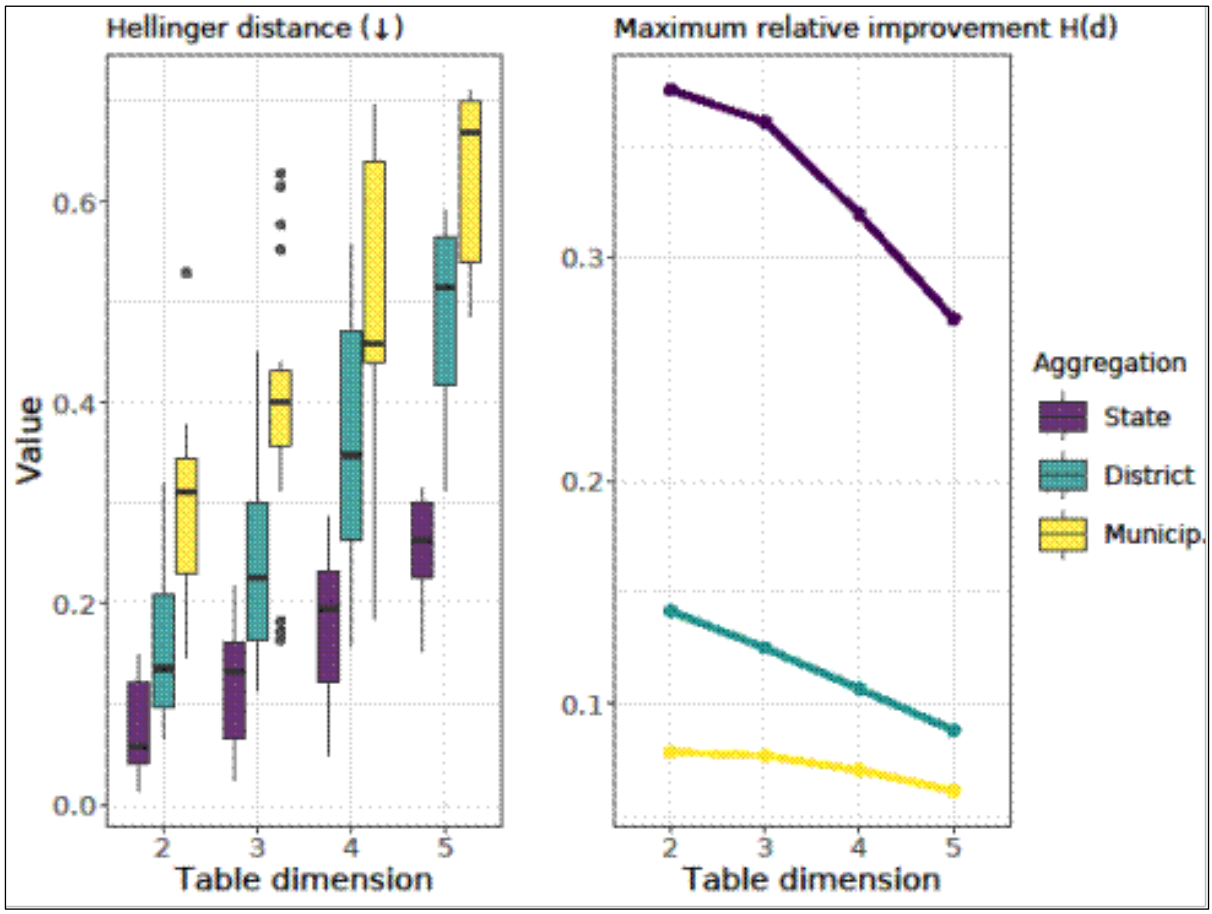


Figure 3: Comparing original and synthetic tables. It shows the for one synthesizer configuration. The right plot shows the maximum improvement over all synthesis setting for for the Hellinger distance averaged over all tables of a specific dimensionality. ↓ lower value represents a better utility.

Utility correlation. A core objective of this study is to determine whether microdata-level utility metrics can serve as meaningful advance indicators of the quality of derived aggregated tables — a practically important question for statistical agencies that need to assess synthesis quality before the final tabular structure is fixed.

To examine this, Pearson correlations were computed between each microdata utility metric (pMSE and pMSE ratio) and the Hellinger distances across all 52 tables and all 21 synthesizer configurations. Results for the highest aggregation level (state) are shown in figure 3. The x-axis groups tables by which synthesised variables they contain, and confidence intervals are based on Fisher's z-transformation.

The correlations between pMSE and Hellinger distance are almost entirely driven by whether variable C is included in the table (figure 3, left panel). Variable C is a high-cardinality categorical variable with over 60 distinct levels and strong associations with many other variables in the dataset. Tables containing variable C show statistically significant positive correlations with the pMSE, whereas tables built exclusively from the two numerical synthesised variables A and B show no meaningful correlation, regardless of table dimensionality. This pattern suggests that the CART classifier underlying the pMSE computation concentrates its discriminative power on the most expressive variable available — in this case variable C — causing the global pMSE score to disproportionately reflect the synthesis quality of

that one variable while being largely uninformative about the remaining synthesized variables or about tables that do not involve variable C. An additional contributing factor is that variables A and B were binned prior to tabulation, which can conceal distributional changes occurring within individual bins and thereby weaken the detectable association between microdata and tabular quality for those variables.

The pMSE ratio shows no meaningful correlation with Hellinger distances for any table type or dimensionality (figure 4, middle panel). This is not simply a limitation: it reflects the intended effect of normalization, which removes the dependence on specific dominant variables that makes the raw pMSE sensitive to variable C. However, the same normalization also means that the pMSE ratio carries no information about the synthesis quality of specific variable subsets or their tabular representations. It functions as a more balanced global indicator, but precisely because of its balance, it is equally unsuitable as a proxy for table-level utility assessment. Taken together, both metrics highlight a fundamental constraint of using a single global microdata utility score to predict heterogeneous tabular quality: when synthesized variables differ substantially in cardinality and associative structure, no single summary measure can adequately represent the quality of the full range of potential publication tables.

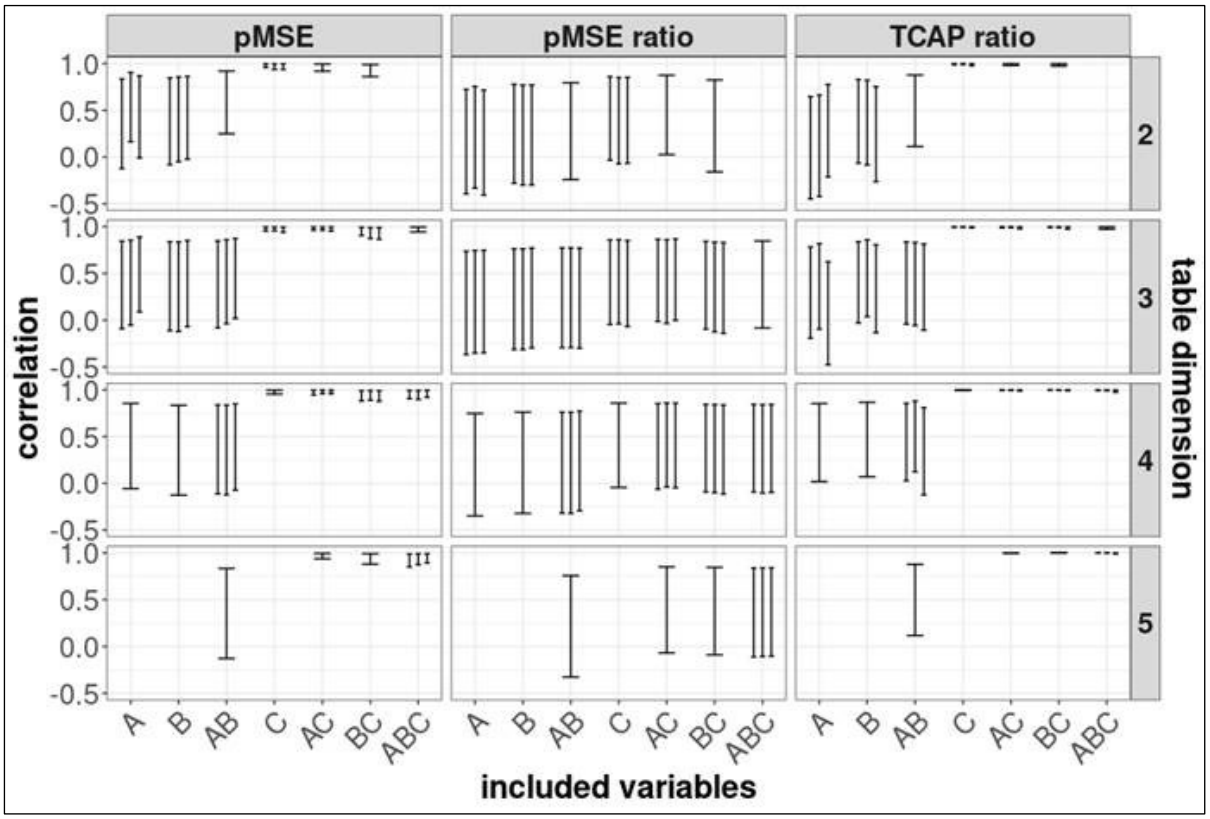


Figure 4: Correlation between microdata-based and table-based metrics for all possible tables. The x-axis describes the table type, denoting which synthesised variables are included in the table. A value of 1 implies maximal correlation between tabular and microdata metrics and the grouping represents the table dimensions

Risk Correlation. The correlation analysis was repeated for the risk dimension, pairing the microdata-level TCAP ratio with the tabular group disclosure (GD) risk measure. Given that municipality-level tables are the most granular and therefore the most exposed to disclosure events, correlations are reported for this aggregation level (figure 4, right panel).

The results mirror the pattern observed for utility: the inclusion of variable C in a table is again the decisive factor. For tables containing variable C, the TCAP ratio shows correlations close to 1 with the tabular GD risk measure across all dimensionalities. For tables built only from the binned numerical variables A and B, confidence intervals are wide and no consistent correlation is detectable, indicating that changes in synthesizer configuration did not produce noticeable differences in the GD risk of those tables.

The near-perfect correlation for variable C tables is not coincidental. The TCAP ratio, as defined at the microdata level, detects precisely those synthetic records whose key combination leaves no ambiguity about the target value — a condition that is equivalent to the definition of a group disclosure event in a contingency table, regardless of the specific tabulation. This equivalence means that the TCAP ratio effectively captures all GD events that could appear across any possible cross-tabulation of the selected variables, making its correspondence with the tabular GD measure theoretically expected rather than empirically surprising.

Although the TCAP ratio achieves near-perfect correlation with the tabular GD risk measure in this setting, this result requires careful interpretation before drawing practical conclusions. The TCAP ratio is computed at the microdata level across all possible key-target combinations simultaneously, meaning it aggregates disclosure risk over the complete space of potential cross-tabulations. Its denominator — the total count of synthetic group disclosure events — is fixed and reflects the entire variable space, irrespective of which tables a statistical agency will actually publish.

In practice, agencies publish only a selected subset of possible tabulations. A GD event that appears in the TCAP ratio's denominator may correspond to a table that is never released, contributing to the measured risk without ever posing a real publication risk. As the number of variables in the dataset grows, the proportion of possible tabulations that will be published shrinks, and the gap between the TCAP ratio's risk estimate and the true publication risk widens accordingly

3.3 Portugal

3.3.1 Statistical Domain and Input Data

The demand for Statistics Portugal microdata for research purposes has been increasing over the past several years. Secure-use files have attracted significant interest. These are defined in Article 2 of Commission Regulation (EU) No 557/2013 of 17 June 2013, as "confidential data for scientific purposes to which no further methods of statistical disclosure control have been applied".

Access to these highly detailed data can only be granted through a secure access system, referred to as a safe centre environment or research data centre (RDC). Research outputs are subject to prior review by Statistics Portugal staff to ensure that they do not contain confidential information. This prior review and assessment of the risk of confidential information disclosure is referred to as output checking. With the growing demand for access to secure-use files and the expanding availability of data and infrastructure, increasing human and technological resources are required to meet output checking requests.

One of the most frequently requested data sources by researchers accessing the RDC is the Integrated Business Accounts System (in portuguese, Sistema de Contas Integradas das Empresas - SCIE), which corresponds to the national response to the Structural business statistics (SBS). SCIE results from a process of statistical data integration based on administrative data, with an emphasis on Simplified Business Information (IES). This data is complemented with information collected from sole proprietors and independent workers from the Ministry of Finance and from Statistics Portugal's Business Register. This integration ensures broad sectoral coverage and enriches the dataset with detailed financial and categorical information relevant to business statistics. The dataset includes a mix of numerical variables—such as income, expenditure, investments, and disinvestments—and categorical fields covering economic classifications and structural characteristics.

3.3.2 Purpose of the Synthetic Data

Enhancing the efficiency and robustness of microdata access requires both administrative improvements and technical innovations. While expanding secure facilities and human resources is beneficial, such measures alone cannot keep pace with rising demand. Technical tools, particularly those that reduce avoidable interactions with secure data, are crucial for scalability. Synthetic structural datasets serve this purpose by offering a preparation file that replicates the structure of the confidential dataset without risking disclosure.

The primary added value of synthetic datasets lies in their ability to support code development outside the RDC. Researchers can write, test, and refine scripts using a non-confidential file that mirrors the original dataset's variables, formats, and structures. This approach minimises time spent within the secure environment correcting technical errors and reduces the frequency of output submissions that require checking. It also encourages reproducible workflows, as well-tested code is less prone to interruptions during secure analysis.

3.3.3 Methods for Synthetic Data Generation

Two R packages, synthesizer and synthpop, were used to generate multiple synthetic versions of an original dataset in order to compare how different synthesis approaches and parameter settings influence the resulting data. For each package, a key parameter was varied, and for each parameter configuration 100 independent synthetic replicates were produced.

The synthesizer package employs a multi-stage approach aimed at replicating the empirical distributions and rank-based dependence structures of the original data. Numerical variables are generated using inverse transform sampling of the empirical quantile function, ensuring accurate reproduction of univariate distributions. Categorical variables are synthesised through sampling with replacement. Dependence is then restored by aligning the rank order of synthetic variables with that of the original data, with an optional parameter (rankcor) introducing controlled randomness when set below 1. Three rankcor values – 0, 0.5, and 1 – were tested.

In contrast, synthpop follows a sequential conditional modelling strategy in which each variable is synthesised conditional on others already generated. The approach uses random sampling for the first variables and relies on Classification and Regression Trees (CART) for subsequent ones, allowing non-parametric modelling of complex conditional structures. The key parameter tested was the synthetic dataset size (k), set to 25%, 15%, or 5% of the original number of records. These smaller datasets help assess whether downscaling affects structural integrity while reducing disclosure risks.

3.3.4 Methods for Utility Measurement

The evaluation of synthetic data utility focused on its capacity to support error-free code development, rather than its ability to reproduce analytical results. The synthetic dataset must faithfully reflect the structure of the confidential dataset, including variable names, formats, types, and coding schemes. Structural consistency is essential because researchers rely on these features when constructing models, merging data sources, or applying transformations.

A central aspect of utility involves ensuring the presence of all categories and codes found in the original dataset. Since the purpose of the synthetic data is to support pre-analysis programming, missing categories could lead to failed tests or incomplete validation of scripts. Therefore, complete category representation is required even when the distributions themselves are not preserved. Finally, the number of records in the synthetic dataset may be smaller than in the original file, as long as structural and categorical fidelity is maintained. This allows for lightweight yet fully functional datasets that remain safe and efficient for users.

3.3.5 Methods for Privacy Assessment

The privacy assessment examined both identity and attribute disclosure risks, using four indicators implemented in the *disclosure()* function of the synthpop package. Identity disclosure was evaluated using replicated uniques (repU), which measures the proportion of patterns that remain uniquely identifiable in both the original and synthetic data. Low values indicate that unique combinations have not been carried forward, thereby reducing re-identification risks.

Attribute disclosure risks were analysed using DiSCO, DiSDiO, and the Targeted Correct Attribution Probability (TCAP). DiSCO examines whether a target attribute – here, turnover – is correctly reproduced for records with matching quasi-identifiers. DiSDiO limits this assessment to cases already disclosive in the original data, helping distinguish risks introduced by the synthesis process itself. TCAP evaluates how frequently synthetic records with the same Nomenclature of Territorial Units for Statistics (NUTS) III region and five-digit activity code correctly reflect the original turnover value. Together, these indicators offer a comprehensive view of how much sensitive information could potentially be inferred from the synthetic data.

3.3.6 Preliminary Results

A total of 600 synthetic datasets were generated using the synthpop and synthesizer packages under different parameter configurations (k for synthpop and rankcor for synthesizer). For each configuration, 100 simulation runs were performed to evaluate both data utility and disclosure risk.

Utility was assessed by comparing the distributions of the synthetic datasets with the original data using measures of categorical overlap and numeric range overlap. Overall, the synthetic datasets produced by both methods showed high utility, particularly regarding categorical overlap with average values ranging from 97.4% to 99.9%; as for the numerical overlap, values are slightly smaller and vary between an average of 51.1% and 84.9% across parameter settings.

As expected, methods that preserved stronger dependence structures – such as synthesizer with rankcor = 1 and synthpop with k = 0.25 – yielded synthetic data that more closely resembled the original patterns. Preliminary indicators reflect this: repU remained relatively low across all settings but reached an average of 2.61% in synthesizer with rankcor = 1; synthpop with k = 0.25 resulted in an average repU of 1.06%, which suggests decreasing the size of the synthetic dataset might be advantageous with respect to identity disclosure. With synthesizer, DiSCO and TCAP increased with dependence strength (DiSCO up to an average of 2,51% and TCAP up an average of 2,76%, when rc = 1). On the contrary, both DiSCO and TCAP declined with increasing size of synthetic data in synthpop.

These results highlight the trade-off between structural realism and disclosure protection, while showing that useful, compliant synthetic datasets can be generated at different points along this spectrum. Continued evaluation using additional risk measures, alternative key and target variables, and further methodological settings will help refine practical guidance for producing synthetic data that effectively balances analytical usefulness and privacy protection.

3.4 Spain

3.4.1 Statistical Domain and Input Data

The area of interest focuses on generating a synthetic census population at the individual microdata level that is consistent with the aggregated information at the census section level from Spain's Atlas of Income (Atlas de la Renta). Several data sources, all of which are available on the Spanish National Statistics Institute (Instituto Nacional de Estadística, INE) website, are used as inputs:

- **Encuesta de Condiciones de Vida (ECV) 2022:** Microdata with detailed information on income and sociodemographic variables (age, sex, marital status, nationality, employment situation, educational level, etc.) of the population residing in Spain;

- **Censo de Población 2022:** Aggregated data on population distribution by age, sex, and census tract used for micro regional demographic distribution and calibration;

- **Censo de Viviendas 2021:** Household counts by census tract, used for micro regional distribution of the synthetic population;

- **Atlas de la Renta de los Hogares 2022:** Information on average income by census section, used to calibrate the synthetic population. Internal files with information on the number of households and persons by section, as well as distribution by age and sex groups, are also used to ensure territorial and demographic coherence.

3.4.2 Purpose of the Synthetic Data

The main objective is to generate a synthetic census file that:

- includes individual information on income and sociodemographic variables: province code (CPRO), household identifier (HID), person identifier (pid), total household income (HHINCOME), average income per person in the household (INCOME), age (AGE), sex (SEX), household size (HHSIZE), census tract (CSEC), marital status (MARITAL), nationality (NATION), employment status (STATUS), current employment situation (SITUPA), last employment situation (SITUPU), and educational level (EDU, grouped by first digit);
- respects the total income by census section provided by the Atlas;
- maintains statistical coherence with the available real sample;
- ensure the privacy of the generated data. In this project, all sources used so far are public, so at this stage privacy should not be a major concern.

3.4.3 Methods for Synthetic Data Generation

The generation process follows four main phases:

1. **Synthesis Phase:** using the calibrated ECV sample, relationships between variables are modeled, and a first version of the synthetic population is generated (without census section assignment yet). Machine learning models are employed for sequential variable generation:
    - for basic variables: the *direct* method of the *simStructure* function is used;

- for AGE: we first generate the AGE_GROUP variable (five-year age groups) using the *simStructure* function, and then, with provided matching break intervals, we use the *simContinuous* function to generate AGE based on AGE_GROUP, with Pareto distribution on the oldest age groups. The purpose of this approach is to faithfully represent the age distribution of the real population. If we used the *simStructure* function directly for AGE, only the age values present in the sample would appear in the synthetic population, and the overall age distribution would be highly unrealistic;
- for the rest of categorical variables: XGBoost (via *simCategorical* function) showed the best performance after evaluating several options;
- for the numerical variable INCOME: A multinomial model based on logistic regression with multiple categories is used. The *multinom* method segments the continuous variable into intervals, with values randomly generated following: Uniform distribution within each interval and Pareto distribution for intervals above a threshold;
- models are trained at the Nomenclature of Territorial Units for Statistics (NUTS) level 2 (Autonomous Communities) using CPRO as a regressor to capture territorial differences;
- the *simPop* R package is used for generation, and *synthpop* for utility metrics.

2. **Calibration Phase**: households are assigned to census tracts within each province to respect:
    - Age and sex distribution per tract (approximately)
    - Total income per tract (exactly as published in the Atlas)

    - First, *simInitSpatial* randomly assigns sections within provinces respecting population and household totals;
    - Then, a Simulated Annealing algorithm (*calibPop*) moves households between census tracts to better adjust marginal distributions of sex and age groups;
    - Finally, sample calibration (*calib* function from *sampling* package) adjusts virtual weights to achieve exact total income targets per census section;
    - This calibration process is sandwiched into the synthesis phase. It is performed after the basic variable generation, to avoid duplicating exact household values, since the *calibPop* function does in fact duplicate some households to achieve calibration.

3. **Cleaning Phase:** inconsistencies are identified and corrected, such as:
    - Edit 1: coherence of marital status among adults in the same household;
    - Edit 2: detection of high-income values in individuals under 16 years old;
    - Edit 3: households composed exclusively of minors

4. **Privacy Control Phase:** although the inputs are public, controls for extreme income values (outliers) are applied. If extreme values coincide between synthetic and real populations within a section, the case is relocated to another similar section. If they do not coincide, no changes are made.

3.4.4 Methods for Utility Measurement

Quality assessment of synthetic data uses:

- **pMSE (Propensity Score Mean-Squared Error):** A metric that evaluates similarity between real and synthetic data by combining both files with an indicator variable. If the indicator cannot

be accurately predicted, the files are considered indistinguishable. Normalized pMSE values below 10 are associated with good synthesis. In our case, most bidimensional pMSE values lie between 0 and 5. The use of this metric does only make sense when applied to a version of the synthetic population that has not gone through any calibration process;

- **Correlation matrices:** Comparison of variable relationships between original and synthetic data;
- **Income distribution by sex and age:** Comparison of average INCOME.

3.4.5 Methods for Privacy Assessment

The approach is considered particularly low risk due to several factors:

- input data is fully published in the (Instituto Nacional de Estadística) INE website;
- only one numerical variable (INCOME) could potentially generate outliers. Since ECV data do not capture the highest incomes and household averaging is applied, outliers are few and easily located;
- census tracts are randomly assigned initially and then redistributed via Simulated Annealing. Even if the model replicates a high-income pattern from a specific section in the sample, the synthetic version would almost certainly be assigned to a different section, enhancing privacy;
- if privacy concerns persist, the option of omitting the census tract variable or providing it at a less disaggregated level (neighbourhoods or group of nearby tracts) is considered.

3.4.6 Preliminary Results

3.4.6.1 Generation of synthetic data from ECV sample

As discussed, the first step in generating the synthetic dataset consists of the sequential simulation of variables using different methods: direct approaches for structural variables, XGBoost for categorical variables, and a multinomial method for income. As mentioned, this procedure is carried out using the R package simPop.

Once generated, the objective is to evaluate the quality of the synthetic dataset by assessing how closely it resembles the original data while preserving its key statistical properties. To do so, we use the pMSE metric (via synthpop R package). The XGBoost and multinomial models were selected after testing several alternative approaches, including random forests, decision trees, and multiple regression models. Among all the candidates, these methods provided the best performance in terms of pMSE metrics, yielding synthetic data that is more difficult to distinguish from the original dataset. In brief, this method combines the original and synthetic datasets, assigns a binary indicator to distinguish them, and estimates the probability that each record belongs to the synthetic data. The pMSE measures how close these probabilities are to the true proportion of synthetic records. Lower pMSE values indicate greater similarity, as the two datasets become harder to distinguish.

The following figures presents a summary of the standardized pMSE for the different variables.

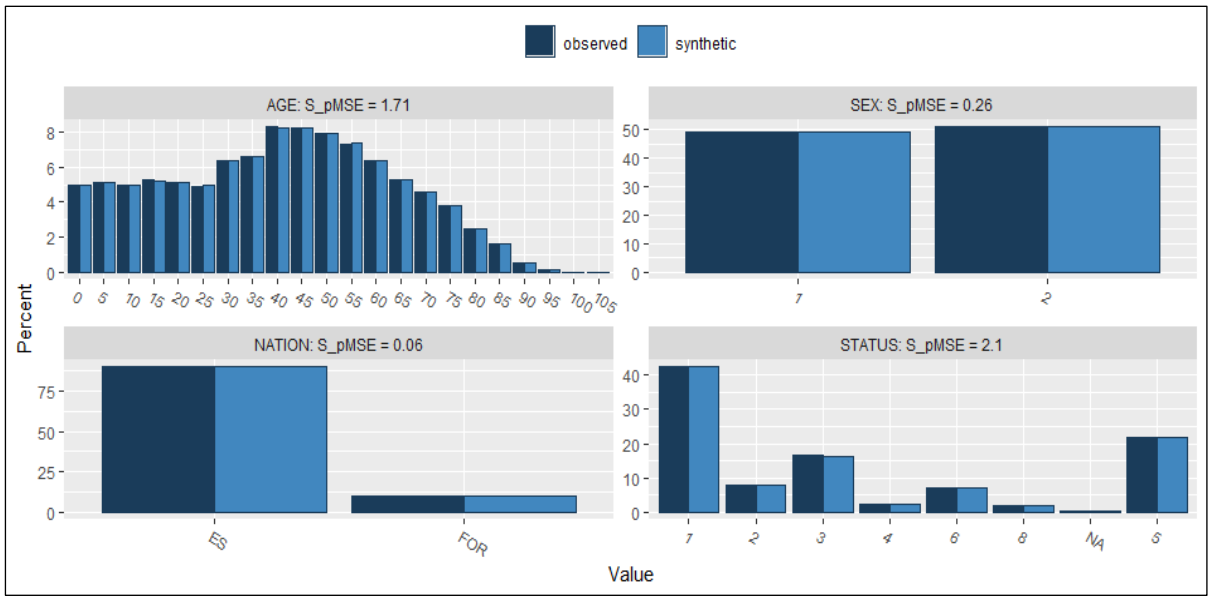


Figure 5: Standardized pMSE for variables AGE, SEX, NATION and STATUS

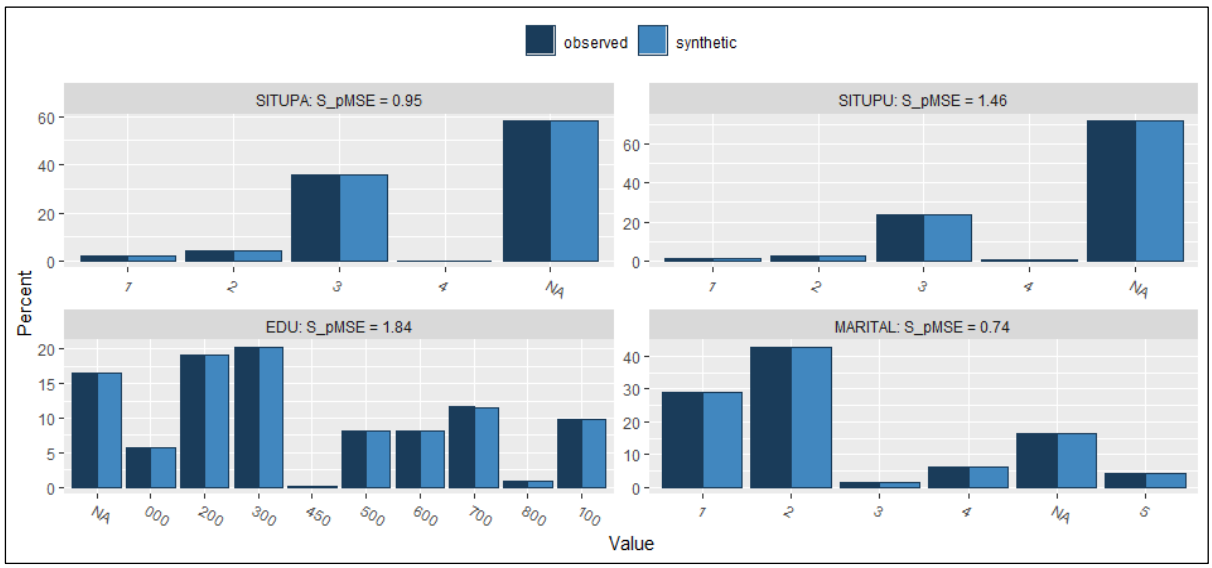


Figure 6: Standardized pMSE for variables SITUPA, SITUPU, EDU and MARITAL

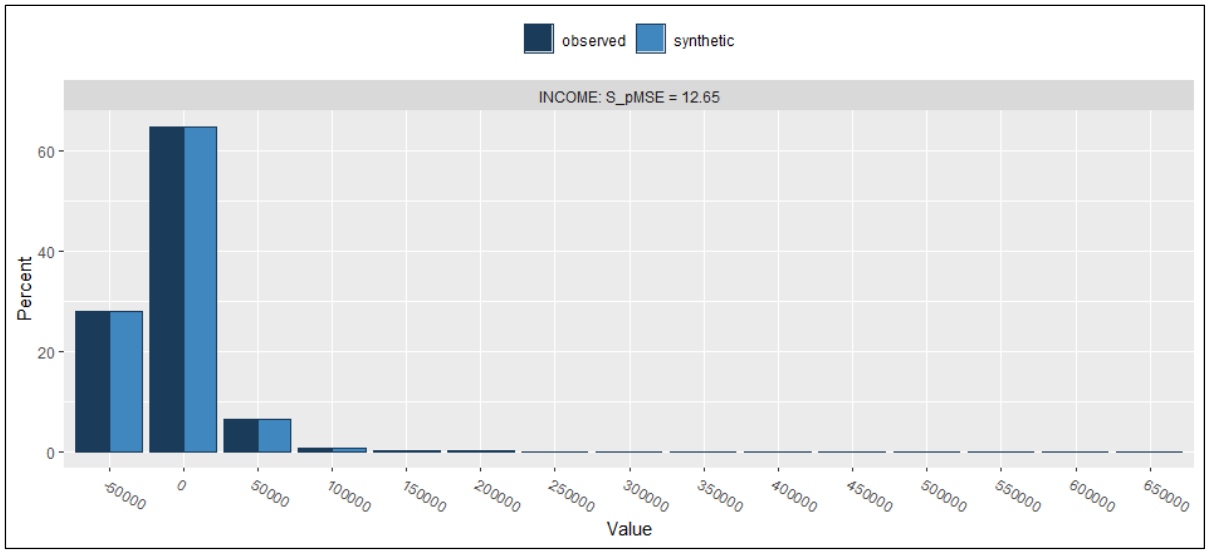


Figure 7: Standardized pMSE for variable INCOME

Similarly, bidimensional pMSE measures are presented in figure 8.

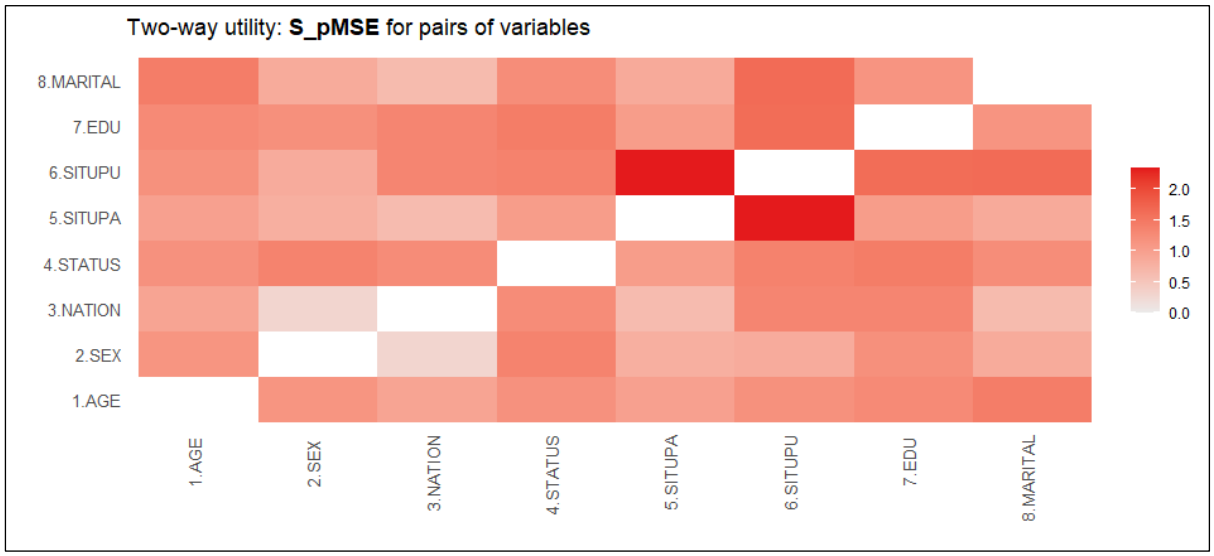


Figure 8: Two-way utility: S_pMSE for pairs of varuables

After synthesis, utility for specific variables, measured by individual and pair-wise standardized p-MSE, was found to be satisfactory, with most values below 5 for categorical variables. For the continuous variables, AGE and INCOME, we found this metric to be less informative, as the limited sample size leads to a poor representation of their underlying distributions. Older individuals are underrepresented in the sample.

In addition, the relationships between demographic variables were examined graphically and were found to closely replicate the weighted distributions observed in the original data. As our population aims to replicate the real age, sex and income distribution at the census tract level, we also analyze these distributions and compare them to the actual values (see figure 9).

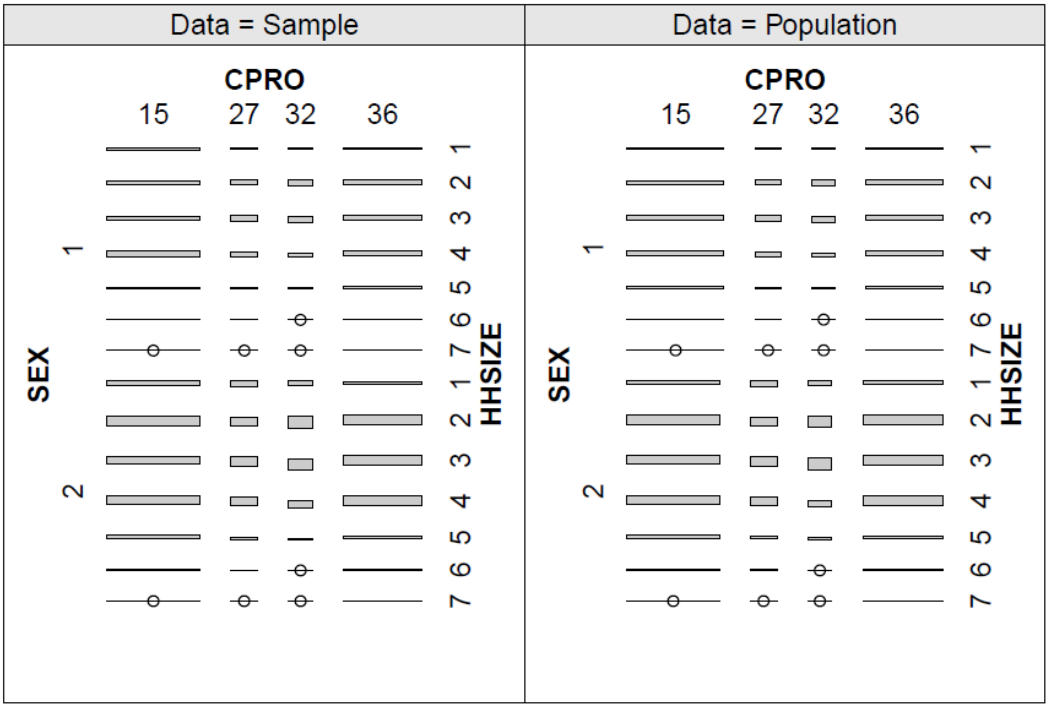


Figure 9: Comparison of the joint distributions of sex, household size (HHSIZE), and province of residence (CPRO) in the sample and in the reference population.

3.4.6.2 Calibration Phase

One of the key objectives of this project is to ensure that the data is consistent with the information published at the census-tract level in the INE's webpage. This is a coherence requirement that is not always imposed in synthetic population generation.

These adjustments are carried out using simulated annealing and sampling calibration techniques, which introduce a slight deviation from the distributions obtained during the initial generation phase. However, as can be observed, the overall distributions remain largely unchanged.

Figure 10 shows the distribution of the relative population differences for each census tract by sex and age group. Most values are concentrated around zero, indicating that the majority of tract-age-sex combinations are very similar between the synthetic population and the original sample. It should be noted that in very small tracts with few individuals, even minor changes can result in relatively large proportional differences.

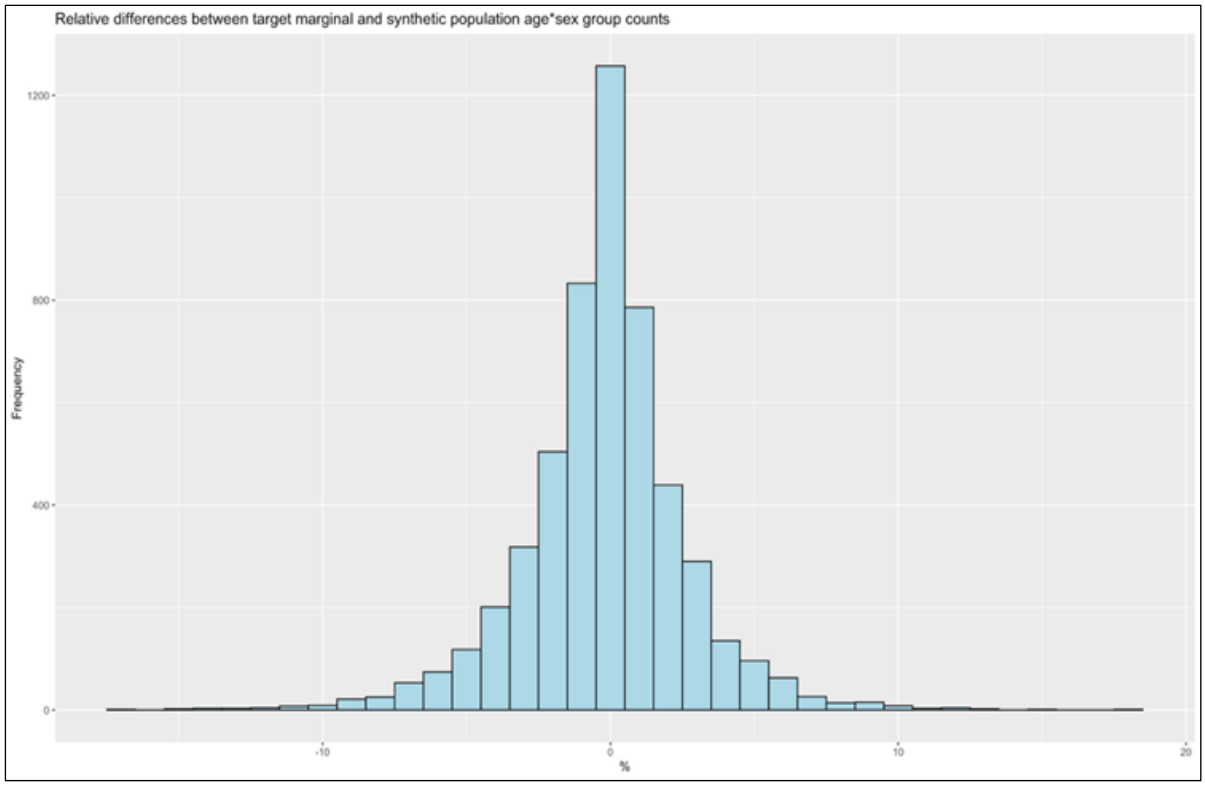


Figure 10: Distribution of tract-level relative differences between target and synthetic population counts for each age–sex group.

Household income is calibrated so that the average personal income per census tract exactly matches the real value (see figure 11)

```
## # A tibble: 6 x 9
##   CSEC       NPERS TOTAL_POB TOTAL_ATLAS     DIFF PDIFF AVG_GWeight MAX_Gweight
##   <fct>      <int>     <dbl>       <dbl>    <dbl> <dbl>       <dbl>       <dbl>
## 1 1500101001   929 13135207.   13135207. -1.30e-8  100.       1.10         1.38
## 2 1500101002   727 10205790.   10205790.  2.24e-8  100.       1.01         1.16
## 3 1500102001   735  9254998.    9254998.  9.31e-9  100.       0.970        1
## 4 1500102002   678  8884254.    8884254.  7.45e-9  100.       1.04         1.16
```

Figure 11 - Comparison between target and calibrated synthetic population totals and household-income calibration weights by census tract.

After calibration, we also analyze the income distribution by age group and sex, which turns out to be very similar to real values (see figure 12).

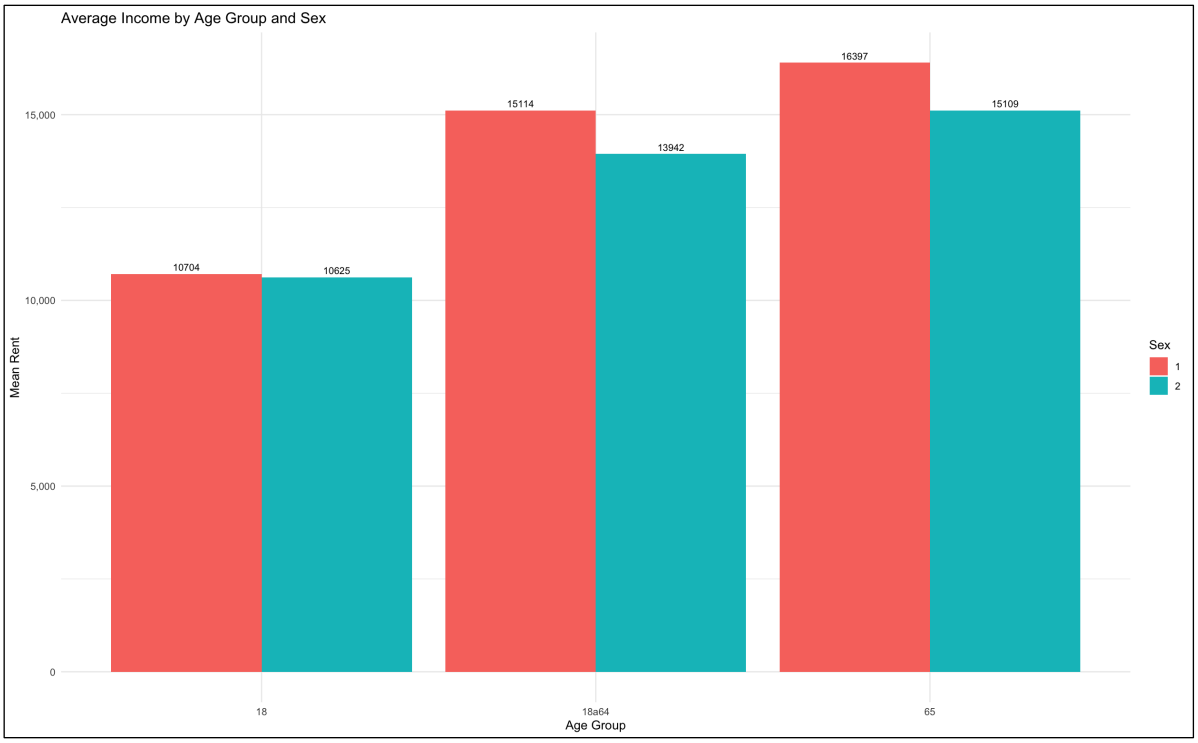


Figure 12: Average personal income by age group and sex after calibration.

We can also compare the correlation heatmaps of the variables for both the original sample and the synthetic population after the calibration phase (see figure 13). The differences are minimal, indicating that the synthetic population has not lost quality after the calibration adjustments.

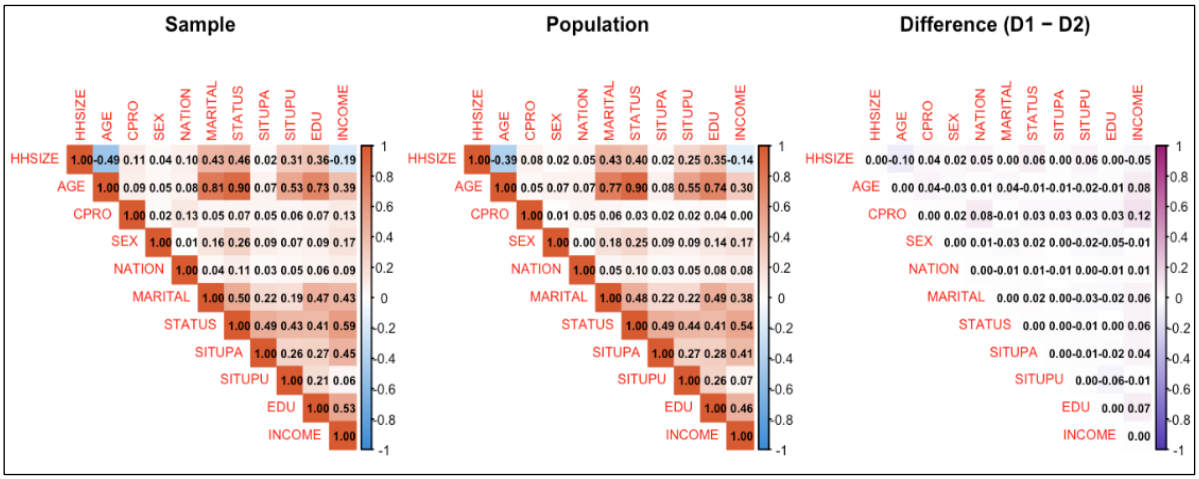


Figure 13: Correlation heatmaps for the original sample and the calibrated synthetic population, with the matrix of pairwise differences.

3.5 Poland

3.5.1 Statistical Domain and Input Data

The area of interest of the Polish contributors includes selected aspects of social statistics. Due to availability, the Polish team used microdata from the Study of Human Capital (BKL), conducted by the Polish Agency for Enterprise Development since 2009 with the support of the Jagiellonian University in Kraków. BKL data are used to monitor the demand for competent staff. The BKL project is financed from the European Social Fund under the Operational Programme Knowledge Education Development (OPKED) 2014–2020.

The data for the synthesis exercise came from the latest available edition of BKL for 2021. The BKL survey consists of two staggered biannual surveys: the main one, conducted on a sample of 4000 respondents and a mid-term one, held a year later, involving a sample of 1500 respondents. The main survey is made up of 2 subsamples: a panel sample with 1500 respondents, and a cross-sectional sample of about 2500 adult respondents aged 18-69, collected using a two-stage stratified sampling design, with strata defined by subregion (NUTS3), place size, sex and age group, which we used for our study. The original dataset contains over 910 variables, of which only 12 were selected since they are usually found in most social surveys conducted to produce official statistics. More information about the survey methodology and results can be found at https://www.parp.gov.pl/component/site/site/bilans-kapitalu-ludzkiego.

In the future, the Polish team plan to use microdata from other social surveys, such as the census or the Labour Force Survey. This, however, requires access to efficient IT environment enabling the processing of very big datasets.

In addition to BKL, simulated data were used to test some more original solutions. The assumption was to make artificial variables resemble real ones typically found in social surveys in terms of range and empirical distribution of variables like sex, education, etc.

3.5.2 Purpose of the Synthetic Data

The main purpose of the work was to work out efficient solutions enabling the creation of synthetic microdata from various surveys which are safe while providing maximum utility for end users. One of the tools tested in the study was differential privacy with different adjustments of the sensitivity function.

3.5.3 Methods for Synthetic Data Generation

One major part of the study involved testing the performance of different synthesis methods available in the synthpop R package using a cross-sectional sample of 2529 adult respondents from the BKL survey. Since the original dataset contained two key variables with a very high percentage of missing values (61.5% for respondent income and 45% for household income), imputation was conducted before the next stage. 6 synthesis methods available in the synthpop R package were used (*parametric, cart, ctree, catall + cart*, *iterative proportional fitting (IPF) + CART* and *cart* stratified by sex, denoted as *strat*) in two scenarios: for 1 set and 10 synthetic sets.

Although **differential privacy** is designed mainly to protect aggregated data and descriptive statistics, it can also be used to generate synthetic microdata consisting of various type variables. We proposed

and tested a method of microdata synthesis based on the exponential mechanism for categorical variables and the Laplace (and alternatively also Gaussian) mechanism for continuous ones; we also assessed total disclosure risk and information loss. For categorical variables the utility function (exponential mechanism) was established individually for each object based on relative frequencies of particular categories. The sensitivity function was defined as a reciprocal of the minimum relative frequency for categories of a given variable, if positive, and 100 otherwise. However, for continuous variables sensitivity was defined in two ways. The first one was uniform for all objects and based on relative ranges. In the second option, sensitivity was set individually for any object as four-thirds of the greater of the normalized absolute deviations of observations from extreme values of a given variable. This option makes it possible to better capture the impact of changes in the privacy level while avoiding unnecessary homogeneity and reducing the risk of overestimating a particular feature.

That synthesis exercise was conducted using an artificial dataset of 150 units described by 7 categorical and 7 continuous variables. The values of categorical variables were generated from the binomial distribution with preselected parameters and the values of continuous ones – from the multivariate normal distribution. The experiment was replicated 200 times.

Differential privacy mechanisms were found to be good tools for generating synthetic data. However, the reliability and utility of synthetic data generated in this way depends largely on specific DP parameters. For instance, the smaller the epsilon, the lower the disclosure risk (fewer objects in the synthesized data correctly matched with the original ones) but the greater information loss; conversely, the greater the epsilon, the greater the risk of disclosure but the smaller the information loss. In the case of reasonable parameterization, the individualization of sensitivity is more advantageous: the smaller epsilon, the greater the role of uniform sensitivity. The only important drawback was the computation time (over 3 hours).

3.5.4 Methods for Utility Measurement

In addition to the utility measures included in the R packages for data synthesis (mainly S_pMSE), there are some useful indices for assessing information loss for microdata resulting from the application of the SDC methods, especially perturbative ones.

In the `sdcMicro` R package there are two functions to measure of the utility of synthetic data. The first one, IL_variables, is used to measure the disturbance of distributions, which is calculated by means of different distance functions depending on the class of the variable. It is the arithmetic mean of individual distances computed for particular observations. The formula for calculating the individual distance between an original and synthetic observation depends on the measurement scale: for nominal variables it is to equal 1 if there is no change and 0 otherwise, for ordinal variables – the normalized number of categories by which both observations differ, and for continuous – the normalized cyclometric function (arcus tangent) of the absolute value of the difference between the original and synthetic value. This ensures that the final result falls within the [0,1] interval and can be interpreted as an estimate of information loss expressed as a percentage. The formula can also be used to determine the percentage information loss for individual variables. It is worth noting that given the binary nature of distances in categorical variables and the flattening of the cyclometric function near its horizontal asymptotes, the estimate of information loss may be slightly overestimated. However, this makes it easier to identify problems in this area. The second function, `IL_correl`, measures the impact of synthesis on the intensity of correlations between continuous variables in the original and synthetic data. It is calculated using normalized distances between diagonal entries of the inverse

correlation matrices for original and synthetic data. Its values range from 0 to 1 and have analogous interpretation as the measure for distributions[1].

The synthesis exercises showed that both measures can be effectively applied to assess the utility of synthetic data by comparing original and synthetic data. They should be used together, as both aspects (distributions, correlations) are equally important for the user.

3.5.5 Methods for Privacy Assessment

Methods of assessing disclosure risk depend on the synthetization method, but they provide comparable results. Basically, they are based on classical rules like *k*-anonymity or *l*-diversity. However, due to artificial nature of some synthetic data such simple measures can overestimate real privacy threats. This is why for such applications other methods should be considered. In the case of differential privacy for continuous variables, sensitivity is defined by specific statistics (established by accounting for user expectations) and distances between values of such statistics in original and synthetic data. It is an integral part of DP mechanisms. Whereas the distance formula is arbitrarily associated with a specific DP mechanism, the sensitivity function should be determined by a competent SDC specialist. The same formula can be used for all records or an more individualized approach can be selected. Our experiments showed that in optimal circumstances (i.e. assuming the best possible trade-off between minimization of disclosure risk and maximization of data utility) individually determined sensitivity gives better results, especially in terms of losses on distributions.

One can also consider using a complex measure proposed by Młodak (2025), which is based on classical rules and models of cooperative games, but accounts for risk concerning variables regardless of their measurement scales and can be applied to partial synthetic data or other forms of SDC. Nevertheless, the problem of efficient disclosure risk assessment still requires deeper studies.

3.5.6 Preliminary Results

3.5.6.1 Generation of synthetic data for BKL

The main results of the synthesis exercise based on the BKL data can be summarised as follows:

- utility for specific variables, measured by standardized ratio propensity mean squared error (S_pMSE), was found to be the best for *catall* and cart, with each yielding the lowest values for 3 or 4 of the 12 variables (depending on the scenario), while cart was the best performing method for the whole set (min=1.98, mean=2.37, max=2.71),

[1] More details on these methods can be found in the following papers: Młodak A., Pietrzak M., Józefowski, T. (2022), The trade-off between the risk of disclosure and data utility in SDC: A case of data from a survey of accidents at work, *Statistical Journal of the IAOS*, 38:1503–1511; Pietrzak M., Józefowski T., Klimanek T., Młodak A.(2022), An Optimized Selection of Statistical Disclosure Control Methods – A Case Study Involving Microdata from the Polish Survey of Accidents at Work. In: Jajuga, K., Dehnel, G., Walesiak, M. (eds) *Modern Classification and Data Analysis. SKAD 2021*. Studies in Classification, Data Analysis, and Knowledge Organization. Springer, Cham, pp. 63–78.

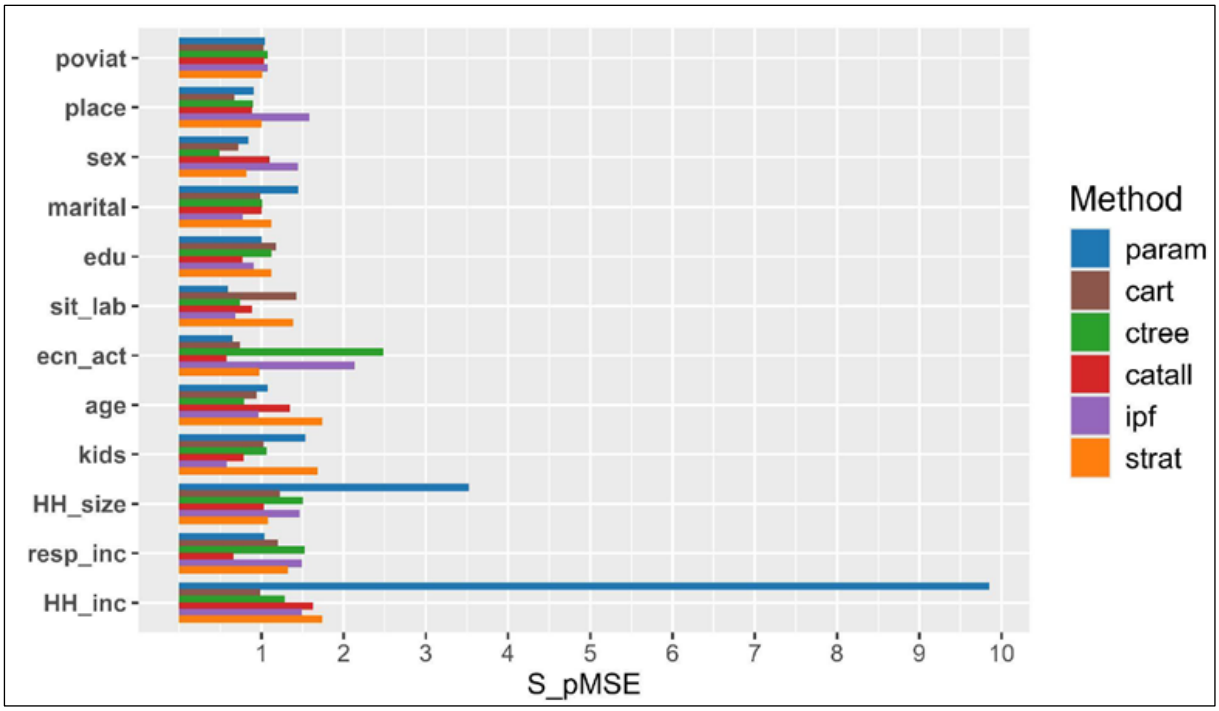


Figure 14: Comparison of variable-level standardized ratio propensity mean squared error (S_pMSE) across different synthetic data generation approaches.

Table 8 - Overall standardized ratio propensity mean squared error (S_pMSE) by synthesis approaches for one and ten synthetic datasets.

	1 set	10 sets		
Method		**Min**	**Mean**	**Max**
param	2.674	2.333	2.670	2.942
cart	1.982	2.018	2.373	2.713
ctree	2.102	2.077	2.471	2.888
catall	2.573	1.922	2.508	2.917
ipf	2.588	2.407	2.578	3.071
strat	2.536	1.982	2.364	2.741

- as regards information loss, measured using two functions implemented in the sdcMicro package, differences between the methods were generally minimal for individual variables, though it is worth noting that for 5 variables (marked in red) values of information loss (IL) exceed 80%. Results below 50% (marked in blue and green) were obtained for variables with relatively few levels or those where only a few levels accounted for the majority of observations.

Table 9 - Minimum and maximum variable-level information loss values across synthesis methods for one and ten synthetic datasets.

	1 set				10 sets			
	Min		Max		Min		Max	
Variable	method	value	method	value	method	value	method	value
poviat	catall	0.992	strat	0.996	catall	0.992	param	0.996
place	param	0.770	strat	0.793	param	0.770	strat	0.774
sex	ctree	0.480	ipf	0.506	ctree	0.480	ipf	0.506
marital	ctree	0.539	strat	0.553	ctree	0.539	strat	0.557
edu	param	0.861	catall	0.879	param	0.861	catall	0.879
sit_lab	cart	0.476	ctree	0.496	strat	0.474	ctree	0.496
ecn_act	strat	0.210	ctree	0.231	strat	0.221	ctree	0.231
age	param	0.896	ctree	0.906	strat	0.893	ctree	0.906
kids	ipf	0.470	param	0.750	ipf	0.470	param	0.750
HH_size	param	0.499	ipf	0.513	param	0.499	ipf	0.513
resp_inc	ctree	0.991	catall	0.993	strat	0.991	catall	0.993
HH_inc	param	0.984	catall	0.993	param	0.984	catall	0.993

- the results for the disturbance of distributions are rather poor and practically the same for all methods, regardless of the scenario. In contrast, the results for the impact of synthesis on correlations between numeric variables are very good, though the differences between the methods are slightly bigger, with the exception of the *parametric* approach for the first scenario, where the correlation-based utility measure (CN) value is higher by one order of magnitude (0.148-0.159), and the *ipf* method, where it is lower by one order of magnitude (0.003).

Table 10 - Distribution disturbance (DD) and correlation-based utility measure (CN) by synthesis method for one and ten synthetic datasets.

	1 set		10 sets	
method	DD	CN	mean_DD	mean_CN
param	0.689	0.159	0.688	0.019
cart	0.687	0.030	0.688	0.019
ctree	0.687	0.023	0.688	0.026
catall	0.689	0.026	0.689	0.023
ipf	0.689	0.003	0.689	0.029

- specific utility in the case of model-based inference for a linear model resp_inc ~ place + sex + age + edu + ecn_act + HH_inc, was tested using the compare function from the synthpop package, which provides the user with 4 measures: mean confidence interval overlap (MCIO), mean absolute standardized coefficient difference (MASCD), Mahalanobis distance ratio for lack-of-fit (MDR) and lack-of-fit test and p-value (LoF)

Table 11 - Mean confidence interval overlap (MCIO), mean absolute standardized coefficient difference (MASCD), Mahalanobis distance ratio (MDR), and lack-of-fit statistic (LoF) by synthesis method..

	1 set				10 sets			
Method	MCIO	MASCD	MDR	LoF	MCIO	MASCD	MDR	LoF
param	0.54	1.80	2.50	69.96	0.80	0.79	23	644
cart	0.66	1.35	5.09	142.61	0.76	0.95	24	681
ctree	0.61	1.54	7.16	200.6	0.68	1.20	51	1422
catall	0.68	1.26	2.85	79.86	0.79	0.81	6.6	185
ipf	0.65	1.35	2.14	60.06	0.81	0.73	11	296
strat	0.70	1.19	2.46	68.89	0.72	1.10	31	881

- adjusted R^2 was the best for the **param** method:

Table 12 - Adjusted R-squared values for the linear model fitted to the original and synthetic datasets.

	1 set	10 sets		
Method	Value	Min	Mean	Max
orig	0.480	0.480	0.480	0.480
param	0.663	0.578	0.641	0.650
cart	0.437	0.409	0.438	0.453
ctree	0.333	0.333	0.391	0.423
catall	0.529	0.437	0.509	0.526
ipf	0.490	0.453	0.496	0.503
strat	0.458	0.362	0.407	0.415

More detailed analyses can be conducted by examining outputs of the `summary` function, which shows inferences to coefficients and standard errors that would be obtained from the original data. Standardized coefficient differences can be interpreted as a measure of the effect size, which quantifies the practical importance of discrepancies between synthetic and original regression coefficients by indicating how many original standard errors (SEs) the synthetic estimate is away from the observed coefficient. For example, in 2nd scenario, there were between 7 (*ctree*) and 11 (*param, catall*) statistically significant coefficients (with p-values below 0.01) including the intercept, 2 place categories, sex, several education categories, economic activity and household income (HH_inc). In some cases, there is evidence of distorted relationships with respect to the observed data. The worst example of this tendency again involves synthetic values of HH_inc, generated by the *parametric* approach, where the standardized coefficient difference was equal to -6.95, with the CI overlap of -0.772 indicating no overlap but an evident gap between CIs. Another problematic variable was ecn_act_yes; its standardized coefficient difference exceeded 2.0**.** for all methods except for *catall* (in 4 cases it was higher than 3.0) and there was either no CI overlap, or it is very small or moderate (0.025-0.439). The *strat* method was the least efficient in preserving relationships: SDC values and confidence interval (CI) overlaps were poor for 4 significant coefficients, including 200-499.9 thous and eduMBA. The only method where SDC values for all statistically significant coefficients were below 2.0 and CIs largely overlapped (over 0.7) was the *catall* method, which means that the same predictors that were important in the original model remain stably significant and are not distorted. Another method that produced synthetic data that look suitable for regression-based inference was *ipf*. Two of its significant coefficients were biased (HH_inc: 3.82 and ecn_act_yes: - 2.12) but only one (though statistically insignificant) coefficient had a different sign than the original one, unlike *catall,* which had

as many as 5. The figure below shows a comparison of intervals for the model fitted to data synthesised by catall and original data:

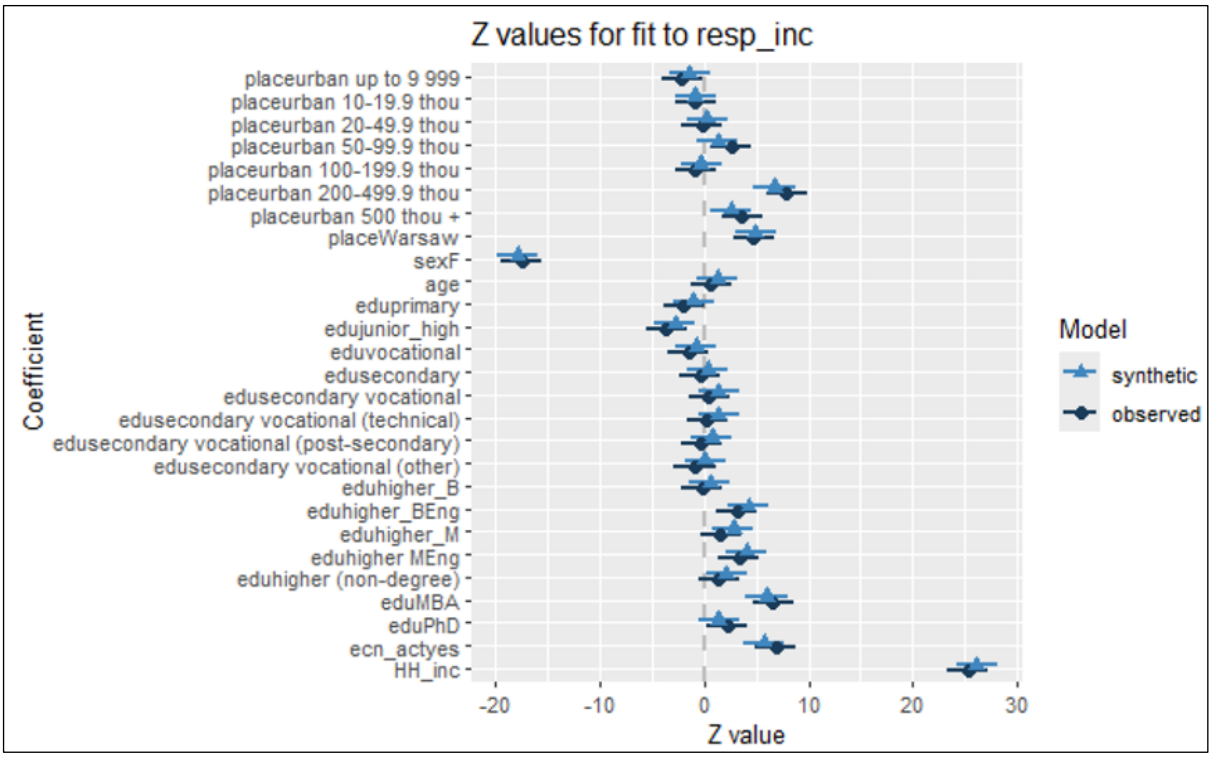


Figure 15: Comparison of the coefficient Z-values estimated from the observed and synthetic data for the resp_inc model.

3.5.6.2 Use DP for generating synthetic data

As indicated earlier, the generation of synthetic data based on the Differential Privacy rule uses exponential mechanism for categorical variables and Laplace (alternatively: Gaussian) mechanism for continuous variables. The latter alternative was chosen because of some different properties. That is, Laplace mechanism is regarded as a simple and most popular tool to application of differential privacy to continuous variables, also in multivariate variant and uses ℓ_1- global sensitivity. On the other hand, the Gaussian mechanism is based on ℓ_2- global sensitivity.

In our experiment the artificial dataset consists of 7 categorical and 7 continuous variables. The values of categorical variables are generated from the binomial distribution $B(n_i, p_i), i = 1,2,\dots,7$, with $n = (n_1, n_2, \dots, n_7) = (7,1,4,14,2,6,9)$ and $p = (p_1, p_2, \dots, p_7) = (0.3,0.5,0.1,0.7,0.3,0.4,0.8)$. These variables are denoted by $V_1, V_2, \dots, V_7$, respectively. The parameters of these distributions were selected so that the individual variables reflect the range - and to a large extent also the distribution – of categories of various typical variables appearing, for example, in national censuses, such as: sex, age groups, type of place of residence, level of education, etc. The values of continuous variables are generated from the multivariate normal distribution $N(\mu, \Sigma)$, where $\mu = (0,15, -3,7,5,1, -6)$ and

$$\boldsymbol{\Sigma} = \begin{bmatrix} 2 & 0.1 & 0 & 0 & 0 & -0.2 & 0 \\ 0.1 & 3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ -0.2 & 0 & 0 & 0 & 0 & 14 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

These variables are denoted by $C_1, C_2, \dots, C_7$, respectively. The expectations and covariances model several variables with relatively large diversification, as is often observed in practice. In our experiment we have generated a dataset containing $n = 150$ objects. The simulation was repeated 200 times.

One of the most important problems was optimal choice of the parameters of used DP mechanisms. After testing several variants, for exponential mechanism we have set $\varepsilon \coloneqq 0.3$. As candidates, for the vector of all possible levels of particular categorical variables was used. The utility function, necessary for defining the mechanism was defined individually for each object, i.e. $u_i = (u_{i1}, u_{i2}, \dots, u_{i7})$ where $u_{ij} = f_j(c)$ if the original value of j -th variable for i -th object (x_{ij}) is c and $u_{ij} = 0.01$ otherwise. $f_j(c)$ is here a relative frequency of category c of the j -th variable, $j = 1,2, \dots ,7$, $i = 1,2, \dots , n$. Some important problem concerns here the optimal definition of the sensitivity function. its form should depend on most expected or most popular descriptive statistics which can be produced using the protected data. Thus, for defining sensitivity let $\xi_j \coloneqq \min_{i=1,2,\dots,n} \left(f\left(x_{ij}\right)/n\right)$, where $f\left(x_{ij}\right)$ is the absolute frequency of x_{ij}. Hence, if $\xi_j > 0$ then we place the value of sensitivity as $1/\xi_j$, otherwise the sensitivity amounts to 100, $j = 1,2, \dots ,7$, $i = 1,2, \dots , n$. This choice reflects the sensitivity of counting units possessing a given feature inversely related to its frequency of occurrence and gives appropriate meaning in this context also to zero frequencies.

In the case of continuous variables, for Laplace and Gaussian mechanisms again privacy budget as $\varepsilon \coloneqq 0.3$ was assumed. For the Gaussian option, the additional parameter δ was established as 0.01. The target function and the sensitivity was defined in the same manner for all algorithms. The function for i -th object was its vector of observations of continuous variables $(x_{i1}, x_{i2}, \dots , x_{i7})$, $i = 1,2, \dots , n$. As regards the sensitivity, two variants of such quantity were applied. First of them was unform for all objects. That is, it was defined as the vector $(r_1, r_2, \dots , r_7)$, where

$$r_j = \frac{\max_{k=1,2,\dots,n} x_{kj} - \min_{k=1,2,\dots,n} x_{kj}}{n}, \quad j = 1,2, \dots ,7.$$

Therefore, the sensitivity is here regarded as a typical part of range of a given variable per object. In the alternative approach the sensitivity was established separately for each record according to greater absolute distance of a given observation from minimum and maximum of relevant variable.

$$\frac{4 \max\left\{x_{ij} - \min_{k=1,2,\dots,n} x_{kj} , \max_{k=1,2,\dots,n} x_{kj} - x_{ij}\right\}}{3 \cdot n},$$

$i = 1,2, \dots , n$.

The risk assessment was based on differences between synthesized and original data. That is, we have looked for such objects in synthetic dataset which can be matched with some original. More precisely, for any original object i we find an object l in the synthetic dataset that $l = \arg\min_{u=1,2\dots,n} \sum_{j=1}^{7} d\left(x_{ij}, x_{uj}^*\right)$ (where x_{uj}^* – value of j –th variable for u –th object after synthetization). The final result is the percentage of objects for whom the best match is himself (i.e. $l = i$). The

distance d is defined according to the Gower's distance formula except for the distance between observations of continuous variables which is defined as $d(x_{ij}, x_{uj}^*) = \frac{2}{\pi}\arctan|x_{ij} - x_{uj}^*|,\ i, u = 1,2, \dots, n\ ,\ j = 1,2 \dots ,7$ (as in the case of measure of information loss proposed and implemented by Młodak, Pietrzak and Józefowski (2022)).

The whole algorithm for such study was written in the R environment. More precisely, we have used the following packages:

- `DPpack` (application of differential privacy: `ExponentialMechanism`, `LaplaceMechanism` and GaussianMechanism);
- `sdcMicro` (assessment of information loss: `Il_variables` – for all variables on distribution and `Il_correl` – for continuous variables on correlation);
- `rcompanion` (`cramerV` – information loss on dependencies for categorical variables based on Cramer's V indices).

In the last case, for any pair of categorical variables a relevant contingency table was constructed, and the value of independency test was computed. It was next used to compute the Cramer's V coefficient. On such basis the matrix of such coefficients was determined. The final measure is computed analogously as in `IL_correl`, i.e. as a normalized distance between diagonal entries of inversed Cramer's V matrix.

Figure 16 shows the histograms of distribution of estimated disclosure risk in particular repetitions of simulation assessed based on correct fits of objects in original and synthetic files.

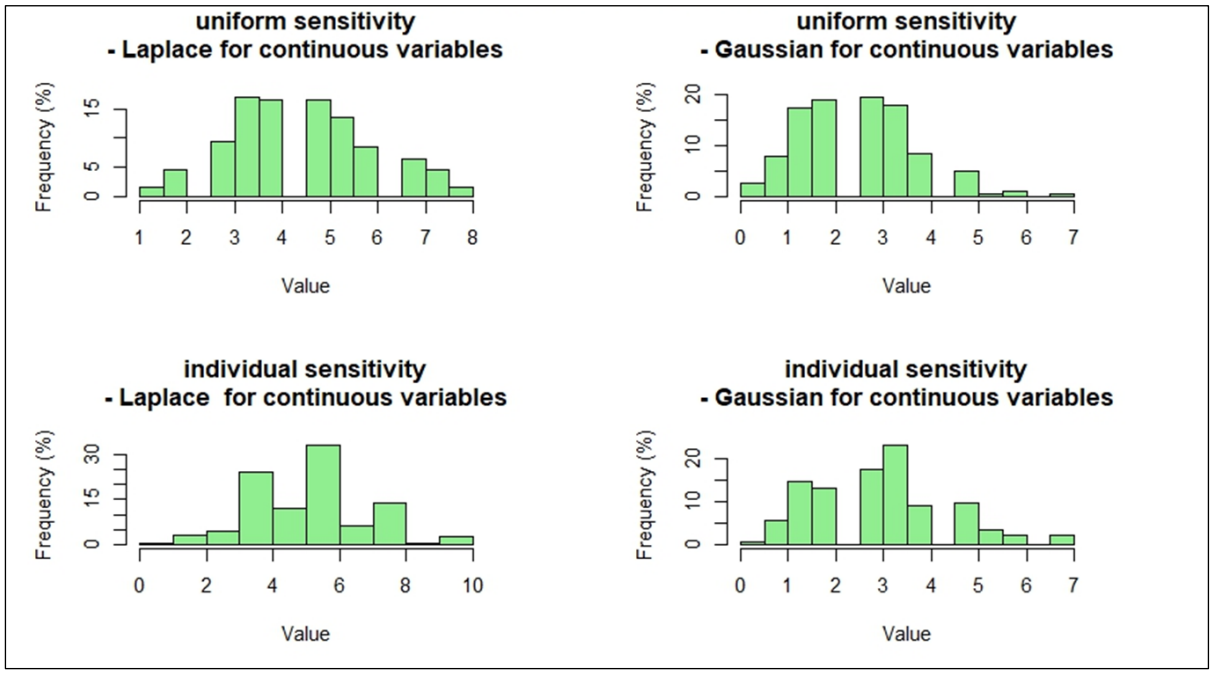


Figure 16: Frequency histograms for percentage of correct fits between original and synthetic datasets.

As one can see, the estimated risk is usually low. Slightly better results are obtained when using Gaussian mechanism for continuous variables – the relevant values are more concentrated around 0%-3%. On figures 17 – 20 the histograms of loss on distribution of variables for all and on correlation for continuous variables for particular types of DP-mechanisms and sensitivity are presented.

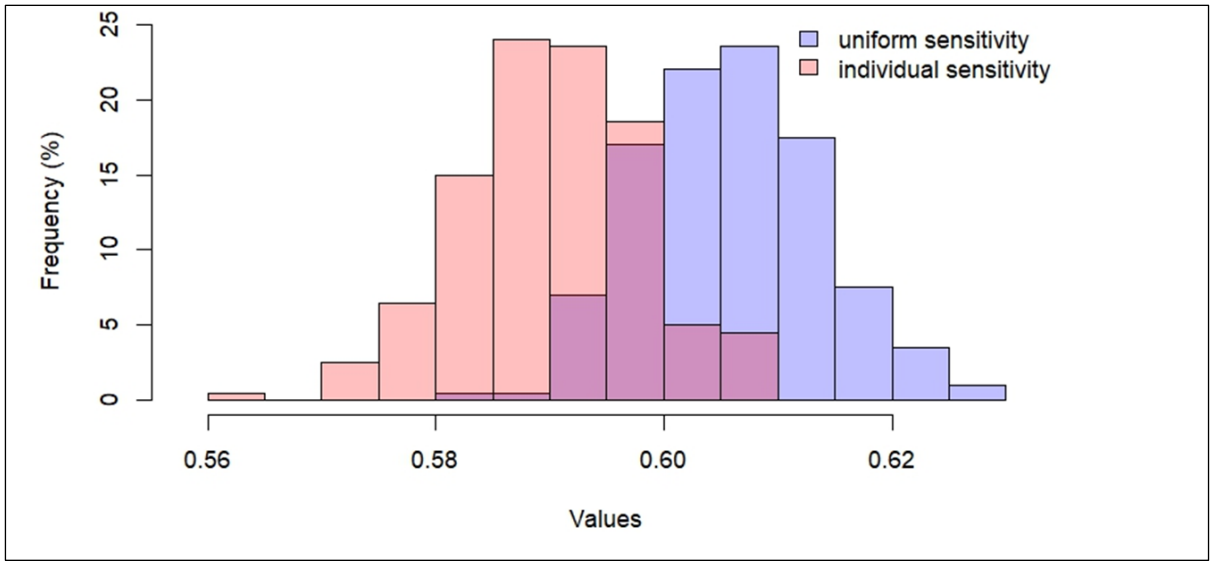


Figure 17: ***Comparison of frequency histograms for loss on distribution – Laplace mechanism for continuous variables.***

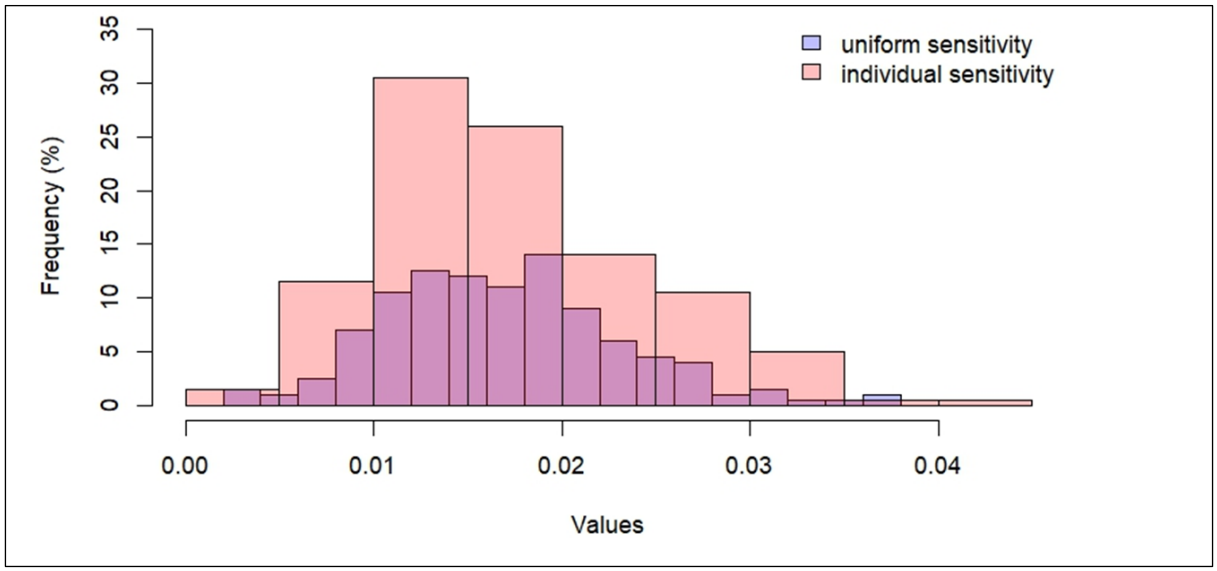


Figure 18: Comparison of frequency histograms for loss on correlation for continuous variables – Laplace mechanism.

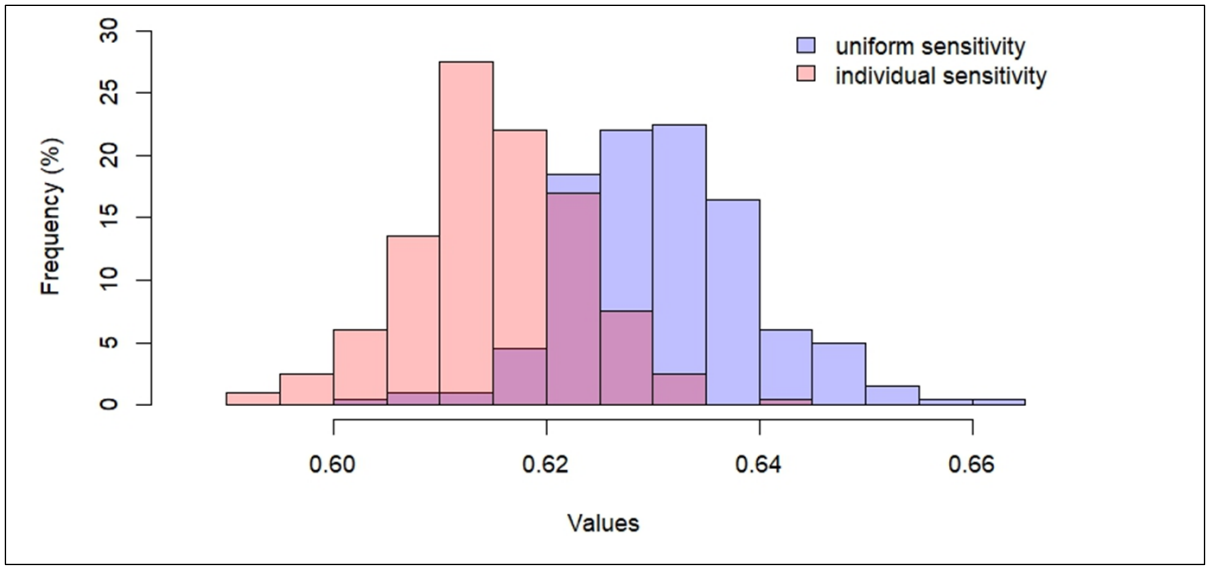


Figure 19: Comparison of frequency histograms for loss on distribution – Gaussian mechanism for continuous variables.

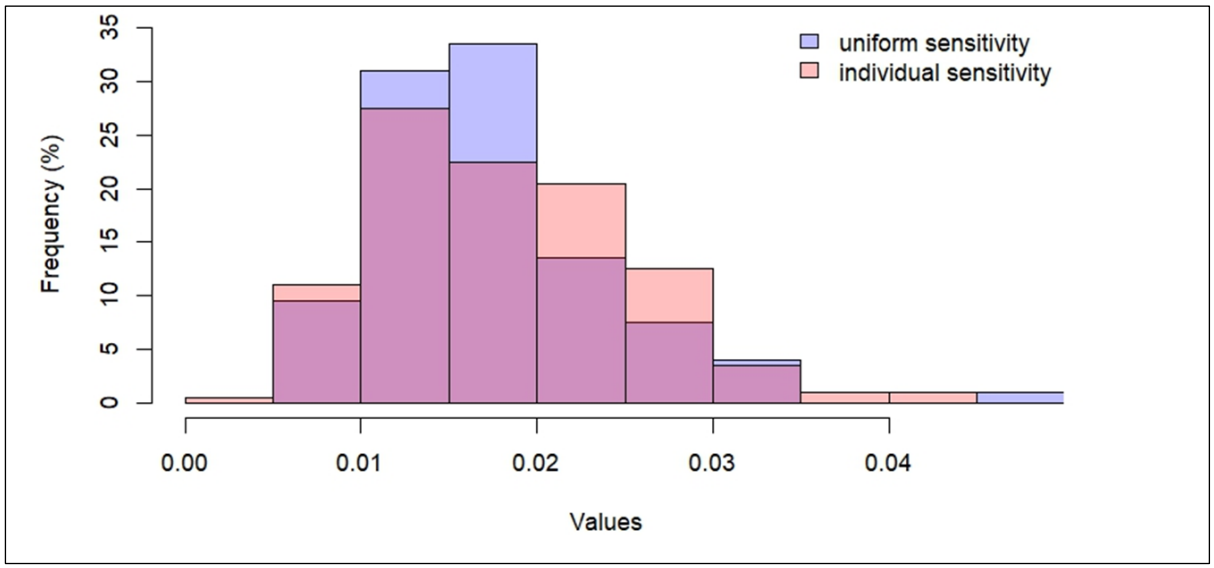


Figure 20: Comparison of frequency histograms for loss on correlation for continuous variables – Gaussian mechanism.

Therefore, the individualization of sensitivity significantly reduces the loss on distribution in comparison with the uniform option in this context. In the case of correlation, the differences between results of applying both approaches are similarly low, although the Laplace DP-mechanism seems to give analysed values more concentrated on interval [0.01, 0.02] for individual sensitivity than for uniform one. Figure 21 provides frequency histograms for distribution of information loss based on Cramer's V coefficient for categorical variables. It shows also that the utility of synthetic data is high and the loss does not exceed 10%, being typically between 5.5% and 7.5%.

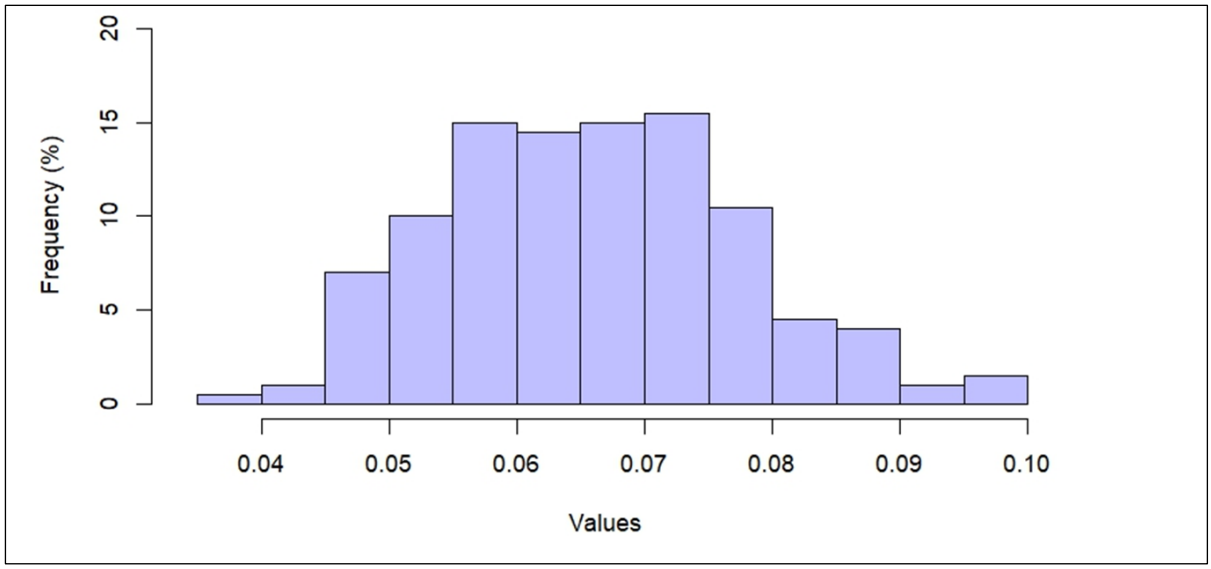


Figure 21: Frequency histogram for information loss on connections between categorical variables measured on the basis of the Cramer's V coefficient.

Figures 22-25 show histograms of information loss for distributions of particular variables depending on sensitivity and DP-mechanism for continuous variables.

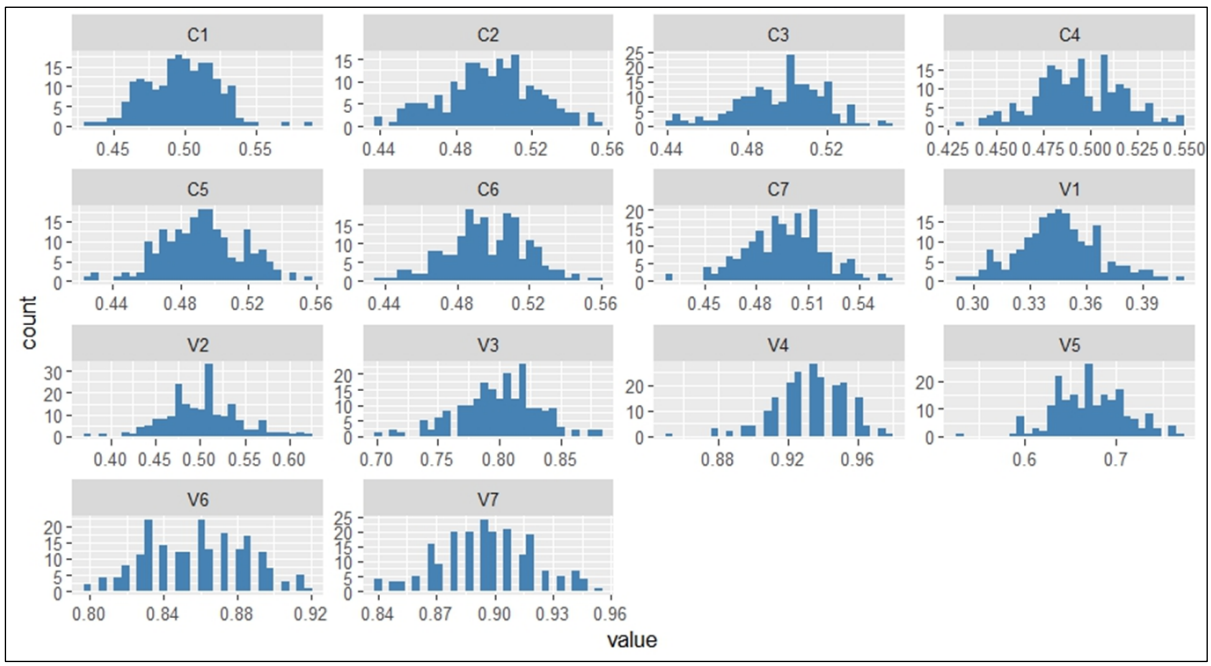


Figure 22: Information loss – uniform sensitivity and Laplace mechanism for continuous variables.

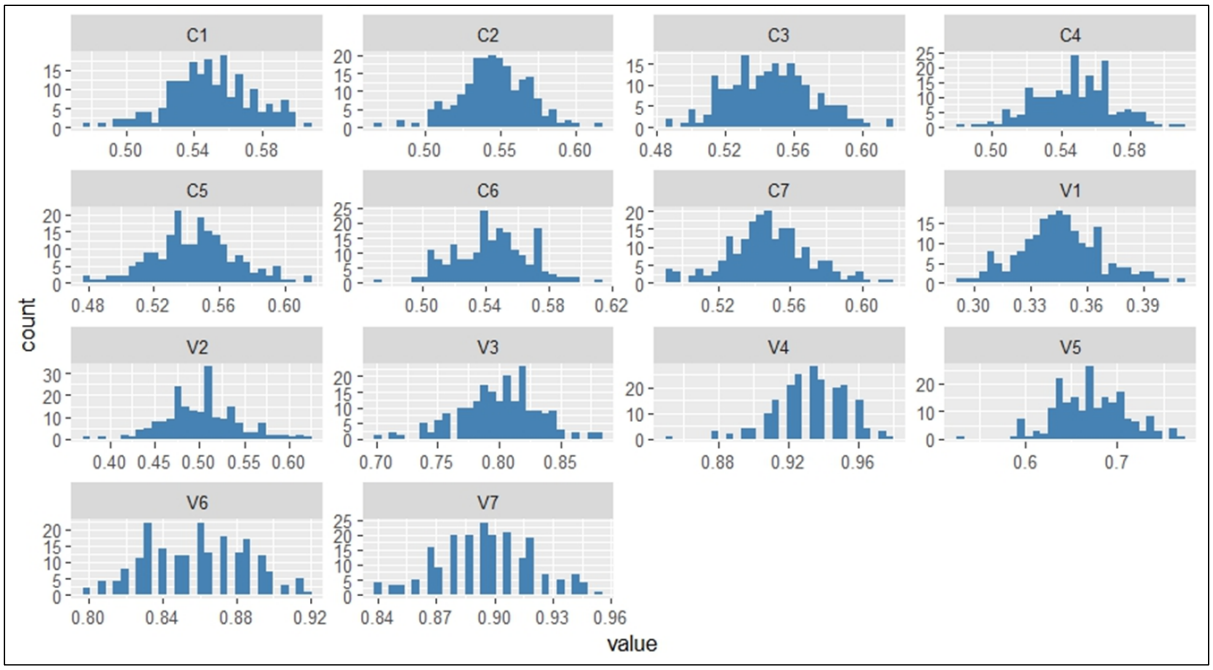


Figure 23: Information loss – uniform sensitivity and Gaussian mechanism for continuous variables.

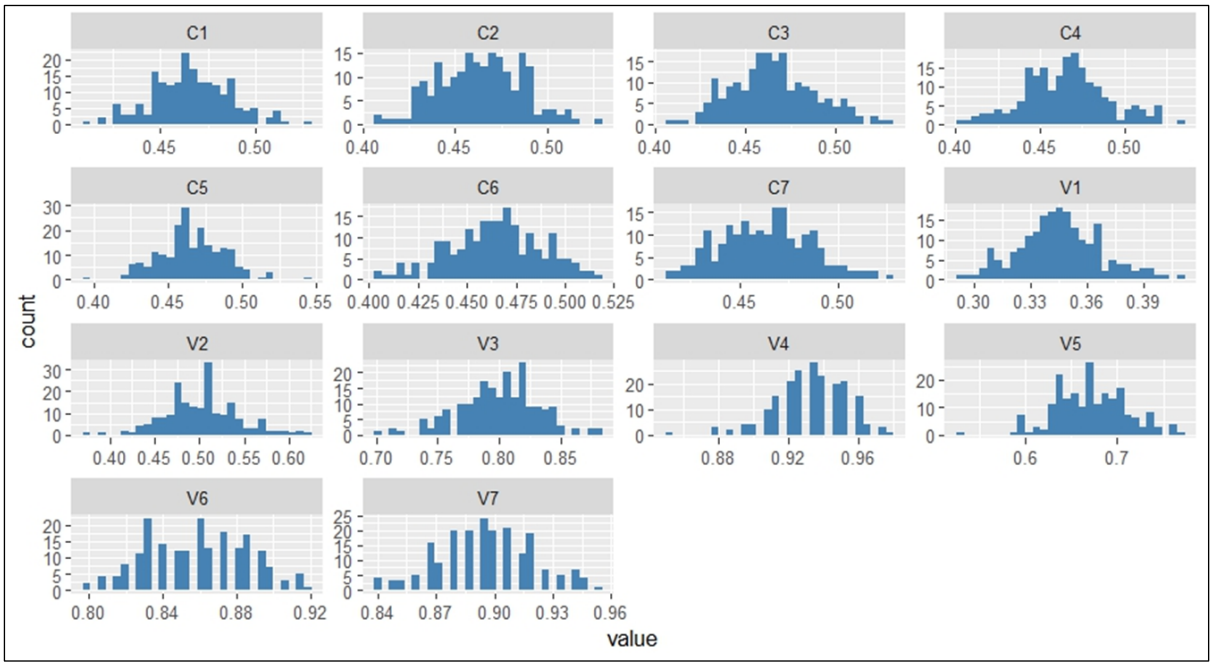


Figure 24: Information loss – individual sensitivity and Laplace mechanism for continuous variables.

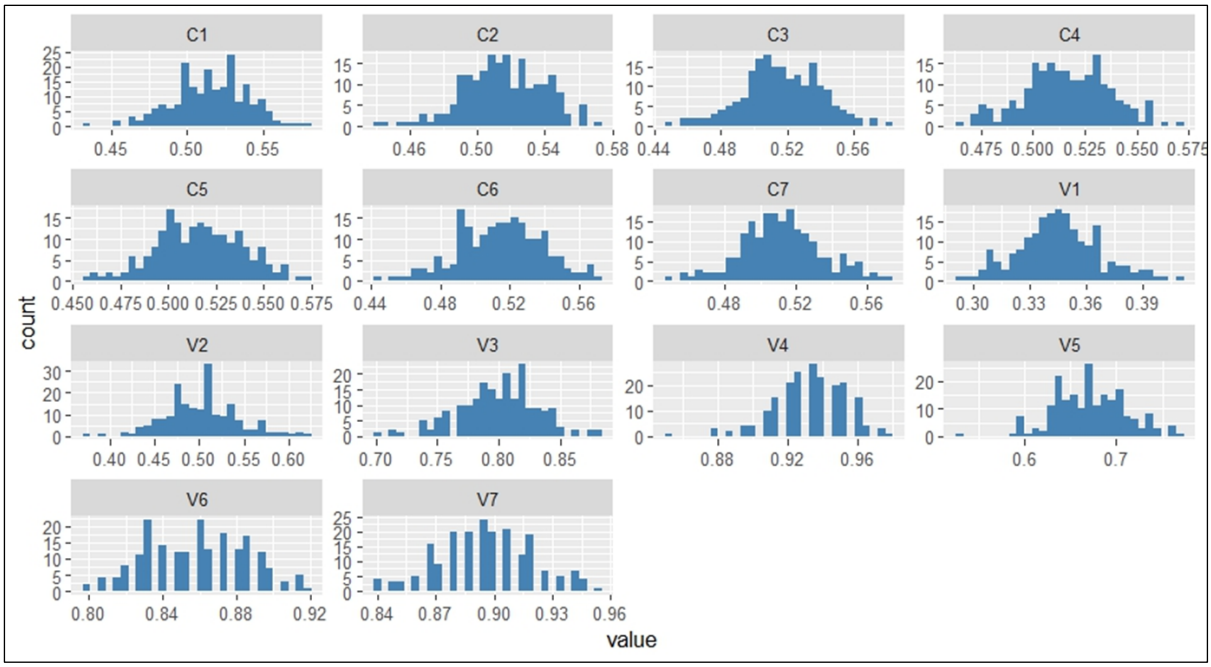


Figure 25: Information loss – individual sensitivity and Gaussian mechanism for continuous variables.

In the case of continuous variables, individualizing sensitivity therefore reduces the level of information loss. It should also be noted that due to the flattening of the cyclometric function on its "tails" close to horizontal asymptotes, information loss can be overestimated. However, this allows for better detection of potential problems. In the case of categorical variables, variables whose values are expressed on a nominal scale play a significant role in the loss value (of the seven considered, for practical reasons we assumed that there are six such variables – the only variable whose data are expressed on an ordinal scale is variable V1). In such a case, the indicator under consideration is based primarily on whether the synthesized value differs from the original value, and nothing more. Hence, the relatively large loss values provide an indication of the scale of change, but not necessarily the level of distortion in the descriptive statistics.

3.6 Italy - Use Case 1 - Synthetic Data Generation for Official Survey Microdata

3.6.1 Statistical Domain and Input Data

The empirical analysis presented in this study is based on data from the Trips and Holidays survey conducted by the Italian National Institute of Statistics (Istat). The survey is implemented as a module of the Household Budget Survey (HBS) and collects information on tourism mobility of residents in Italy, including both leisure and business trips undertaken domestically and abroad. The dataset contains information on several aspects of travel behavior, such as duration of trips, destination, organization of accommodation and transport, as well as selected demographic characteristics of the travelers.

For the purposes of this study, the Public Use Files (PUF) derived from the Trips and Holidays survey were used as the primary data source. These datasets are publicly available and have already undergone standard confidentiality protection procedures applied by the statistical authority. Their public accessibility ensures the reproducibility of the experiment, allowing the methodological framework to be replicated without requiring access to confidential microdata.

The original PUF dataset contains 22,404 records collected between 2019 and 2023 and includes 45 variables describing different aspects of tourism activity. For the experimental analysis presented in this work, a subset of the data was selected in order to simplify the modeling process and reduce sparsity issues in categorical variables.

In particular, the analysis focuses on the year 2023, selected to avoid potential distortions associated with the COVID-19 period. After the data preparation and variable selection process, the final dataset used in the study consists of 4,932 observations and 14 variables.

These variables are divided into two groups according to their measurement type: numerical variables and categorical variables reported in the following.

Table 13 - Numerical variables included in the dataset

Variable	Description	Categories/Values
ANNRIF	The reference year when the vacation ended.	Year value
MESE	The month of return from the vacation.	1–12
GGINIZ	The starting day of the month was when the vacation began.	1–31
DURATA	The duration of the stay, measured in days.	Number of days
NPART	The number of participants on the trip (e.g., 4 for a couple traveling with their two children).	Integer value
ESPE_CO	Consumption of expenditures, expressed in euros (€).	Euros (€)
ESPE_GIO	Expenditures for leisure or recreational activities, expressed in euros (€).	Euros (€)

These variables describe the timing and duration of trips as well as expenditure-related aspects and are essential for reproducing realistic quantitative distributions in the synthetic datasets.

Categorical variables (see Table 14) describe demographic characteristics and contextual aspects of travel.

Table 14 - Categorical variables included in the dataset

Variable	Description	Categories / Values
SESSO	Gender of the individual	1 = Male; 2 = Female
ETA10	Age group	0–14; 15–24; 25–34; 35–44; 45–54; 55–64; 65–74; 75+
STACIV4	Marital status	1 = Single; 2 = Married; 3 = Separated/Divorced; 4 = Widowed
PNASC	Country of birth	1 = Italy; 2 = Other country
ISTR4	Education level	Aggregated education categories
DEST_IE	Trip destination	1 = Within Italy; 2 = International
ORGALL	Accommodation organization	1 = Direct booking; 2 = Agency/tour operator; 3 = No booking; 4 = Don't know / No answer
ORGTRA	Transport organization	1 = Direct booking; 2 = Agency/tour operator; 3 = No booking; 4 = Don't know / No answer

These variables were defined using aggregated categories (for example age groups or education levels) to limit sparsity problems and facilitate the generation of synthetic data that preserve the joint distribution of the original dataset.

Overall, the resulting dataset provides a balanced structure combining demographic, temporal, and travel-related information. This structure makes it suitable for evaluating different synthetic data generation methods while preserving the main statistical characteristics of the original survey data.

3.6.2 Purpose of the Synthetic Data

Synthetic data are artificially generated datasets designed to reproduce the statistical structure, and relationships present in real data without containing information about real individuals. By preserving key statistical properties such as distributions, correlations, and multivariate patterns, synthetic data can support data analysis while reducing the risk of disclosing sensitive information. Within the classification framework introduced in section “Purpose of synthetic dataset”, this use case is aligned with category (e): “perfect synthetic data / digital twin for safe microdata sharing.” The objective of this category is to generate synthetic datasets that closely reproduce the statistical characteristics of the original data, enabling most analytical tasks to produce results comparable to those obtained from real micro data.

In this experiment, the Public Use Files (PUF) from the Trips and Holidays survey are used as input data and treated as a proxy for confidential microdata. Synthetic datasets are generated using several data generation methods and subsequently evaluated in terms of both statistical utility and privacy risk. The goal of the use case is therefore to explore whether modern synthetic data generation techniques can support secure microdata sharing, preserve the main analytical properties of the original dataset while limit potential disclosure risks.

3.6.3 Methods for Synthetic Data Generation

We experimented with various approaches to generate synthetic data, including traditional Machine Learning techniques such as Random Forest and XGBoost, a statistical resampling algorithm like Synthetic Minority Over-sampling Technique (SMOTE) and generative models like VAE, CTGAN and DP-CTGAN.

Random Forest and XGBoost

Random Forest (Breiman, 2001) and XGBoost (eXtreme Gradient Boosting) (Chen et al., 2016) are both tree-based ensemble machine-learning methods, typically used to perform classification and regression tasks. In this work, we used them to generate a synthetic dataset that preserves the multivariate relationships present in the original data (Reiter, 2005). The process operates column-by-column and rests on the idea that each feature can be modelled as a function of all the remaining features.

Since standard Random Forest and XGBoost implementations operate natively on numerical inputs, all categorical features are first transformed using target encoding (or equivalent numeric encoding). This step allows the models to ingest both continuous and categorical predictors within a unified numerical feature space.

The folloqing is a description of the process of Random Forest, which will also apply to XGBoost. For each column in the dataset, the algorithm temporarily designates that column as the target variable and treats all other columns as input features. Depending on the nature of the target, if it is categorical, the system trains a *RandomForestClassifier*; if it is numeric, the system trains a *RandomForestRegressor*. In both cases, the model learns how the selected column depends on the remaining predictors by fitting an ensemble of decision trees to the conditional distribution:

$$X_j \mid \{X_1, \ldots, X_{\{j-1\}}, X_{\{j+1\}}, \ldots, X_p\}$$

This approach is conceptually aligned with methods used in multiple imputation, synthetic microdata generation, and non-parametric conditional modelling. Following training the Random Forest model for a designated column, predictions are made for each row in the dataset. These predictions serve as the synthetic version of that feature. This procedure replaces the original column with a fully synthetic counterpart that is statistically consistent with the relationships learned by the Random Forest model. The algorithm repeats this process for every column in the dataset. By cycling through all features, the method generates a complete synthetic dataset. This process ensures that each column is reconstructed through predictions made by a dedicated random forest model, while preserving the multivariate dependencies between variables, capturing nonlinear interactions through decision-tree partitioning, and generating a final synthetic dataset that contains non-copied values. This approach produces a synthetic dataset that reflects complex relationships, including nonlinearities and interactions, while mitigating the risk of directly reproducing records from the original data.

SMOTE (Synthetic Minority Over-sampling Technique)

SMOTE is a statistical resampling algorithm used to handle class imbalance in supervised machine-learning datasets. It was introduced by (Chawla et al., 2002) and has become one of the most widely used techniques for generating synthetic training samples. Class imbalance occurs when one class (the minority class) is under-represented than the majority class. This often leads to models that are biased toward the majority class and perform poorly on minority-class predictions. SMOTE creates

artificial data points for minority classes that are not duplicated but new, synthetic observations generated by interpolating between a real sample and one of its nearest minority-class neighbors. k-Nearest Neighbors (kNN) is the mechanism SMOTE uses to figure out which samples to interpolate between. kNN ensures that the synthetic data is generated within the same neighborhood, preserving the feature relationships of the minority region.

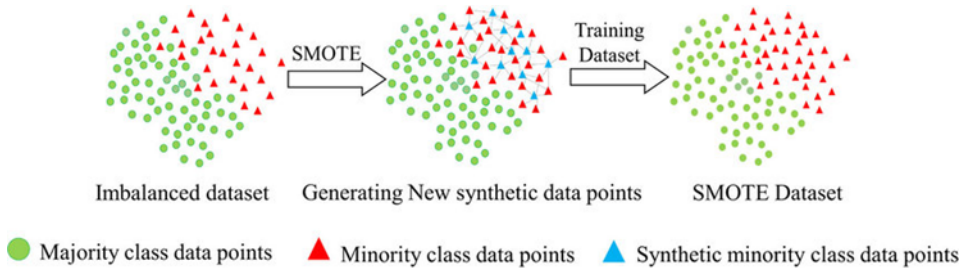


Figure 26: Illustration of the SMOTE process for generating synthetic data points.

This method prevents simple duplication of existing records, but it assumes the target is categorical, and all features are numeric. SMOTE-NC (SMOTE-Nominal Continuous) is a refinement of SMOTE intended for use with datasets that contain both numerical and categorical features. In this approach, numerical variables are interpolated in the same way as in SMOTE, while categorical variables are assigned the most frequent value (mode) among neighboring samples. This allows the generation of fully synthetic data while preserving the significance of categorical attributes.

VAEs (Variational Autoencoders)

VAEs (Kingman et al., 2014) and their tabular-specific evolution, TVAEs, are powerful generative AI tools for creating synthetic data (Wu et al., 2023) that mimics the statistical patterns of real datasets while protecting sensitive information. They combine principles from neural networks and probabilistic graphical models to learn a compressed latent representation of complex data distributions for reconstructing original data. Although VAEs were originally developed for image generation, they have become increasingly popular for tabular synthetic data, especially in domains such as finance, healthcare, cybersecurity, and customer analytics. A tabular VAE follows the same probabilistic framework but adapts the architecture to the characteristics of structured data.

A VAE consists of two jointly trained neural networks. The Encoder (*inference model*) transforms input data *X* into latent variable *Z*. Unlike standard autoencoders that map input to a single fixed point, VAEs map input to a probability distribution, usually defined by a mean and variance. The Decoder (*generative model*) reconstructs the original data X by mapping a sampled latent Z back to the data space. For tabular data, the decoder typically includes multiple output heads, one per variable type: numeric and categorical.

A VAE is trained by maximizing the Evidence Lower Bound (ELBO), which provides a tractable approximation to the otherwise intractable marginal log-likelihood of the data. ELBO consists of a reconstruction term and a Kullback-Leibler (KL) divergence term. The reconstruction term pushes the decoder to accurately recreate the input data from latent representations, while the KL divergence encourages the encoder's approximate posterior distribution to remain close to a chosen prior, typically a standard normal distribution. This combination ensures that the model learns meaningful latent structure rather than memorizing the training samples.

During training, the encoder produces the mean and variance of the approximate posterior distribution for each input. To keep the sampling operation differentiable, the VAE applies to the reparameterization trick, expressing the latent variable as a deterministic function of the encoder outputs plus random noise drawn from a normal distribution. This enables optimization through gradient-based methods. The decoder then reconstructs the data from the latent sample, and the resulting reconstruction error and KL penalty jointly define the ELBO objective that the model optimizes.

Because this process shapes a smooth, continuous latent space, new synthetic data can be generated by sampling latent points from the prior distribution and decoding them. When the decoder is designed with appropriate likelihood functions for different variable types - such as Gaussian outputs for numerical features and *softmax* distributions for categorical features - the VAE can generate high-fidelity synthetic tabular data that reflects the statistical properties and dependencies present in the original dataset.

CTGANs (Conditional Tabular GANs)

Conditional Generative Adversarial Networks for Tabular data (CTGAN) is a deep-learning–based generative model (Xu et al. 2019) designed to learn and reproduce the joint distribution of rows in tabular datasets, which typically combine continuous and categorical variables with complex statistical properties. Standard GAN architectures developed for image or text data often fail to model tabular distributions due to non-Gaussian, multi-modal continuous columns and severely imbalanced categorical columns

The CTGAN architecture incorporates a conditional generative adversarial framework where both *generator* and *discriminator* receive conditioning information indicating the value or category being modeled. This design directly addresses the imbalance inherent in categorical features by forcing the generator to learn how to produce realistic samples even for underrepresented categories. In addition, CTGAN applies a mode-specific normalization procedure to continuous columns, using techniques such as kernel density estimation to better represent multi-modal distributions rather than assuming unimodal Gaussian behavior. This allows the model to learn high-fidelity continuous distributions that are typically poorly modeled by traditional generative approaches.

A further innovation is the “training-by-sampling” strategy, which rebalances the frequency of categories during training and ensures that the discriminator is exposed to a representative distribution of categorical values. This improves the stability of adversarial training and prevents mode collapse, which is a frequent issue when modeling sparsely represented categories.

DP-CTGANs (Differential Privacy CTGANs)

Differential Privacy CTGAN (DP-CTGAN) is a privacy-enhanced extension of CTGAN, a conditional generative adversarial model designed for synthesizing realistic tabular data. DP-CTGAN builds on this foundation by incorporating differential privacy into the training process, ensuring that the model learns global patterns without exposing or memorizing any individual record from the original dataset. To achieve this, DP-CTGAN applies mechanisms such as gradient clipping and noise injection during adversarial training, limiting the influence that any single datapoint can have on the learned generator and discriminator. This modification preserves CTGAN’s ability to learn complex distributions while

providing formal privacy guarantees that protect against membership inference and reconstruction attacks. Once trained, the differentially private generator can produce synthetic tabular rows that reflect the statistical relationships and structure of the source data, but without revealing information attributable to specific individuals.

DP-CTGAN is therefore well suited for sensitive domains - such as finance, healthcare, and telecommunications - where synthetic data must protect against membership inference, attribute inference, or reconstruction attacks. DP-CTGAN must balance privacy and data quality: stronger differential-privacy guarantees require injecting more noise into the training process, which can reduce the realism and fidelity of the generated synthetic data.

3.6.4 Methods for Utility Measurement

The utility of the generated synthetic datasets is evaluated by assessing how well they reproduce the statistical characteristics of the original data while remaining suitable for analytical tasks. Evaluating synthetic data utility is a key step in assessing whether generated data can support meaningful analysis while preserving the main statistical properties of the original dataset (El Emam et al., 2020; van der Schaar et al., 2021). To obtain a comprehensive evaluation, different complementary approaches are used, focusing on descriptive statistics, distributional properties, and predictive performance.

The first group of methods evaluates statistical similarity between the original and synthetic datasets by comparing basic descriptive indicators, such as column-wise means and standard deviations. These comparisons provide an initial indication of whether the synthetic data correctly reproduces the central tendencies and variability observed in the original data. The analysis is performed using the Table Evaluator framework in Python, which facilitates automated comparisons of descriptive statistics and visual diagnostics (Patki et al., 2016; SDV Documentation, 2024).

A second group of methods focuses on distribution-based diagnostics. In particular, the empirical distributions of the variables in the real and synthetic datasets are compared using cumulative distribution functions, which allow a visual inspection of potential differences in the shape of the distributions. In addition, correlation heatmaps are used to compare the correlation matrices of the real and synthetic datasets. Differences between these matrices provide an indication of how well synthetic data preserve the multivariate relationships between variables.

Finally, a machine learning–based evaluation is conducted to assess whether the synthetic data preserves the predictive relationships present in the original dataset. In this test, a Random Forest classifier is trained on the synthetic dataset and evaluated on a test set derived from the real data. Predictive performance is measured using standard indicators such as accuracy, F1-score, and recall. If the synthetic data adequately captures the statistical structure of the original dataset, the predictive performance obtained using synthetic data should approach the performance achieved when training the model on the real data (Xu et al., 2019; van der Schaar et al., 2021).

3.6.5 Methods for Privacy Assessment

At this stage of the experiment, the assessment of privacy risks associated with the synthetic datasets generated is still at a preliminary and exploratory stage. The objective is not to provide a definitive

evaluation of disclosure risk, but rather to introduce a first set of basic indicators that can support an initial understanding of potential privacy issues related to the generation of synthetic data.

The current implementation is largely inspired by the privacy evaluation framework provided within the Synthetic Data Vault (SDV) ecosystem, in particular the privacy metrics described in the *sdmetrics* documentation for synthetic data evaluation (Patki et al., 2016; Xu et al., 2019; SDV Documentation, 2024). These metrics provide standardized procedures to assess the similarity between real and synthetic datasets and to identify situations in which synthetic records may be excessively close to real observations.

In this first implementation, the analysis focuses mainly on distance-based indicators and simple attack-based evaluations.

The first group of measures is based on the concept of distance to the closest record (DCR), which evaluates how close each synthetic record is to its nearest counterpart in the original dataset used for training. The underlying idea is that synthetic data should reproduce the statistical structure of the original data while avoiding the reproduction of individual observations. Extremely small distances between real and synthetic records may indicate a potential risk of information leakage, as discussed in the literature on synthetic data disclosure risk assessment (El Emam et al., 2020).

In addition to distance-based measures, we also implemented a simplified membership inference attack scenario. In this setting, a classifier is trained to determine whether a given record was part of the training dataset used to generate synthetic data. The goal of this experiment is not to simulate a fully realistic attack, but rather to obtain an initial indication of whether the synthetic data could potentially reveal information about the presence of specific records in the original dataset. Membership inference attacks have been widely studied in the context of machine learning privacy risks (Shokri et al., 2017).

Finally, the analysis also includes the computation of a Disclosure Protection indicator, implemented through the SDV *sdmetrics* framework. This metric evaluates the extent to which the synthetic dataset may allow an attacker, who knows a subset of variables, to infer information about other attributes present in the original data.

Given the exploratory nature of the current work, the results are mainly intended to provide preliminary insights and to support the design of more comprehensive privacy risk assessment methodologies in future developments of the project. The preliminary results of this initial evaluation are presented in the following section.

Further work will therefore focus on expanding the set of privacy metrics considered and on analyzing more systematically the trade-off between privacy protection and statistical utility.

3.6.6 Preliminary Results

3.6.6.1 Utility Metrics

Evaluation of the Similarity to the Initial Dataset

Synthetic dataset quality was assessed using the TableEvaluator module from SDV, a tool that facilitates visual comparison with the original data. This module provides several visualizations that compare synthetic data against the original training data. Each plot highlights a different aspect of data similarity.

Mean and Standard Deviations Comparisons

Evaluation of the means and standard deviations of each column in the real and synthetic datasets. This plot compares the means and standard deviations of each column in the real and synthetic datasets. It offers a quick evaluation of whether the synthetic data maintains the overall statistical properties of the original dataset. In figure 27, we can see that all the methods show collinearity between the standard deviations and means across all features. The Differential Privacy–CTGAN method is the only approach that exhibits a lower correlation between feature means and standard deviations. This effect results from the constraints introduced by differential privacy, which impose a trade-off between model utility and privacy protection according to the specified privacy budget.

Cumulative Sums

The cumulative sum plot illustrates the running total of values for each column. This visualization allows the identification of potential systematic differences in trends or cumulative distributions between the synthetic and the real datasets. Using the *TableEvaluator* tool, it is possible to compute the cumulative sums (CumSums) for each feature within the considered dataset. A cumulative sum (commonly referred to as cumsum) represents the running total of a sequence of values. This visualization shows the accumulated values for each column and helps identify potential systematic differences between the real and synthetic datasets. In other words, the cumulative sum of a distribution is obtained by progressively adding each new value to the total of all preceding values.

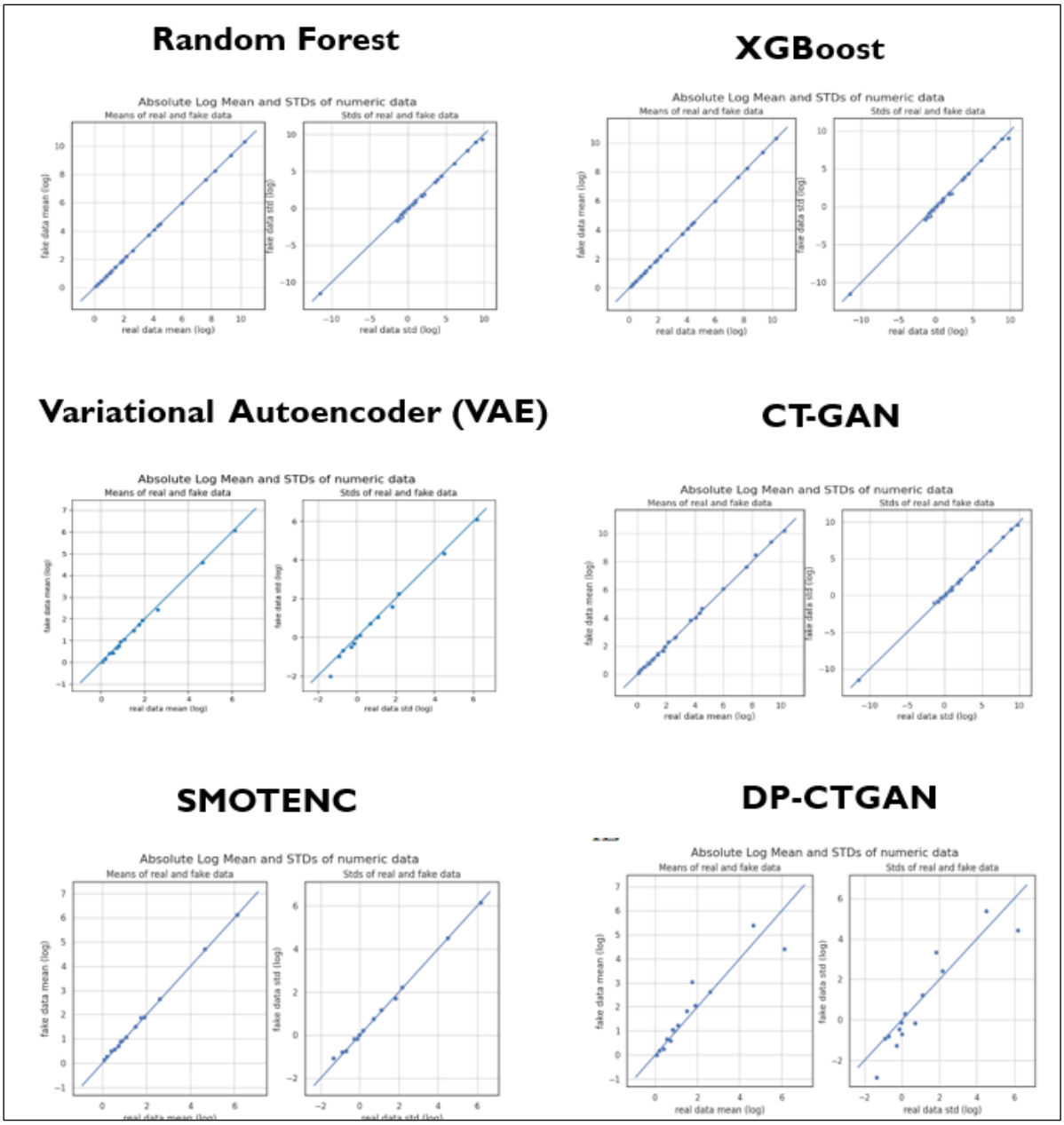


Figure 27: Means and standard deviation of original and synthetic distributions.

If you see figure 27, in the original distribution, the values represent the observed data and are not cumulative for sums of previous values. The accumulation occurs only in the cumulative sum representation, where each value is obtained by adding it to the total of all preceding values. From the cumulative distributions, it can be observed that less effective methods, such as Random Forest and XGBoost, tend to perform worse on categorical variables. In these cases, the models often fail to accurately reproduce the categories, that is, the domain values associated with those variables. Moreover, even for continuous variables, these methods often produce distributions that do not overlap with those of the real data. In contrast, more effective methods show an almost perfect overlap between the cumulative distributions of real and synthetic datasets. Therefore, the distribution-based approach provides an immediate visual indication of whether the synthetic data accurately reproduces the statistical properties of the original dataset.

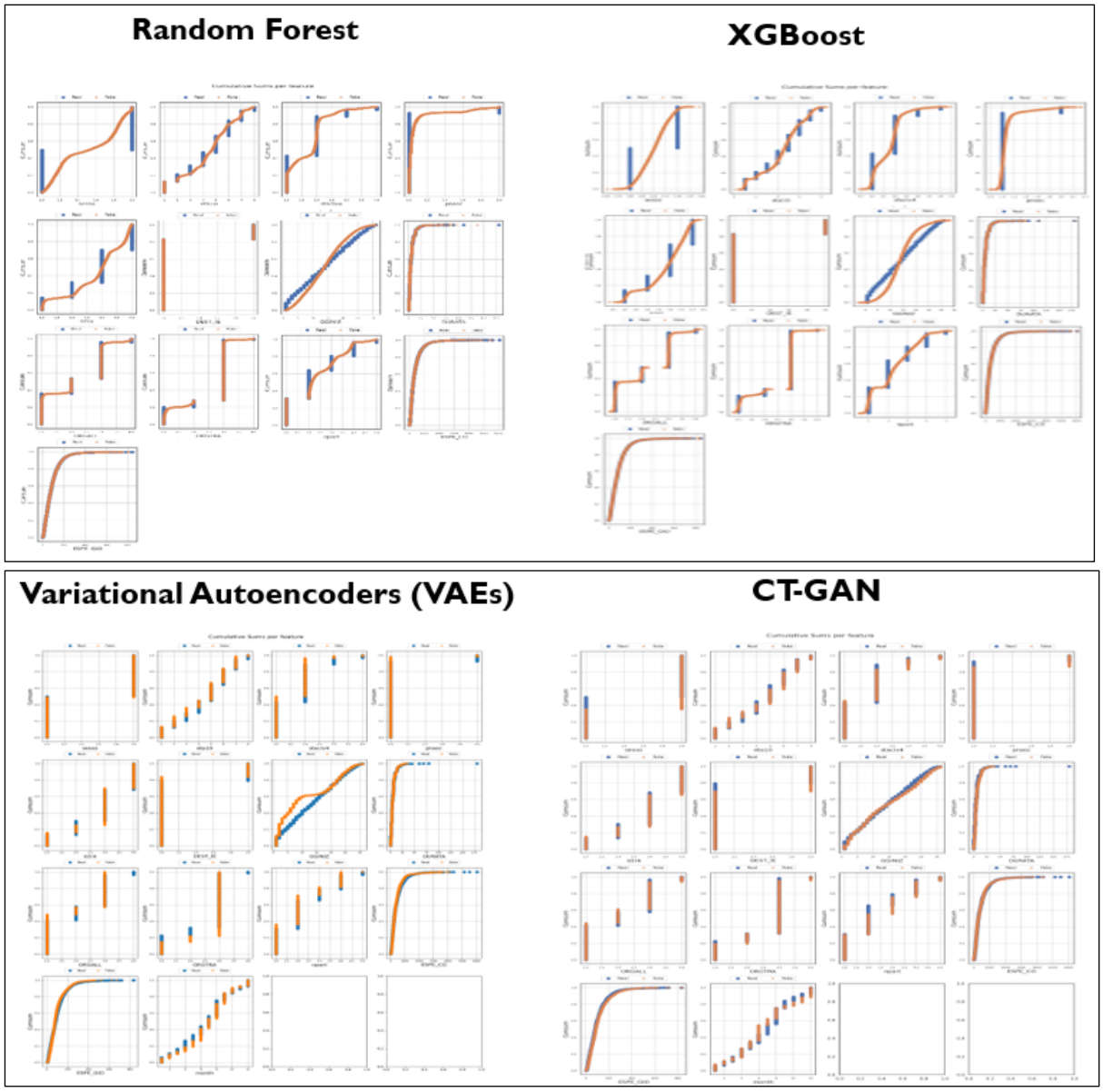


Figure 28: Cumulative Sums of original and synthetic distributions.

Correlation Heatmaps

This heatmap shows the differences in correlations between all pairs of variables in the real and synthetic datasets. It helps evaluate whether synthetic data accurately preserves the relationships and interactions among variables. A Correlation Heatmap is another visualization used to highlight differences in relationships between variables. To create it, we first compute the correlation matrices for both the original and the synthetic datasets and then calculate the difference between them. Larger differences suggest that the synthetic data generation method is less accurate in preserving the original relationships between variables. In figure 29, the correlation maps are depicted.

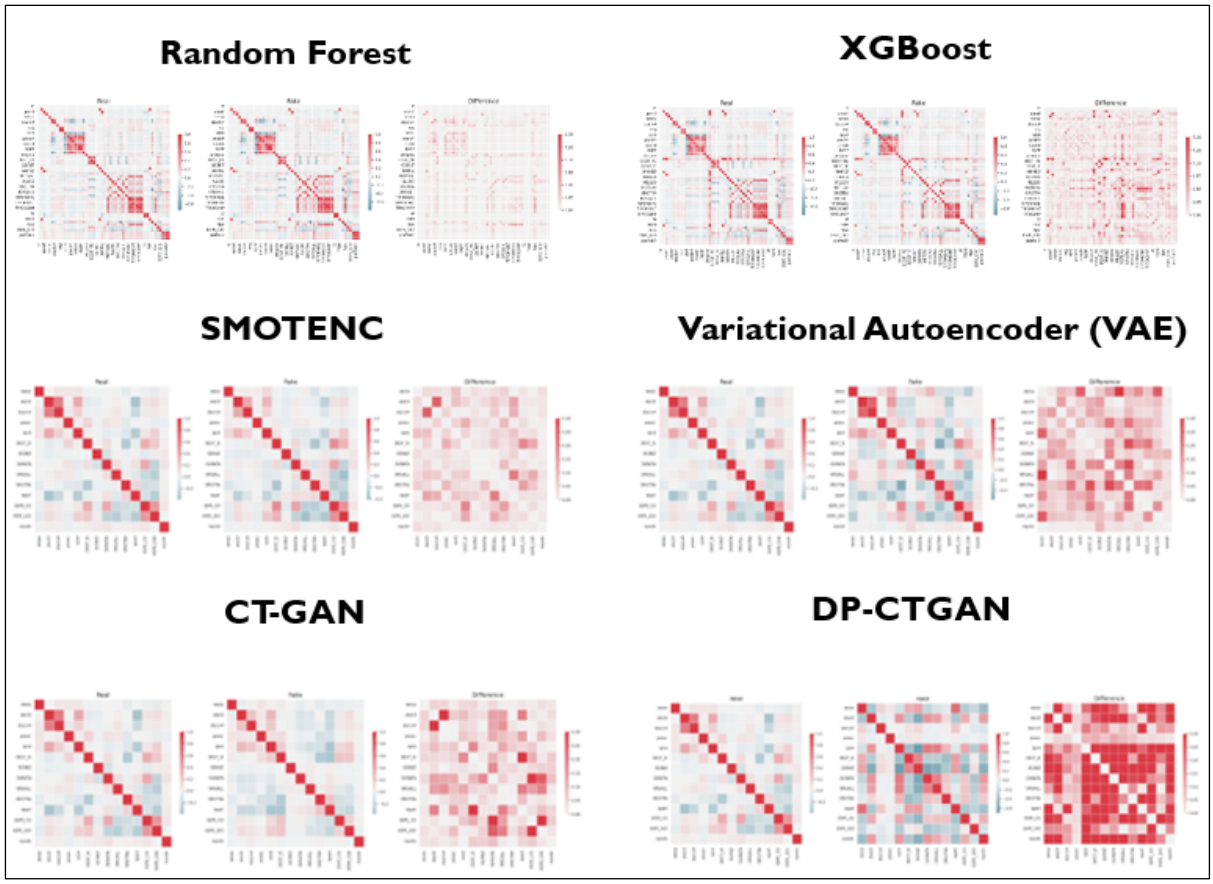


Figure 29: Correlation Heatmaps of original and synthetic distributions.

Machine Learning Test

In the Machine Learning test, we evaluated model performance (accuracy, F1-score, and recall) on a 20% test set using a Random Forest classifier. The classifier is trained twice: first on the original dataset and then on the synthetic dataset, with "*motvac*" used as the target variable. The same classifier configuration is used for all methods to ensure a fair comparison. In theory, if the synthetic data generator successfully reproduces the statistical characteristics of the original dataset, the model's performance when trained on synthetic data should be close to the performance obtained when training on the real data. If there is a large gap between the two, it indicates that the synthetic data does not adequately capture the underlying patterns of the original dataset.

In table 15, we can observe that (except for SMOTENC which tends to replicate the data almost exactly) the most effective methods appear to be the deep learning–based approaches, such as CTGAN and VAE. These methods achieve high accuracy while still producing results that are not identical to those obtained with the original dataset, which is expected. During the synthetic data generation process, some loss of the original data statistical properties is unavoidable, so a small difference in performance is normal.

Table 15 – Machine learning test metrics

Model	Accuracy	F1-Score	Recall
Original Data	0.964516848	0.964516848	0.964516848
RF	0.660685592	0.660685592	0.660685592
XGB	4.46349e-05	4.46349e-05	4.46349e-05
SMOTENC	0.992813783	0.992813783	0.992813783
DP-CTGAN	4.46349e-05	4.46349e-05	4.46349e-05
VAE	0.892385288	0.892385288	0.892385288
CTGAN	0.896134619	0.896134619	0.896134619

3.6.6.2 Privacy Metrics

Disclosure Protection

To evaluate the privacy level of the generated synthetic data, we calculated the Disclosure Protection metric. This measure assesses the risk of re-identifying individuals in the synthetic dataset when certain attributes are already known. The principle behind disclosure protection is the following: if an attacker knows the values of a set of attributes K for an individual i in the original dataset, they may try to infer the values of the sensitive attributes S from the synthetic dataset. A DP score close to 1 indicates a high level of protection against re-identification, whereas a score close to 0 signals a greater risk of sensitive information being disclosed. In table 16, we can see the results of this test. The Disclosure Protection (DP) results show that most synthetic data generation methods provide moderate privacy protection, with DP values slightly above 0.5. SMOTE (0.514) and Random Forest (0.505) are close to the baseline, indicating that an attacker's ability to infer sensitive attributes is roughly comparable to random guessing. Deep learning methods perform slightly better: VAE (0.537) and CTGAN (0.564) achieve higher DP scores, suggesting stronger protection against attribute disclosure while still preserving some data structure. The XGB model obtains a DP value of 1, indicating perfect protection, although this may also suggest that the generated data does not preserve meaningful relationships with the original dataset. This is in excellent agreement with utility metrics. DP-CTGAN also obtains a DP value of 1, indicating a very high level of protection against attribute disclosure. This result is consistent with the use of differential privacy during the training process, which introduces calibrated noise and limits the influence of individual records on the generated data. Consequently, the synthetic dataset produced by DP-CTGAN makes it extremely difficult for an attacker to infer sensitive attributes from known information. Overall, CTGAN and VAE appear to offer the best balance between privacy protection and data utility.

Table 16 – Disclosure Protection results

Method	DP Score
SMOTE	0.5137
RF	0.5048
XGB	1.0000
VAE	0.5367

CTGAN	0.5642
DP-CTGAN	1.0000

DCR (Distance to Closest Record) Baseline Protection

The DCR Baseline metric evaluates the similarity between a synthetic sample and two randomly selected subsets of the real dataset, providing a reference distance for comparison. It is used to determine whether the synthetic data is excessively similar to the real data, which may indicate potential overfitting or memorization by the generative model. A baseline value of approximately 0.5 is expected when comparing random subsamples. If the DCR metric is significantly higher than this value, it may indicate that the model has memorized portions of the training data. In table 17, the DCR Baseline Protection results show that most synthetic data generation methods produce values below the expected baseline of 0.5, indicating that the synthetic samples are not excessively similar to the original dataset and therefore do not exhibit strong memorization effects. In particular, Random Forest (RF) (0.247), VAE (0.261), and SMOTE (0.279) show the lowest scores, suggesting a lower risk of reproducing real records. CTGAN (0.338) remains below the baseline as well, indicating a reasonable balance between data realism and privacy protection. In contrast, XGB (0.723) significantly exceeds the expected baseline, suggesting that the generated data may be too similar to the original dataset and could potentially indicate memorization of training data. DP-CTGAN (0.976) shows an even higher DCR value, indicating that the samples generated are very distant from the original dataset. This behavior is consistent with the use of differential privacy during training, which introduces calibrated noise to limit the influence of individual records. Therefore, the model strongly reduces the risk of memorizing training samples, although such a high distance may also suggest a potential reduction in the similarity between synthetic and real data distributions in agreement with utility metrics too.

Table 17 – DCR Baseline Protection results

Method	**DCR Score**
SMOTE	0.2791
RF	0.2466
XGB	0.7231
VAE	0.2614
CTGAN	0.3377
DP-CTGAN	0.9763

Membership Inference Attacks (MIA)

Membership inference attacks (MIA) are a common approach for assessing privacy leakage in generative models. Their objective is to determine whether a specific record was included in the training dataset used to generate synthetic data. If an adversary can reliably distinguish members (records used during training) from non-members (records from validation or test sets), this suggests that the synthetic data, particularly in certain contexts (i.e., healthcare datasets) may expose sensitive information. MIA risk scores close to 0.5 indicate little to no privacy leakage, whereas scores closer to 1 suggest a higher probability that an attacker could identify records that were part of the training dataset from the synthetic data. In table 18, we first have kNN-based Membership Inference showing

risk scores between 0.242 and 0.302 across all methods. These values are well below 0.5, indicating low membership inference risk. In other words, the attacker has a limited ability to determine whether a record belongs to the training dataset. Additionally, the ROC AUC score is approximately 0.495 for all models, which is essentially equivalent to random guessing (0.5). This suggests that the synthetic datasets do not reveal meaningful signals that allow the attacker to distinguish training records from unseen ones. Among the models: CTGAN shows the lowest risk score (0.242), suggesting slightly stronger privacy protection. VAE shows the highest risk (0.267), though the difference remains small. DP-CTGAN shows a risk score of 0.302, which is slightly higher than the other models but still well below the critical threshold of 0.5, indicating that membership inference remains difficult even when differential privacy is applied. Overall, the kNN-based attack indicates strong privacy protection for all methods. The Membership Classification attack reports risk scores around 0.80 for all models. Basically, the MIA evaluation suggests that none of the synthetic data generation methods leak significant membership information. The ROC AUC values close to 0.495 confirm that the attacker cannot distinguish training samples from non-training samples better than random guessing. Among the tested models: CTGAN appears to provide the strongest membership privacy protection. VAE shows slightly higher risk but remains within a safe range. SMOTE, RF, and XGB exhibit similar privacy behavior. DP-CTGAN shows comparable behavior, with a classification risk of approximately 0.801, indicating that the application of differential privacy does not introduce additional membership leakage. Overall, the results indicate that all generation methods provide robust protection against membership inference attacks, with very limited privacy leakage.

Table 18 – Membership Inference Attacks results

Method	kNN Risk	ROC AUC	Classification Risk
SMOTE	0.2555	0.4953	0.8000
RF	0.2588	0.4953	0.8000
XGB	0.2629	0.4953	0.8000
VAE	0.2668	0.4953	0.7995
CTGAN	0.2418	0.4953	0.8023
DP-CTGAN	0.3022	0.4953	0.8010

3.7 Italy - Use Case 2 - Toy World: a Benchmark to Assess Utility/Privacy Trade-off)

3.7.1 Statistical Domain and Input Data

In the first of the two Italian PoCs, we designed a “toy world” as a fully controlled experimental environment to support the systematic evaluation of the privacy–utility trade-off in synthetic data generation.

An implicit but central assumption underlying the interest for synthetic data is the separability between *population-level information* and *unit-level information*. Population-level information refers to the structural patterns and statistical dependencies that characterize the overall distribution of the data (e.g. correlations, conditional patterns, etc.), while unit-level information concerns record-specific details that may allow the identification, reconstruction or memorization of individual observations. The core objective of synthetic data generation is to preserve population-level information while

preventing the leakage of unit-level information, which implies that the two components can be separated, at least in principle.

In real-world datasets it is generally impossible to know exactly what population-level information is present in the data, which complicates the task of assessing whether a model has learned structural patterns or has encoded information about specific individuals. For this reason, the *toy world* framework does not rely on operational survey data but instead represents an artificial statistical domain constructed specifically for methodological experimentation, where the full data-generating process is explicitly known and controlled, and where population-level information is fully known and clearly separable from unit-level information.

The overall experimental workflow is illustrated in figure 30:

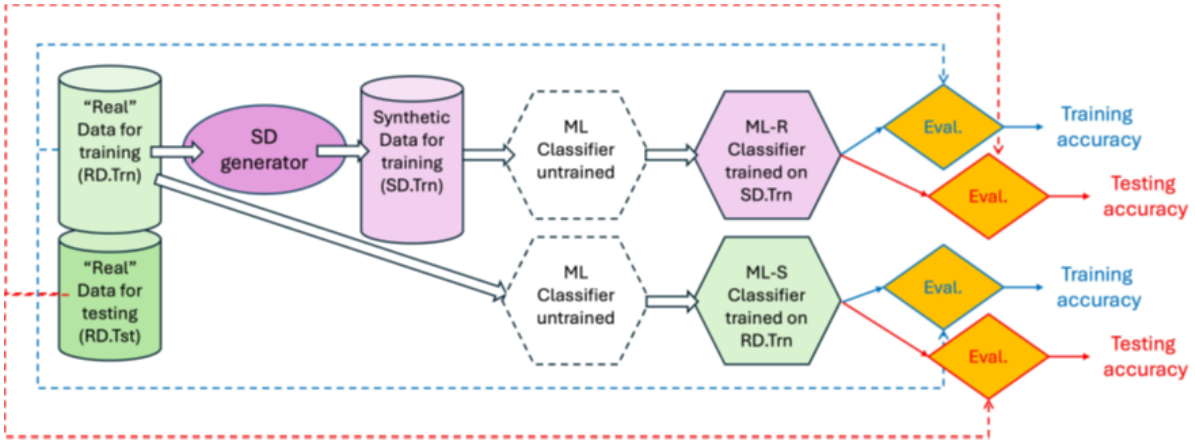


Figure 30: Toy world workflow.

The framework consists of two parallel branches departing from the same source dataset (RD – "real data"). The source dataset is randomly split into a training set (RD.Trn) and a test set (RD.Tst), typically using an 80:20 ratio. In the benchmark branch, classifiers are trained directly on RD.Trn and evaluated on both RD.Trn (*training accuracy*) and RD.Tst (*test accuracy*). In the synthetic-data branch, RD.Trn is first used to train a synthetic data generator, which produces a synthetic training dataset (SD.Trn). The same classifier architecture is then trained on SD.Trn and, again, evaluated on RD.Trn and RD.Tst. Because the classifier models and evaluation procedures are identical across the two branches, differences in performance can be attributed exclusively to the use of synthetic rather than source data.

In both branches, the same classifier architectures and evaluation metrics are used. This controlled comparison makes it possible to quantify the utility loss (if any) introduced by synthetic data generation, while also monitoring potential signs of memorization, i.e., retaining unit-level information beyond population-level information.

The source dataset RD consists of individual records (baseline configuration: *n=10,000*). Each record includes a set of discrete variables as may be encountered in the context of health data collection. We refer to a reference *toy scenario* where the source data refers to the patient records from a hospital. The design mimics a realistic microdata environment while remaining fully artificial.

The variables can be conceptually grouped into three categories:

- standard demographic variables of the kind that are typically recorded at triage (e.g. age, gender, civil status, residence, occupation). These variables are assumed to be potentially available to potential attackers;
- auxiliary variables, collected by the hospital but not easily observable by actors external to the hospital. Their availability to potential attackers depends on the assumed attack scenario;
- a sensitive variable (or attribute) representing the diagnosis received by the hospital. This is the variable targeted by potential attackers.

In *Toy World,* all variables are generated independently according to predefined probability distributions. This ensures full control over marginal structures and the absence of statistical correlations between the input variables. The sensitive attribute (diagnosis) is then assigned based on a function of three selected variables – in the baseline configuration one demographic and two auxiliary variables. More precisely, the diagnosis depends deterministically on the three variables for a fraction *1-r* of records, while for the remaining fraction of *r* records the diagnosis is generated randomly between all possible values. The parameter r ∈ [0,1] determines the proportion of records for which the diagnosis value is completely independent on, and cannot be predicted from, the other variables.

The diagnosis variable is defined over an alphabet of 4 values. The parameters of the toy world are tuned to produce diagnosis value with approximately equal frequencies. That means a blind guess would be correct with probability approximately equal to 0.25, while a perfect classifier with full knowledge of the input variables would reach an accuracy of approximately *1-r.*

When *r* = 0, the relationship between predictors and target is fully deterministic; for increasing values of *r*, part of the structural signal is progressively replaced by random assignments. This parameter therefore regulates the global "predictability" of the target attribute, thus enabling systematic testing of synthetic data generators and classifiers in different scenarios.

3.7.2 Purpose of the Synthetic Data

The objective of the Italian Toy World use case is to investigate the feasibility of generating synthetic datasets that closely replicate the statistical properties of the original microdata while preventing the disclosure of unit-level information. In the classification framework introduced in Section Purpose of Synthetic Dataset, this use case is therefore aligned with the category of "perfect synthetic data / digital twin for safe microdata sharing."

The ambition of this category is to produce synthetic datasets that can act as a statistical "digital twin" of the original data, preserving the multivariate relationships between variables so that most statistical analytical tasks produce results comparable to those obtained from the real data. In this perspective, the objective of the experimentation is to evaluate, within a controlled environment, the feasibility of synthetic datasets as a mechanism for useful and safe microdata sharing.

The Toy World framework provides a controlled experimental setting where the full data-generating process is known. This makes it possible to distinguish clearly between population-level information,

corresponding to structural statistical relationships present in the data, and unit-level information, corresponding to details related to specific observations that may lead to disclosure risks.

3.7.3 Methods for Synthetic Data Generation

The Toy World synthetic datasets were generated using three modelling approaches commonly adopted in the synthetic data literature for tabular data. The objective was to evaluate the capacity of different modelling strategies to reproduce the statistical structure of the reference dataset under controlled experimental conditions.

The methods considered are:

- Conditional Tabular Generative Adversarial Networks (CTGAN);
- Tabular Variational Autoencoders (TVAE);
- an incremental synthetic data generation approach based on XGBoost inspired by the popular SimPop tool.

CTGAN extends the traditional Generative Adversarial Network (GAN) framework (Goodfellow et al., 2014) by incorporating a conditional generation mechanism specifically designed for tabular data (Xu et al., 2019). Moreover, the CTGAN framework incorporates a conditional generation mechanism that improves the handling of categorical variables and imbalanced distributions. This feature is particularly relevant in official statistics contexts, where discrete variables and structured multivariate dependencies are common. The model learns an implicit representation of the joint distribution of the training data and generates synthetic records by sampling from a learned latent representation.

TVAE is based on the variational autoencoder architecture introduced by Kingma and Welling (2014), adapted for tabular data generation within the SDV framework.

The model maps the original data into a latent space and reconstructs synthetic observations by sampling from learned probability distributions. Compared to GAN-based approaches, VAE-type models generally offer greater training stability, although their ability to reproduce strong multivariate dependencies may differ.

In addition to the deep learning–based models, a synthetic data generation procedure based on incremental XGBoost was implemented replicating the sequential synthesis strategy used in the popular SimPop tool.

The main idea is to approximate the joint distribution of the variables through a sequence of conditional models, generating variables one at a time according to a predefined ordering. Formally, the joint distribution of a dataset with p variables can be factorized as:

$$P(X_1, \ldots, X_p) \approx P(X_1) \prod_{j=2}^{p} P(X_j | X_1, \ldots, X_{j-1})$$

This sequential synthesis strategy is widely used in the synthetic data literature, where the joint distribution is approximated through a sequence of conditional models (Reiter, 2005; Drechsler, 2011; Kowarik et.al,2017).

Under this framework, each variable is generated conditional on the values already synthesized for the preceding variables. This approach allows the method to capture the main statistical dependencies present in the original data while maintaining a flexible modelling structure.

The procedure starts by generating the first variable through bootstrap sampling with replacement from the original dataset, preserving its marginal distribution. For each subsequent variable, an XGBoost model is trained to learn the relationship between the target variable and the variables that have already been synthesized.

Depending on the nature of the variable, different models are used:

- for categorical variables, an XGBoost classifier is trained, and synthetic values are generated by sampling from the predicted class probabilities;
- for numerical variables, an XGBoost regressor is trained and synthetic values are obtained from the model predictions, optionally constrained within the observed range of the original data.

This sequential modelling strategy allows the approach to capture non-linear relationships and interactions between variables while remaining computationally efficient and relatively simple to implement.

Implementation frameworks

The CTGAN and TVAE models were implemented using the Synthetic Data Vault (SDV), an open-source framework developed by the Massachusetts Institute of Technology (MIT) Data to AI Lab for the generation of synthetic tabular data. SDV provides stable and reproducible implementations of deep generative models specifically designed for tabular datasets, including CTGAN and TVAE (Patki et al., 2016; Xu et al., 2019). The framework includes built-in mechanisms for encoding categorical variables and handling mixed data types, ensuring a consistent preprocessing pipeline and facilitating the training of generative models on complex tabular structures.

The use of a consolidated framework such as SDV ensured:

- reproducibility of experiments,
- consistent preprocessing of categorical and discrete variables;
- standardized training procedures across models.

The generative models were trained on the Toy World training dataset (RD.Trn), and synthetic datasets of identical size and structure (SD.Trn) were produced. Hyperparameters were selected to ensure convergence and comparability across configurations. All experiments were conducted under a controlled random seeds to reduce stochastic variability.

No manual post-processing or structural adjustments were applied to the synthetic outputs, allowing a direct evaluation of each method's intrinsic ability to preserve multivariate structure. The comparison between models was carried out under identical experimental conditions, enabling a consistent assessment of their impact on downstream utility and disclosure risk.

In XGBoost implementation, the method was developed in Python using the XGBoost scikit-learn application programming interface (API). At each step, the variables already synthesized are used as

predictors and transformed into a numerical design matrix through one-hot encoding when required. A separate model is then trained for each target variable and used to generate corresponding synthetic values.

The synthetic dataset produced using this approach was subsequently included in the utility evaluation experiments, together with the datasets generated using CTGAN and TVAE.

Our Python implementation of the incremental approach is based on XGBoost, which mirrors the R implementation adopted by SimPop (Kowarik et al., 2017).

3.7.4 Methods for Utility Measurement

In this use case, classification models are used as a methodological tool to quantitatively measure the preservation of relevant statistical structures in synthetic datasets compared to the original data. The underlying methodological assumption is that, if a synthetic dataset correctly reproduces the structural relationships between the variables present in the source data, a supervised classifier trained on it should achieve test accuracy levels comparable to those obtained when trained on real data. Conversely, a degradation in performance indicates a loss of statistically relevant information.

This approach is well aligned with the task-based utility evaluation approach, in line with the methodological objectives of WP13.

To ensure methodological robustness and avoid conclusions that are specific to a single architecture, we have tested multiple classification models, including traditional statistical models, machine learning, and deep learning–based methods.

This distinction reflects different learning paradigms. Traditional statistical models assume an explicit functional relationship between explanatory variables and the target variable and estimate a finite set of interpretable parameters. In contrast, machine learning and deep learning methods adopt a purely data-driven approach, where the structure of relationships is learned from the data without requiring a predefined functional specification.

Within this framework, one traditional statistical model with relatively low complexity was included (Logistic Regression), together with ensemble methods based on decision trees (Random Forest and XGBoost) and a deep learning model characterized by high model flexibility (Multi-Layer Perceptron with four hidden layers). This choice of different methodological families allows us to cover a broad spectrum of model complexity and supports the robustness of empirical findings.

Classification Models

To systematically evaluate the preservation of structural relationships in the synthetic datasets, four classification models representing different learning paradigms were implemented.

The Multinomial Logistic Regression (LR) was adopted as the traditional statistical benchmark. The model was configured with L2 regularization and optimized using the limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) algorithm, with a convergence tolerance of 1×10^{-4} and a maximum of 2000 iterations to ensure stable convergence. Automatic class weighting was applied to

compensate for potential class imbalance. Categorical variables were transformed using one-hot encoding, and a fixed random seed was used to guarantee reproducibility. Multinomial logistic regression provides an interpretable reference model and enables verification of the preservation of linear or additive relationships in the transformed feature space (Bishop, 2006).

Among the Machine Learning–based methods, the Random Forest (RF) was implemented by generating 200 fully grown decision trees. Each tree was trained on a bootstrap sample of the dataset, and at each split a random subset of features of size equal to the square root of the total number of available variables was considered. The splitting criterion was the minimization of Gini impurity. Automatic class weighting was also applied, and out-of-bag estimation was enabled to provide an internal performance assessment (Breiman, 2001; Cutler et al., 2012). Random Forest is well suited for capturing nonlinear relationships and interactions among variables, making it appropriate for assessing the preservation of complex decision patterns.

The XGBoost (XGB) model represents a gradient boosting method based on decision trees trained sequentially, where each new tree is constructed to correct the residual errors of the previous ones, incorporating explicit regularization mechanisms to control model complexity. In the present framework, XGBoost was configured with 100 boosting rounds, a maximum tree depth of 6, and a learning rate of 0.1 to control the incremental contribution of each tree to the final model. The objective was specified for multi-class classification using the *multi-softmax* function with four classes. Labels were converted to a zero-based numerical format (0, 1, 2, 3) in accordance with the library requirements. XGBoost also provides feature importance measures based on the contribution to the reduction of the loss function, enabling additional analysis of structural consistency (Chen & Guestrin, 2016).

Finally, the Multi-Layer Perceptron (MLP) represents the deep learning component of the framework. It was implemented as a feedforward neural network with multiple hidden layers of progressively decreasing size, each using a Leaky Rectified Linear Unit (LeakyReLU) activation function, while the output layer employs a softmax activation for multi-class classification. The network was trained using the Adam optimizer with a learning rate of 0.001 and a batch size of 32 observations per iteration. The categorical cross-entropy loss function was used, and early stopping based on validation performance was applied to mitigate overfitting (Goodfellow et al., 2016; LeCun et al., 2015). Owing to its high expressive capacity, the MLP enables a stringent evaluation of the synthetic dataset under complex modeling scenarios.

Overall, the joint use of a traditional statistical model, tree-based ensemble methods, and a deep neural network allows a multi-level assessment of the informational quality of synthetic datasets and reduces the risk that conclusions are driven by the characteristics of a single architecture.

Evaluation Strategy

Model performance is evaluated by computing accuracy on test sets. The interpretation of results is based on two main elements that are the absolute performance level and the comparison between models trained on real and synthetic data. Alignment between performance on real and synthetic data indicates good preservation of statistical structure; conversely, a systematic reduction in accuracy signals the loss of relevant information in the synthetic dataset. The empirical results of the classification experiments are reported below in paragraph “Preliminary results”.

3.7.5 Methods for Privacy Assessment

In addition to evaluating the statistical utility of the synthetic datasets, the experiment also includes a privacy risk assessment aimed at verifying whether the released synthetic data may allow an attacker to infer sensitive information from the original training dataset.

The evaluation follows an attack-based approach, where privacy risks are assessed by simulating plausible attack scenarios in which an adversary attempts to infer unknown attributes of real individuals using the released synthetic dataset in combination with prior information (e.g., knowledge of some variables). Within this framework, three different attack models were implemented, each representing a different type of adversarial knowledge and inference strategy.

Similar attack-based privacy evaluations are widely adopted in the synthetic data literature and are closely related to record linkage and attribute inference attacks proposed in disclosure risk analysis (Domingo-Ferrer & Torra, 2003; El Emam et al., 2020).

We focus on Attribute Inference Attacks where the attacker's goal is to discover the value of the secret attribute (diagnosis) given (i) prior information about the remaining variables and (ii) full access to the synthetic data.

3.7.5.1 Attack model 0: Training–Test accuracy attack

The first scenario considered is referred to as "Attack Model 0". In this setting, the attacker trains a classifier using the same architecture and learning scheme adopted in the classification task used to assess utility. To assess whether the attack is successful, we compare the training accuracy versus the test accuracy: when the former is substantially higher than the latter, we conclude there has been memorization (for the classification model) and encoding (for the synthetic data) of unit-level information from the training data on top of population-level information. For the sake of completeness, we analyze the training vs. test accuracy gaps both in (i) models trained on real data and (ii) models trained on synthetic data. The results of Attack 0 are reported in the subsequent section, "Preliminary Results".

3.7.5.2 Attack model 1: similarity-based attribute inference

The first attack scenario assumes that the attacker has access to:

- the synthetic dataset (SD) released by the data holder;
- a set of victims' records originating from the real training dataset, for which all variables are known except the attribute of interest that the attacker aims to infer.

The goal of the attacker is therefore to predict the value of the unknown attribute for each target record by exploiting similarities between the target records and the synthetic data.

The attack is based on the following procedure:

- for each target record $\underline{x}$, the attacker computes a dissimilarity metric between the vector of known attributes of the target record and the corresponding attributes of all records y_i (i=1,…,N) contained in the synthetic dataset of N records;

- after calculating the dissimilarity values between the target and each synthetic record, the attacker selects the n<<N synthetic records with the smallest dissimilarity from the target record;
- the unknown attribute is then inferred using a majority vote among the *n* closest synthetic records weighted by their relative dissimilarity (Gower weighted).

We have tested different choices for the dissimilarity metric $L(x,y_i)$. To define a global distance function between two records, we must (a) define a dissimilarity function for each variable, and (b) define a composition rule to compound them.

The Gower distance, a dissimilarity measure, is particularly suitable for datasets containing mixed variables (numeric, categorical, and binary) (Gower, 1971). For numeric variables, the distance between two observations is calculated as the absolute difference between their values normalized by the variable's observed range, as a normalized Manhattan distance. For categorical variables, the distance is 0 if the two values are identical and 1 otherwise. For binary variables, a similar matching rule is applied depending on whether the attribute is symmetric or asymmetric. The overall Gower distance between two records is then obtained as the average of the variable-specific distances, weighted if necessary to account for the relevance of each variable.

The intuition behind this attack is that if the synthetic generator reproduces records that are very similar to individuals contained in the training dataset, the synthetic dataset may contain records that are sufficiently close to real individuals to allow the attacker to infer their hidden attribute.

Distance-based nearest-neighbor inference has been widely studied as a disclosure risk assessment strategy for microdata and synthetic data releases (Domingo-Ferrer & Torra, 2003; El Emam et al., 2020; Yao et.al, 2025).

Baseline and evaluation criterion

In order to distinguish between (i) successful inference due to memorization (encoding) of unit-level information from the training records and (ii) predictions driven by general population-level patterns, the attack results are compared with a baseline scenario.

The baseline is constructed using a set of target test records drawn from the real test dataset, which were not used during the training of the synthetic data generator. In other words, we are again comparing some sort of training accuracy versus testing (baseline) accuracy, where “accuracy” is now referred to the attack model.

The attack is considered successful if the rate of correct attribute predictions for records in the training dataset is significantly higher than the accuracy obtained for the test (baseline) records (BR). A substantial difference between the two results would suggest that the synthetic dataset contains information specific to the training data, indicating a potential privacy leakage.

The attack was implemented in Python as part of the privacy evaluation pipeline developed for the Toy World experiment.

For each target record, the procedure consists of:

- removing the attribute to be inferred from the victim record;
- computing distances between the target record and all records in the synthetic dataset for the remaining variables;
- identifying the n nearest neighbours in the synthetic dataset (we tested for n = 3, 5, 7, 9 and 11);
- predicting the unknown attribute using majority voting among the neighbors;
- compare prediction with true value.

The implementation allows testing of different dissimilarity metrics and neighborhood sizes, enabling systematic exploration of multiple attack configurations. The objective is to verify whether any configuration of the attack leads to significantly higher inference accuracy for training records compared with the baseline. The results of Attack 1 are discussed in the following section, "Preliminary Results".

3.7.5.3 Attack Model 2: Attack with Class-Specific Synthetic Features and Linear Discriminant Analysis (LDA) Enhancement

The goal of the second attack is to discover the unknown attribute (diagnosis class) for each victim record (VR) by exploiting the information contained in the synthetic dataset (SD) and the labeled training records (TR).

The attacker has access to:

- synthetic dataset (SD): 10,000 synthetic samples with labels, released by the data holder;
- training records (TR): A sample of 8,000 records (p=80%) from the real training dataset for which all variables are known, including the attribute of interest (the diagnosis class). These records serve as the attacker's background knowledge to build the attack model;
- victim records (VR): The remaining 2,000 records (1-p=20%) from the real training dataset. For these victims, the attacker knows all variables except the attribute of interest that they aim to infer;
- baseline records (BR): 2,000 real records not included in the training dataset (test records).

The intuition behind this attack is that, if synthetic data SD are encoding unit-level information from the training data, including TR, then a model trained on SD and TR could attempt to learn (discover) the hidden relationship between the two dataset, and then use such information to infer the secret attribute of VR. If the attack is successful, the rate of correct predictions for VR should be significantly higher than for BR.

The attack aims to capture this relationship through a rich set of meta-features that describe a record behavior in the space of synthetic and real data.

The attack procedure consists of three main phases:

1) Hybrid Meta-Feature Extraction

For each record r in TR and VR, a meta-feature vector $\phi(r)$ is constructed that encodes its behavior with respect to SD and TR. The use of meta-features to characterize the properties of datasets and

instances is widely studied in the literature. Reif (2012) proposed a comprehensive set of meta-features to represent datasets in a transfer space, while Leite and Brazdil (2021) demonstrated how similarity between datasets can be effectively measured in this space to enable transfer learning.

The meta-features include:

- *Class-specific synthetic distances (20 features):* by splitting SD into four distinct classes based on the value of the attribute of interest ("diagnosis"), several distance statistics (minimum, mean, median, standard deviation) are calculated for each record *r* from the synthetic examples of each class only. The closest synthetic class, the second closest, and the ratio between the two distances are also recorded. These features capture which synthetic class a record most resembles;
- *Real space behavior (9 features):* minimum, mean, median distances, and standard deviation from TR; proportion of each class among the nearest neighbors in TR and the majority class among real neighbors (*tr_majority_class*);
- *LDA features (13 features):* this is the key experimental component. Linear Discriminant Analysis (LDA) is applied to the entire synthetic dataset SD to project the data into a 3-dimensional space that maximizes separation between the four classes. As argued in the literature, LDA remains a powerful feature extractor even when its statistical assumptions are not fully met (Hastie et al., 2009; Duda et al., 2001). Recent empirical studies confirm its robustness as a dimensionality reduction method (Kampouris et al., 2026; Liu & Guo, 2024). For each record r, the following are calculated: the LDA coordinates, distances and cosine similarities to each class centroid, the closest class in LDA space (*lda_closest_class*), and a confidence score based on the distance ratio;
- *Domain shift metrics (8 features):* ratios between distances to synthetic classes and the average distance to TR; binary indicators signaling whether a record is closer to a specific synthetic class than to real data.

*Additional features (17 features): d*escriptive statistics of TR (size, class distribution) and statistical moments of randomly sampled features.

2) Meta-Classifier Training

The attacker uses the public TR dataset to train an XGBoost model. For each record in TR, there are meta-feature vector $\phi(r)$ and the true class label (the attribute of interest). The model learns to map meta-features to the target class. Hyperparameters: *n_estimators =150, max_depth=4, learning_rate=0.01, subsample=0.5, colsample_bytree=0.5, reg_alpha=0.5, reg_lambda=2.0*.

3) Attack Execution

For each victim record *v* in VR, the attacker extracts $\phi(v)$ and feeds it to the trained meta-classifier, which produces a prediction $\hat{y}_v$ for the unknown attribute. The true labels for VR are never accessed during training or attack; they are held for final evaluation only.

Baseline and Evaluation Criterion

To distinguish between successful inference due to memorization of training records and predictions driven by general population patterns, the attack results are compared to a baseline scenario.

The baseline records (BR) are drawn from the real test dataset that were not used during the training of the synthetic data generator. This dataset consists of 2,000 real samples generated using the same underlying process of the real data.

The attack is considered successful if the rate of correct attribute predictions for records belonging to the training dataset (VR) is significantly higher than the accuracy obtained for the baseline records (BR). A substantial difference between the two results would suggest that the synthetic dataset contains information specific to the training data, enabling the attacker to infer hidden attributes more accurately for records that were part of the original training set, indicating a potential privacy leakage through memorization.

Conversely, if the attack achieves similar accuracy on both VR and BR, this indicates that the model is simply learning to classify real data effectively based on general patterns, rather than exploiting training-set-specific information. In this case, the vulnerability is to generally attribute inference from any real data, not to memorization-based disclosure.

The attack was implemented in Python as part of the privacy evaluation pipeline developed for the Toy World experiment.

For each victim record in VR and each baseline record in BR, the procedure consists of:

- extracting the hybrid meta-feature vector $\phi(r)$ based on class-specific synthetic distances, real space behavior, and LDA projections (where applicable);
- feeding the meta-feature vector to the pre-trained XGBoost classifier;
- obtaining a prediction for the unknown attribute;
- comparing the prediction with the true attribute value.

The implementation allows testing of two experimental configurations:

- NO-LDA (Baseline configuration): uses only the class-specific meta-features (47 features) without LDA enhancement;
- LDA (Enhanced configuration): augments the meta-feature space with LDA-derived projections (67 features).

The objective is to verify whether either configuration of the attack leads to significantly higher inference accuracy for training records (VR) compared with the baseline (BR), thereby satisfying the success criterion for an attribute inference attack that exploits potential memorization in synthetic data. The results of Attack 2 are detailed in the next section.

3.7.6 Preliminary Results

This section presents the preliminary results obtained within WP13, focusing on both utility and privacy risk dimensions. In particular, the analysis investigates: (i) the predictive performance of

classification models trained on real and synthetic datasets, (ii) the potential presence of memorization effects as a proxy for disclosure risk (Attack 0), and (iii) the robustness of the data against attribute inference attacks under different experimental settings (Attack 1 and Attack 2).

The results are evaluated under varying levels of data complexity, including scenarios with deterministic patterns and scenarios with partial randomization, to assess how structural noise affects both utility and privacy risk. Overall, this section provides an initial empirical assessment of the trade-off between data utility and privacy preservation.

Utility Evaluation: Classification Performance Across Real and Synthetic Data

This section reports the empirical results obtained by applying the four classification models to the different datasets considered in the experimentation. Performance is expressed in terms of accuracy, computed on the test set (evaluation of real data not used for training).

The results enable comparison between real and synthetic datasets, considering both deterministic data (for which r=0) and randomized data where the diagnosis is purely random for 40% of the records (r=0.4).

Real data without randomization yields accuracy values close to one for all models, confirming the coherence between the data-generating process and the classification task. Introducing 40% randomization leads to a substantial reduction in performance, as expected.

Regarding synthetic datasets, marked differences emerge across generation methods. In particular, the synthetic dataset generated with the XGBoost approach achieves performance levels close to real data in the non-randomized scenario, whereas CTGAN and TVAE exhibit a more pronounced utility loss. Under the 40% randomization scenario, all synthetic configurations show a further decline in performance, consistent with the expectation that the presence of randomized records outside the core pattern “confounds” the generators and the classificators, makes it more difficult to delineate the core pattern itself. In other words, the increased structural complexity of the data drives the test accuracy relative to the ideal maximum reference.

Table 19 – Classification Results for MLP, Logistic Regression, Random Forest, and Extreme Gradient Boosting

Model training data	Evaluation on real data	MLP (4 hydden layer)	Logistic Regression	Random Forest	XGBoost
Real	Test	1	0,997	0,992	0,993
Real – 40% random	Test	0,546	0,543	0,535	0,546
Synthetic (CTGAN)	Test	0,853	0,854	0,834	0,833
Synthetic (TVAE)	Test	0,657	0,684	0,654	0,66
Synthetic (XGBOOST)	Test	0,964	0,982	0,947	0,947
Synthetic (CTGAN) – 40% random	Test	0,521	0,513	0,481	0,514
Synthetic (TVAE) – 40% random	Test	0,387	0,378	0,365	0,389
Synthetic (XGBOOST) – 40% random	Test	0,558	0,541	0,51	0,55

3.7.6.1 Attack 0 – Memorization and Overfitting Analysis

During the experimentation, it was observed, for instance, that the Random Forest (RF) classifier trained on real data reaches training accuracy values considerably higher than the test accuracy, close to one, indicating a strong ability to memorize the training dataset. In fact, with the chosen set of parameter values, the capacity of the resulting RF model is sufficiently large to memorize completely the entire dataset.

However, when the same RF model is trained on synthetic data, we found no evidence of unit-level data encoding. However, this came at the price of a certain loss of test accuracy, i.e., degradation of utility.

Table 20 – Comparison of Training and Test accuracy across classification models

Model training data	Evaluation on real data	MLP (4 hydden layer)	Logistic Regression	Random Forest	XGBoost
Real	Test	1	0,997	0,992	0,993
Synthetic (CTGAN)	Test	0,853	0,854	0,834	0,833
Synthetic (TVAE)	Test	0,657	0,684	0,654	0,66
Synthetic (XGBOOST)	Test	0,964	0,982	0,947	0,947
Real – 40% random	Test	0,546	0,543	0,535	0,546
Synthetic (TVAE) – 40% random	Test	0,387	0,378	0,365	0,389
Synthetic (XGBOOST) – 40% random	Test	0,558	0,541	0,51	0,55
Synthetic (CTGAN) – 40% random	Test	0,521	0,513	0,481	0,514
Real	Training	1	0,999	1	0,997
Synthetic (CTGAN)	Training	0,866	0,866	0,854	0,857
Synthetic (TVAE)	Training	0,652	0,683	0,647	0,652
Synthetic (XGBOOST)	Training	0,954	0,981	0,953	0,951
Real – 40% random	Training	0,573	0,568	0,995	0,595
Synthetic (CTGAN) – 40% random	Training	0,524	0,506	0,463	0,512
Synthetic (TVAE) – 40% random	Training	0,392	0,376	0,367	0,393
Synthetic (XGBOOST) – 40% random	Training	0,552	0,54	0,515	0,545

3.7.6.2 Attack 1 – Sensitivity to parameter k and model behavior

Figure 31 illustrates the relationship between the VR and BR performance metrics as a function of the parameter k. Overall, both curves exhibit a similar trend, with performance remaining relatively stable across a wide range of k values. No clear divergence between VR and BR emerges, suggesting that the model does not exploit specific structural properties of the target dataset to significantly improve predictive performance. This behavior indicates a limited sensitivity of the attack to the choice of k and provides no strong evidence of successful attribute inference under the tested configurations.

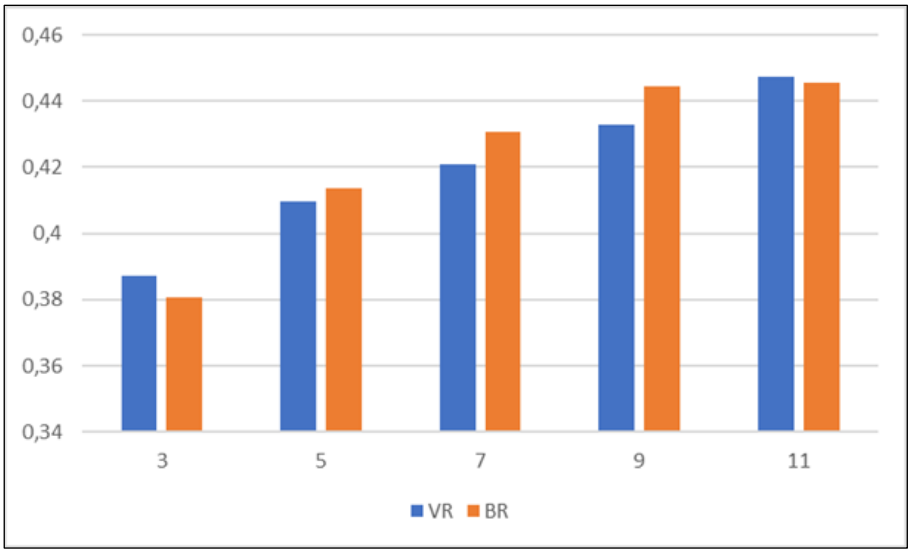


Figure 31: VR vs BR comparison over k parameter.

3.7.6.3 Attack 2 – Attribute Inference

This section extends the analysis of attribute inference risk by comparing different modeling configurations, including a baseline approach, a class-specific setting without dimensionality reduction (NO-LDA), and an enhanced configuration based on Linear Discriminant Analysis (LDA).

Table 21 – Performance comparison – all settings

Configuration	VR Accuracy	BR Accuracy (Baseline)
Previous Approach	50.0%	49.6%
NO-LDA (Class-specific)	52.8%	51.7%
LDA (Enhanced)	55.7%	53.9%

Table 21 reports the overall classification accuracy obtained under the three experimental configurations on the VR dataset and on the baseline dataset BR. The results show a progressive improvement in performance from the Previous Approach (50.0% VR; 49.6% M2) to the NO-LDA configuration (52.8% VR; 51.7% M2), with the LDA configuration achieving the highest accuracy (55.7% VR; 53.9% M2). However, the differences between VR and BR remain small across all configurations, indicating no substantial advantage in predictive performance on the target dataset.

An additional advantage of our experimental design with four distinct diagnosis classes is the ability to perform a granular, class-wise analysis of privacy vulnerabilities. By comparing the VR-BR accuracy gap not only overall but also for each class individually, we can identify whether certain classes are more susceptible to attribute inference attacks than others.

Table 22 – class-wise accuracy comparison

Class	LDA		NO-LDA	
	VR	BR	VR	BR
Class 1	49.9%	52.9%	46.7%	51.7%
Class 2	55.2%	53.2%	46.9%	51.5%
Class 3	56.2%	52.7%	57.2%	51.6%
Class 4	62.0%	56.8%	61.6%	54.3%

Table 22 reports the classification accuracy obtained for the four classes under the different experimental configurations (LDA VR, LDA BR, NO-LDA VR and NO-LDA BR). Overall, the results do not provide clear evidence of a successful attribute inference attack according to the adopted success criterion. Class 1 shows a negative difference in both configurations, meaning that the classifier performs better on the baseline dataset (BR) than on the target dataset (VR).

Positive differences between VR and BR are observed mainly for Classes 3 and 4, but their magnitude remains limited, ranging from 3.5 percentage points (Class 3 with LDA) to 7.3 percentage points (Class 4 without LDA). On average, the difference between VR and BR remains small, equal to 1.8 percentage points with LDA and 1.1 percentage points without LDA, indicating only marginal variations in predictive performance.

Finally, Class 4 consistently achieves the highest accuracy values across configurations (62.0% with LDA and 61.6% without LDA), suggesting that this class is generally easier for the classifier to identify.

3.8 Austria

3.8.1 Statistical Domain and Input Data

The aim of this research analysis is to synthesize the Austrian population as a fully synthetic dataset. The input data is the main sampling frame at Statistics Austria, the so-called **Rich Frame**, used for surveys on households or persons. The rich frame is a pseudonymized version of the central housing and population registers enriched with additional sociodemographic and regional variables. It contains sociodemographic variables on each registered individual as well as household specific variables. Available variables on an individual level include but are not limited to age, gender, highest completed educational, employment status and citizenship. Household size, state of residence, regional classifications, etc. … are available as household specific variables.

3.8.2 Purpose of Synthetic Data

The primary objective of this work is to generate a fully synthetic dataset that includes variables at both the individual and household levels. This requires the creation of synthetic households and their corresponding synthetic individuals. Although the input data cover the entire population of Austria, the analysis is restricted to individuals and households within the NUTS-2 region **AT32** (**Salzburg**). Focusing on this region substantially reduces computation time while still providing a meaningful and diverse test environment. **AT32** encompasses both densely populated urban areas and sparsely populated rural regions, and from a socioeconomic perspective it includes affluent districts within the city of Salzburg as well as more modest areas in the surrounding countryside. This heterogeneity makes the region well-suited for assessing the performance and robustness of the synthetic data generation process on a smaller yet representative subset of the Austrian population.

3.8.2.1 Individual level information

Regarding variables on an individual level the synthetic dataset should hold

- Age: variable containing the age if an individual;
- Sex: variable containing the sex of an individual (male/female);
- Education: highest educational attainment (5 Groups);
- Working status: working status of individual.

Personal yearly income dervied from tax recrods on employment and self-employment and transfers (pensions).

3.8.2.2 Household level information

Regarding variables on an individual level the synthetic dataset should hold:

- HouseholdID: household identifier to distinguish between households;
- HSize: number of individuals living in the same household (~ same HouseholdID);
- POL: political district (6 Groups);
- DEGURBA: Degree of Urbanization (rural areas, towns and suburbs, cities).

3.8.3 Methods for synthetic data generation

The R packages simPop (Templ et al. (2017)) and synthpop (Nowok, Raab, and Dibben (2016)) as well as the Python package Synthetic Data SDK (Krchova et al. (2025)) are used to generate a single synthetic version of the original dataset.

The R package simPop is designed for the simulation of complex synthetic populations based on survey and auxiliary data. The package supports tasks like simulating categorical and continuous variables, generating household structures, correcting age heaping, and calibrating synthetic data to match given population totals. The synthetization workflow follows a sequential conditional modelling strategy. In the analysis presented in this work, household structures were first generated using age and gender as the core categorical variables defining household composition. Subsequent variables were then simulated with the functions simContinuous() and simCategorical(), relying on an XGBoost-based model (specified via model = "xgboost"), see Chen et al. (2019), to capture complex relationships in the data.

The R package synthpop adopts a sequential synthesis strategy similar to that of simPop. In this approach, the first variable is generated by random sampling, while all subsequent variables are synthesized from conditional distributions fitted to the original data using either parametric models or, by default, Classification and Regression Trees (*CART*). This allows for flexible, non-parametric modelling of complex conditional relationships among variables. In contrast to simPop, the synthpop package does not support the generation of household or dwelling structures. Instead, its focus lies on creating synthetic microdata at the record level for statistical disclosure control. Despite this limitation, synthpop is one of the most widely used synthetic data packages on Comprehensive R Archive Network (CRAN), in part due to its user-friendly interface: an entire dataset can be synthesized automatically with a single call to syn(...). For this analysis, data were synthesized using synthpop with CART-based models and the package's default parameter settings.

The Python package Synthetic Data SDK provides a high-fidelity, privacy-preserving framework for generating synthetic data across a wide range of data types. Central to the SDK is the concept of synthetic data generators, which can be trained on tabular or language-based datasets and subsequently used to produce synthetic samples of arbitrary size. The package offers flexible modelling capabilities, including so called TabularARGN for structured data, differential privacy options, automated quality assurance, and conditional sampling controls. Like the simPop package the Synthetic Data SDK has the capabilities to synthesise individauls living in households or dwellings. Altough the Synthetic Data SDK offers parameters to finely tune the TabularARGN in this analysis the default parameter settings where used. Furhtermore we did not have a GPU available and needed to split the population by regional variables and synthesise each geographic subpart of *AT32*.

3.8.4 Methods for Utility Measurement

The quality assessment of the synthetic data consists of two main components. First, a series of distributional comparisons between the original input data and the synthetic data is conducted. These include checks of first-order distributions, which serve as a general sanity check of the overall fidelity of the synthetic dataset, as well as more detailed household-level distributions, which assess whether the synthetic data correctly capture the internal composition of households. The comparisons are performed both visually and quantitatively, using the Hellinger distance as a measure of distributional similarity. In addition to distributional validation, the internal consistency and integrity of each synthetic record are evaluated using a set of predefined editing rules. These rules range from simple variable-level constraints, such as:

- **Age must be a non-negative value and not exceed 110,** to more complex relational constraints, for example:
- **The oldest individual within a household must be at least 15 years old.**

The final quality metric is derived from the proportion of synthetic records that violate any of these rules, providing a clear indication of the structural reliability of the generated dataset.

3.8.5 Methods for Privacy Assessment

For the privacy assessment, we examined both identity disclosure risk and attribute disclosure risk. Because the dataset is fully synthetic, identity disclosure is generally considered the less critical threat. To quantify it, we apply the concept of replicated uniques, which captures the proportion of records whose identifying variable patterns remain unique in both the original and the synthetic data.

Attribute disclosure risk is assessed using the **Risk of Attribute Prediction-Induced Disclosure in Synthetic Microdata** (RAPID) (Templ, Thees, and Müller (2026)). This measure evaluates how well an adversary could infer sensitive attributes of real individuals by training predictive models on the released synthetic data. For continuous sensitive variables, RAPID calculates the share of records for which the model's predictions fall within a specified relative error tolerance. For categorical sensitive variables, RAPID uses a baseline-normalized confidence score that reflects how much more confident an attacker is in the predicted true class compared to what would be expected from overall class prevalence. An R package for *RAPID* can be found on github: https://github.com/qwertzlbry/RAPID.

3.8.6 Preliminary Results

Initial results indicate that distributions can be reproduced quite well by all methods. Comparison of selected distributions are shown in figures 32 - 35.

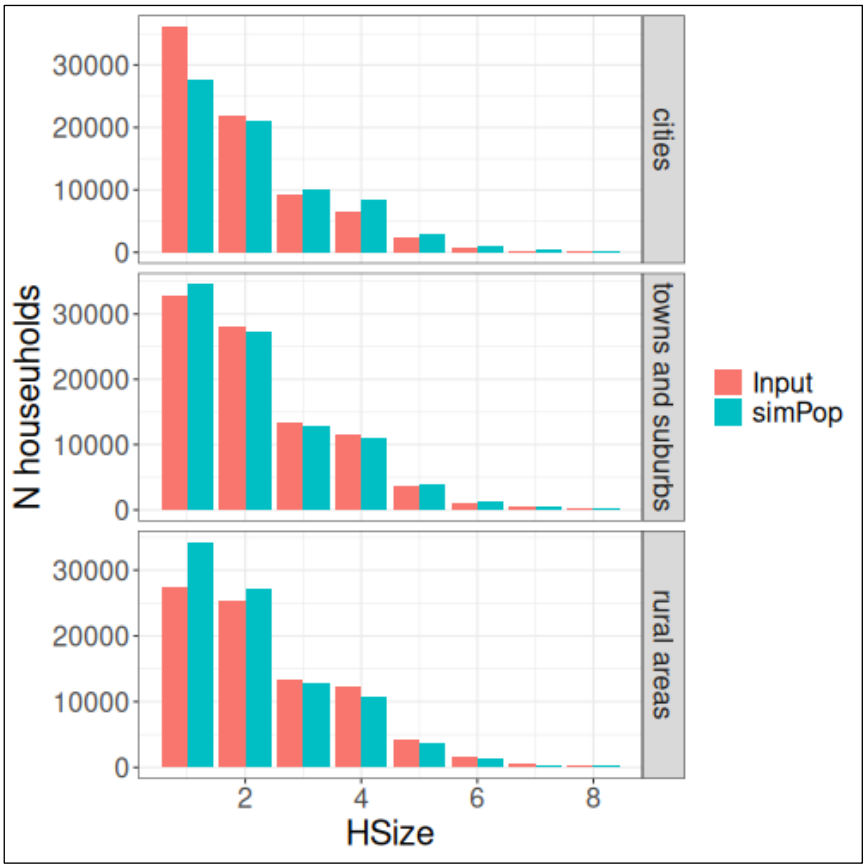


Figure 32: Number of households by household size and district.

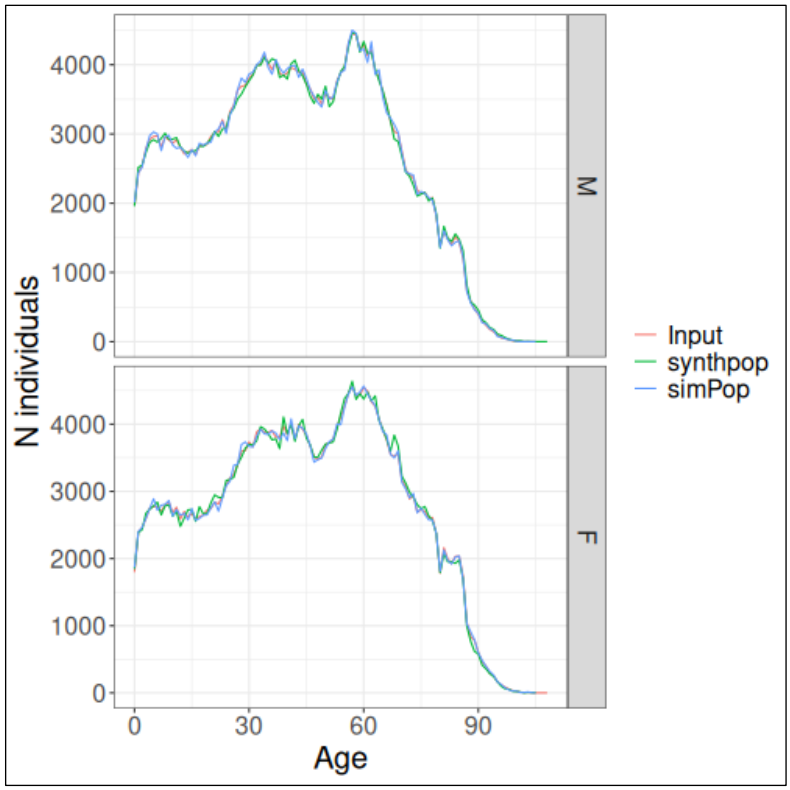


Figure 33: Number of individuals by age and sex.

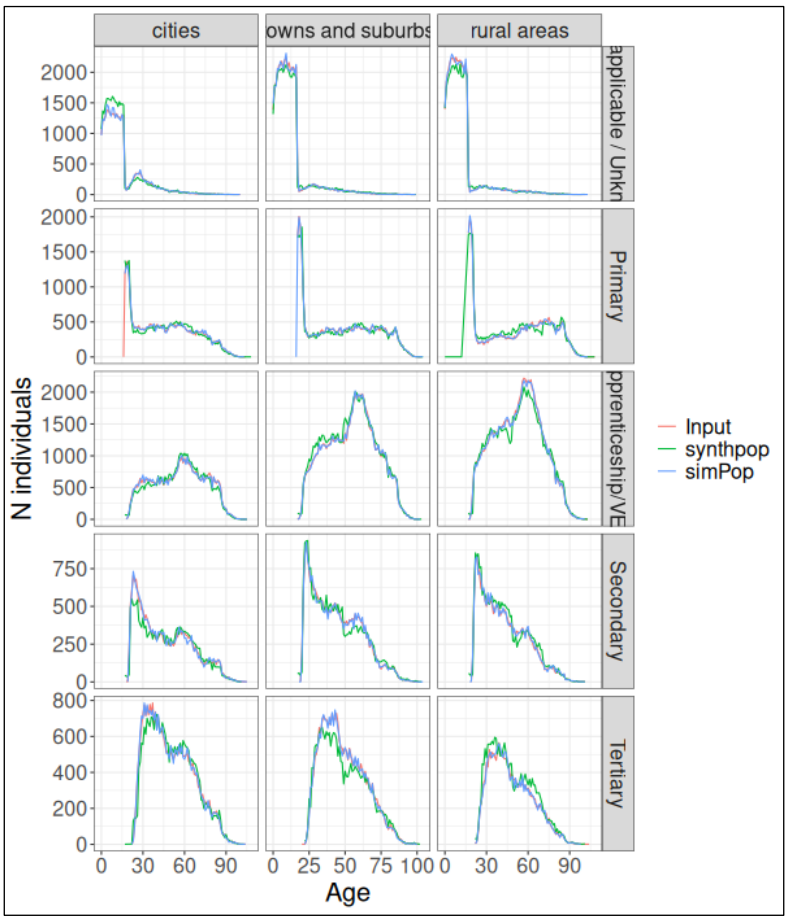


Figure 34: Number of individuals by age, highest educational attainment and degree of urbanization.

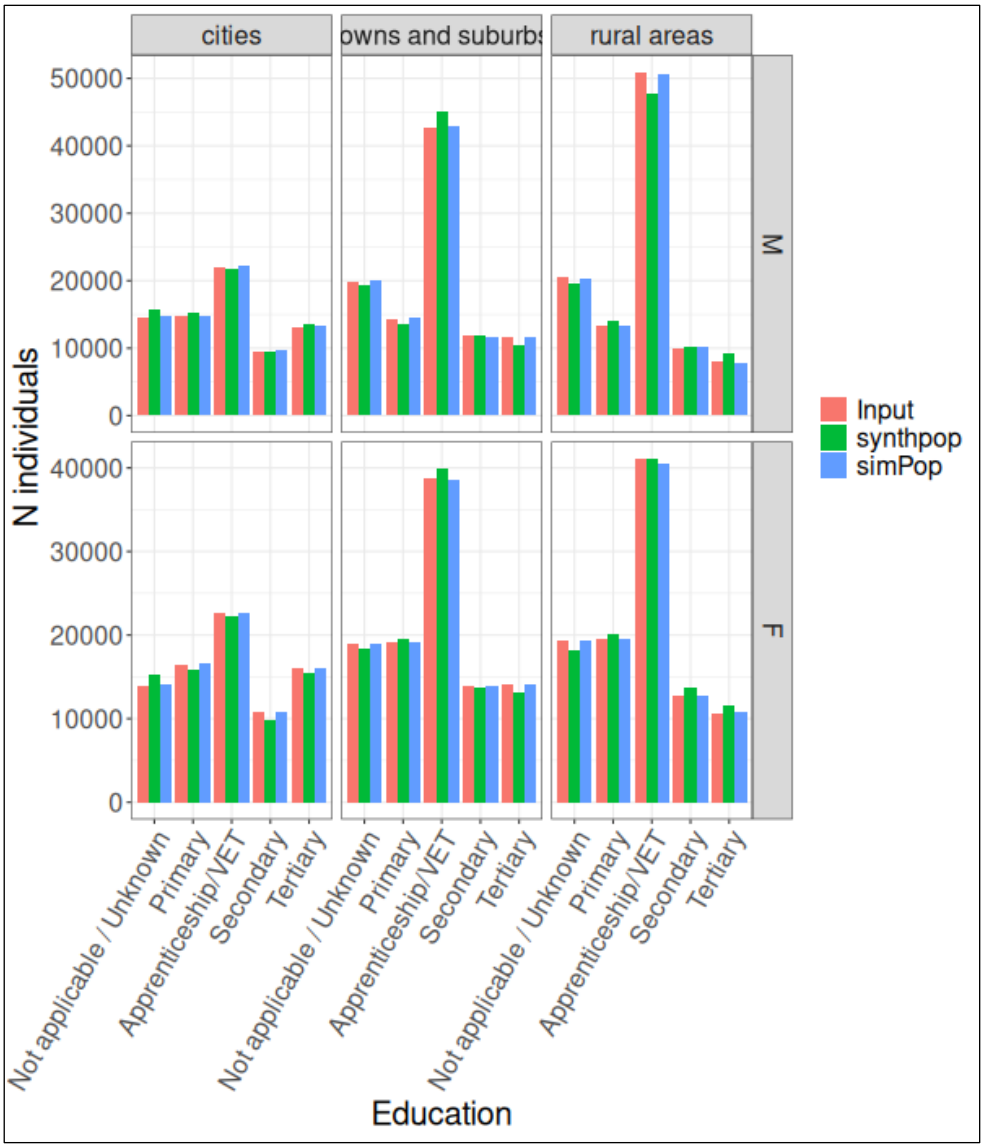


Figure 35: Number of individuals by highest educational attainment, sex and degree of urbanization.

The results for synthpop match the target distribution more closely than the other methods. Looking at the distribution of income (figure 36), the picture is a bit more diverse. The income distribution generated by synthpop matches the input data very closely whereas the results from simPop display not such a good fit. Using a more fine-tuned model with simPop can potentially improve these results.

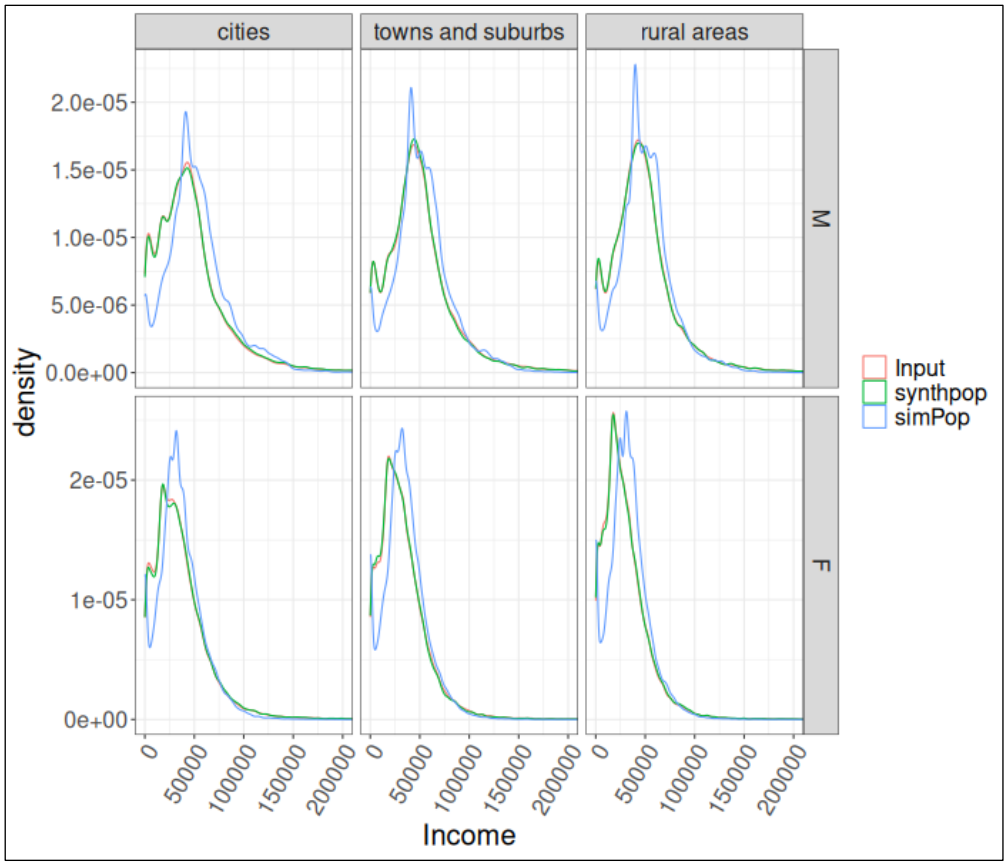


Figure 36: Distribution of income by sex and degree of urbanization.

Table 23 shows the share of replicated uniques, with respect to variables Age, Sex, District and Income. The results show that with the synthpop method the share of replicated uniques is much higher than with the simPop method.

Table 23 – Share of replicated uniques in the synthetic dataset.

Method	Share of repl. uniques (%)
simPop	0.19
synthPop	10.12

Table 24 shows the share of records at risk using Income as sensitive variable with and uninformed Model (Intercept) or Age + Sex + District as quasi-identifiers. It is clear to see, that the use of more Variables has little to no impact on the share of records at risk. Thus, the synthetic data, with respect to these variables, does not leak additional individual specific information.

Table 24 – Share of records at risk (in %), using different modelling variables (Model) and error metrics.

Error Metric	Model	simPop	synthpop
absolute	Age + Sex + District	1.61	1.63
absolute	Intercept	1.35	1.36
symmetric	Age + Sex + District	5.97	6.05
Symmetric	Intercept	4.96	5.10

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5 Code examples

- Example to demonstrate how to evaluate synthetic microdata with respect to utility and disclosure risk**.**
  [**Germany**, https://github.com/AIML4OS/WP13/tree/main/DE]

- Example of the application simPop to generate a synthetic “census-like” population and estimate some utility and risk measures
  [**Austria**, https://github.com/AIML4OS/WP13/tree/main/AT]

- Example of the ISTAT use case based on the “Trips and Holidays” survey dataset. The GitHub repository was created using the SSPCloud template.
  [**Italy**, https://github.com/AIML4OS/WP13_ISTAT_UC1]

- Example of the ISTAT use case focused on the development of a framework to assess the privacy–utility trade-off in synthetic data. The GitHub repository was created using the SSPCloud template.
  [**Italy**, https://github.com/AIML4OS/WP13_ISTAT_UC2]

6 List of Acronyms and Abbreviations

Abbreviation	Meaning
AIML4OS	Artificial Intelligence and Machine Learning for Official Statistics
AI	Artificial Intelligence
AUC	Area Under the Curve
BR	Baseline Records
CART	Classification and Regression Trees
CI	Confidence Interval
CIO	Confidence Interval Overlap
CKM	Cell Key Method
CPU	Central Processing Unit
CRAN	Comprehensive R Archive Network
CTGAN	Conditional Tabular Generative Adversarial Network
DCAP	Differential Correct Attribution Probability
DCR	Distance to Closest Record
DiSCO	Disclosive in Synthetic, Correct in Original
DiSDiO	Disclosive in Synthetic and Disclosive in Original
DP	Differential Privacy
DPI	Data Plagiarism Index
DP-CTGAN	Differentially Private Conditional Tabular Generative Adversarial Network
ECV	Encuesta de Condiciones de Vida
EEC	Enquête Emploi en Continu
ELBO	Evidence Lower Bound
EMR	Expected Match Risk
EU-SILC	European Union Statistics on Income and Living Conditions

Abbreviation	Meaning
FAMD	Factor Analysis of Mixed Data
FCS	Fully Conditional Specification
FMR	False Match Rate
FPR	False Positive Rate
GAN	Generative Adversarial Network
GD	Group Disclosure
GPU	Graphics Processing Unit
HBS	Household Budget Survey
IES	Simplified Business Information / Informação Empresarial Simplificada
IL_correl	Information loss in the correlation structure
IL_variables	Information loss per variable
IMS	Identical Match Share
INE	Instituto Nacional de Estadística
kNN	k-Nearest Neighbours
KL	Kullback-Leibler
LDA	Linear Discriminant Analysis
LFS	Labour Force Survey
LoF	Lack of Fit
LR	Logistic Regression
MAE	Mean Absolute Error
MASCD	Mean Absolute Standardized Coefficient Difference
MCIO	Mean Confidence Interval Overlap
MDR	Mahalanobis Distance Ratio
MIA	Membership Inference Attack
ML	Machine Learning

Abbreviation	Meaning
MLP	Multi-Layer Perceptron
NAF	Nomenclature d’activités française
NNDR	Nearest-Neighbour Distance Ratio
NUTS	Nomenclature of Territorial Units for Statistics
PCA	Principal Component Analysis
pMSE	Propensity Score Mean Squared Error
PoC	Proof of Concept
PUF	Public Use File
RAPID	Risk of Attribute Prediction-Induced Disclosure
RDC	Research Data Centre
RF	Random Forest
RGAN	Recurrent Generative Adversarial Network
RMSE	Root Mean Squared Error
ROC	Receiver Operating Characteristic
SBS	Structural Business Statistics
SDC	Statistical Disclosure Control
SD	Synthetic Data
SDV	Synthetic Data Vault
SDK	Software Development Kit
SCIE	Sistema de Contas Integradas das Empresas
SMOTE	Synthetic Minority Over-sampling Technique
SMOTE-NC	SMOTE for Nominal and Continuous features
SUF	Scientific Use File
TAPAS	Toolbox for Adversarial Privacy Auditing of Synthetic Data
TCAP	Targeted Correct Attribution Probability
TMR	True Match Rate

Abbreviation	Meaning
TPR	True Positive Rate
TR	Training Records
TVAE	Tabular Variational Autoencoder
VAE	Variational Autoencoder
VR	Victim Records
WEAP	Within Equivalence Class Attribution Probability
WP13	Work Package 13
XGBoost	Extreme Gradient Boosting